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[69]	DEBUG, VR HEADERS AND TEST POINTS
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[72]	LABELS AND MOUNTING

GREYBULL
REV 1.0
FAB M
RETAIL

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RULES: (APPLIED WHEN POSSIBLE)
1.) MSB TO LSB IS TOP TO BOTTOM
2.) WHEN POSSIBLE: INPUTS ON LEFT, OUTPUTS ON RIGHT
3.) ORDER OF PAGES=CHIP INTERFACES, TERMINATION, POWER, DECOUPLING
4.) AVOID USING OFF PAGE CONNECTORS FOR ON PAGE CONNECTIONS
5.) LANED SIGNALS ARE GROUPED ON SYMBOLS
6.) TRANSIMITTER NAME USED AS PREFIX WITH RX AND TX CONNECTIONS
7.) SUFFIX V_ IS USED FOR VOLTAGE RAIL SIGNAL NAMES
8.) SUFFIX _DP AND _DN ARE USED FOR DIFFERIENTAL PAIRS
9.) UNNAMED NETS ARE NAMED WITH /2 TEXT SIZE
10.) SUFFIX _N FOR ACTIVE LOW OR N JUNCTION
12.) SUFFIX _P FOR P JUNCTION
13.) SUFFIX _EN FOR ENABLE
14.) 'CLK' FOR CLOCKS, 'RST' FOR RESETS
15.) PWRGD FOR POWER GOOD
16.) REV AND FAB ARE SET USING CUSTOM VARIABLES
TOOLS>OPTIONS>VARIABLES

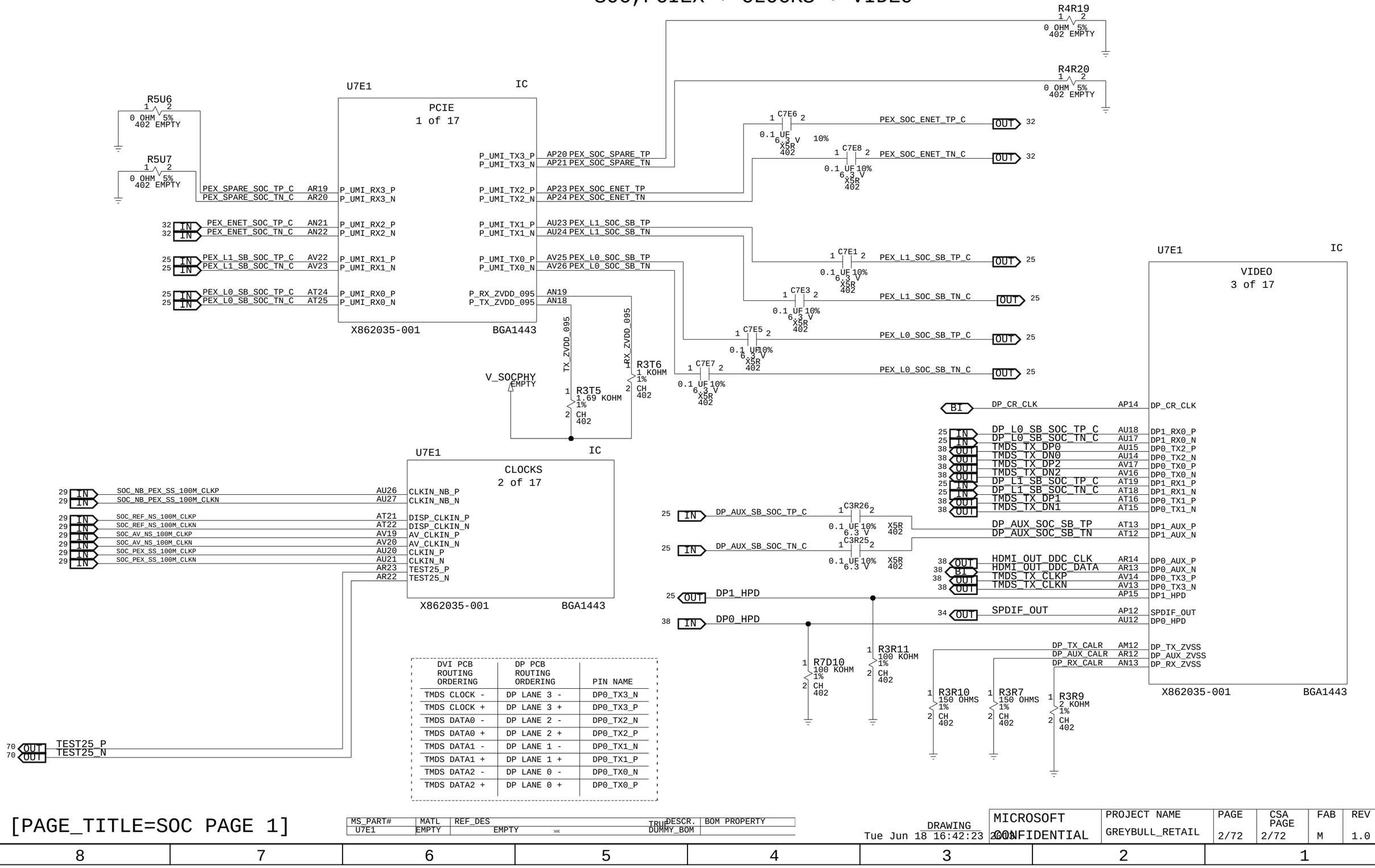
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PROJECT NAME	PAGE	CSA PAGE	FAB	REV
GREYBULL_RETAIL	1/72	1/72	M	1.0

SOC,PCIEX + CLOCKS + VIDEO



DVI PCB ROUTING ORDERING	DP PCB ROUTING ORDERING	PIN NAME
TMDS CLOCK -	DP LANE 3 -	DP0_TX3_N
TMDS CLOCK +	DP LANE 3 +	DP0_TX3_P
TMDS DATA0 -	DP LANE 2 -	DP0_TX2_N
TMDS DATA0 +	DP LANE 2 +	DP0_TX2_P
TMDS DATA1 -	DP LANE 1 -	DP0_TX1_N
TMDS DATA1 +	DP LANE 1 +	DP0_TX1_P
TMDS DATA2 -	DP LANE 0 -	DP0_TX0_N
TMDS DATA2 +	DP LANE 0 +	DP0_TX0_P

[PAGE_TITLE=SOC PAGE 1]

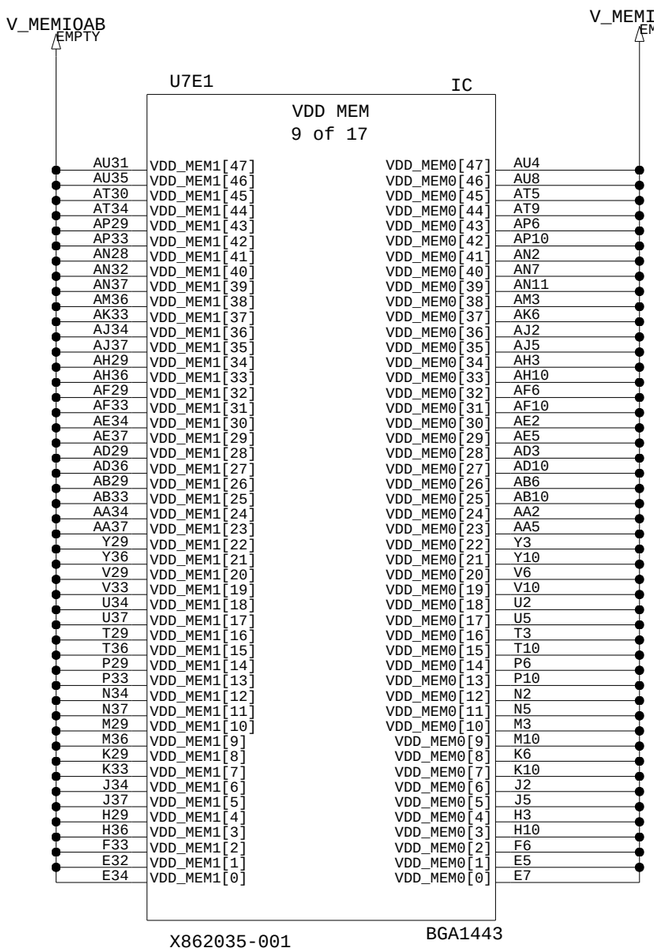
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U7E1	EMPTY	EMPTY	DUMMY_BOM	

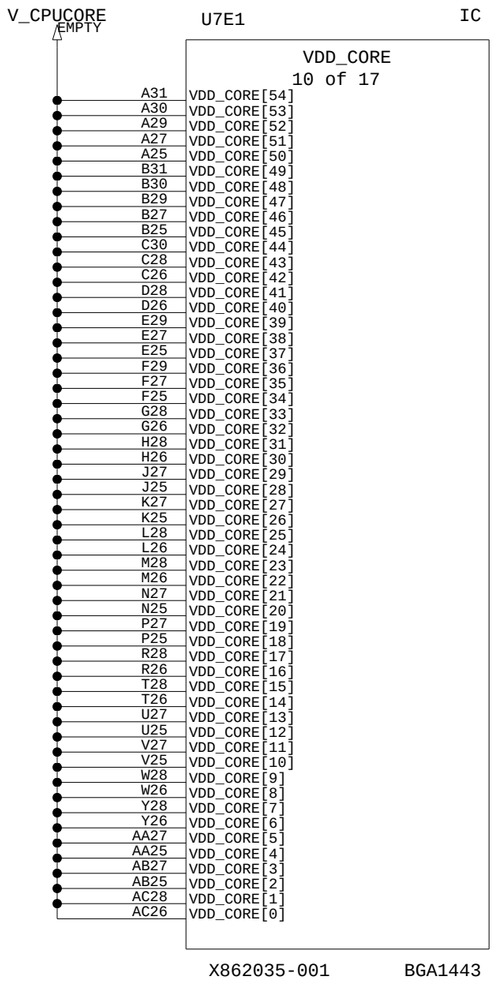
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Tue Jun 18 16:42:23 2008

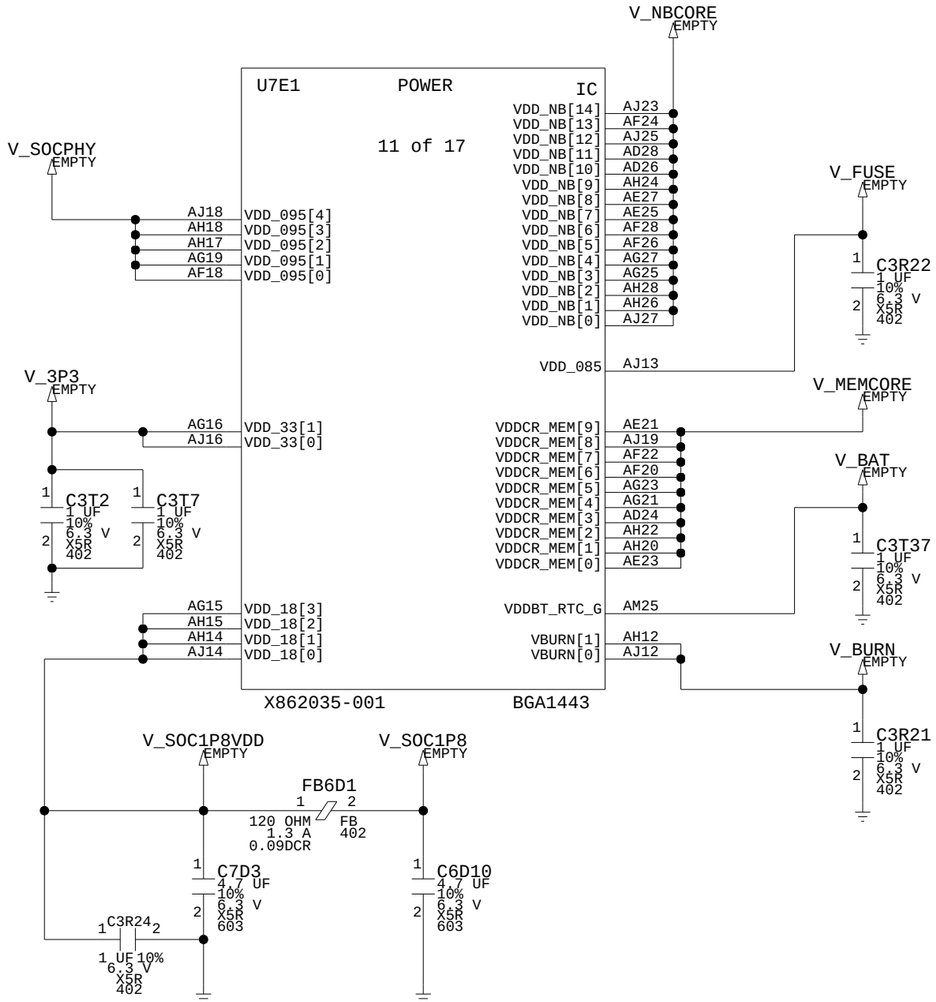
MICROSOFT CONFIDENTIAL

PROJECT NAME	PAGE	CSA PAGE	FAB	REV
GREYBULL_RETAIL	2/72	2/72	M	1.0

SOC POWER, MEM + CPUCORE + MEMCORE + NBCORE + MISC





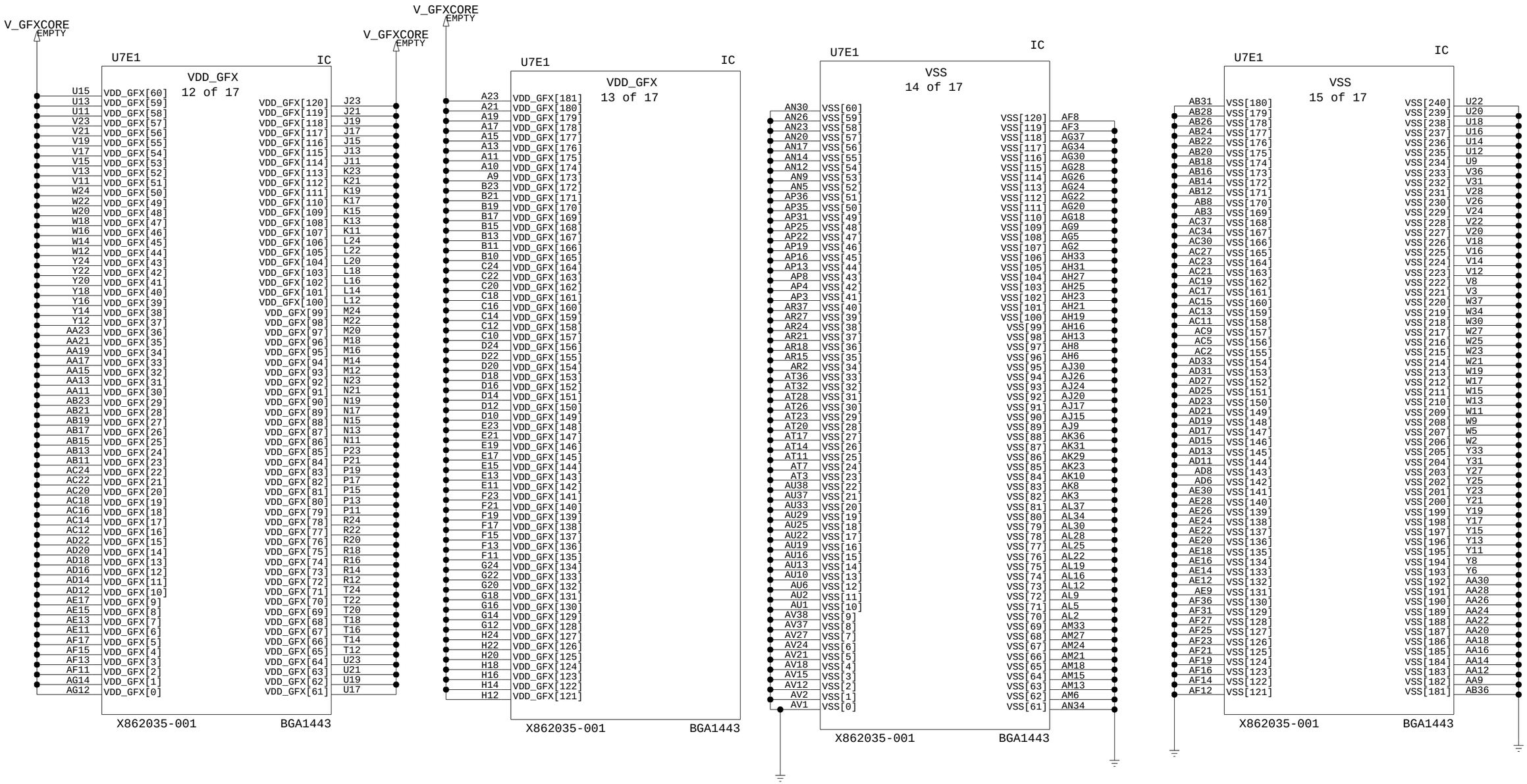


[PAGE_TITLE=SOC PAGE 2]

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MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	3/72	3/72	M	1.0

SOC POWER, V_GFXCORE + VSS



[PAGE_TITLE=SOC PAGE 3]

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Tue Jun 18 16:42:24 2019

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	4/72	4/72	M	1.0

SOC, MEMORY PARTITION C + D

U7E1 — IC — MEM CHANNEL D 6 of 17 — X862035-001 — BGA1443

MD_DQ63	P2	MD_DQ63	B1
MD_DQ62	W1	MD_DQ62	B1
MD_DQ61	N4	MD_DQ61	B1
MD_DQ60	V2	MD_DQ60	B1
MD_DQ59	N1	MD_DQ59	B1
MD_DQ58	W3	MD_DQ58	B1
MD_DQ57	P1	MD_DQ57	B1
MD_DQ56	W4	MD_DQ56	B1
MD_DQ55	Y2	MD_DQ55	B1
MD_DQ54	M2	MD_DQ54	B1
MD_DQ53	Y1	MD_DQ53	B1
MD_DQ52	M1	MD_DQ52	B1
MD_DQ51	T1	MD_DQ51	B1
MD_DQ50	N3	MD_DQ50	B1
MD_DQ49	U3	MD_DQ49	B1
MD_DQ48	R4	MD_DQ48	B1
MD_DQ47	M7	MD_DQ47	B1
MD_DQ46	R6	MD_DQ46	B1
MD_DQ45	M5	MD_DQ45	B1
MD_DQ44	P7	MD_DQ44	B1
MD_DQ43	M4	MD_DQ43	B1
MD_DQ42	R8	MD_DQ42	B1
MD_DQ41	N6	MD_DQ41	B1
MD_DQ40	T4	MD_DQ40	B1
MD_DQ39	L4	MD_DQ39	B1
MD_DQ38	H1	MD_DQ38	B1
MD_DQ37	L3	MD_DQ37	B1
MD_DQ36	G4	MD_DQ36	B1
MD_DQ35	L1	MD_DQ35	B1
MD_DQ34	H2	MD_DQ34	B1
MD_DQ33	K1	MD_DQ33	B1
MD_DQ32	J1	MD_DQ32	B1
MD_DQ31	H5	MD_DQ31	B1
MD_DQ30	K5	MD_DQ30	B1
MD_DQ29	H4	MD_DQ29	B1
MD_DQ28	K4	MD_DQ28	B1
MD_DQ27	G8	MD_DQ27	B1
MD_DQ26	L6	MD_DQ26	B1
MD_DQ25	H7	MD_DQ25	B1
MD_DQ24	L8	MD_DQ24	B1
MD_DQ23	G3	MD_DQ23	B1
MD_DQ22	C1	MD_DQ22	B1
MD_DQ21	G1	MD_DQ21	B1
MD_DQ20	C3	MD_DQ20	B1
MD_DQ19	F1	MD_DQ19	B1
MD_DQ18	D2	MD_DQ18	B1
MD_DQ17	E1	MD_DQ17	B1
MD_DQ16	D1	MD_DQ16	B1
MD_DQ15	C8	MD_DQ15	B1
MD_DQ14	D5	MD_DQ14	B1
MD_DQ13	E8	MD_DQ13	B1
MD_DQ12	G6	MD_DQ12	B1
MD_DQ11	D9	MD_DQ11	B1
MD_DQ10	F5	MD_DQ10	B1
MD_DQ9	B7	MD_DQ9	B1
MD_DQ8	F4	MD_DQ8	B1
MD_DQ7	A3	MD_DQ7	B1
MD_DQ6	D6	MD_DQ6	B1
MD_DQ5	B3	MD_DQ5	B1
MD_DQ4	A7	MD_DQ4	B1
MD_DQ3	D4	MD_DQ3	B1
MD_DQ2	C6	MD_DQ2	B1
MD_DQ1	C4	MD_DQ1	B1
MD_DQ0	A6	MD_DQ0	B1
MD_DQSL_0_P	B5	MD_DQSL_0_P	B1
MD_DQSL_0_N	A5	MD_DQSL_0_N	B1
MD_DQSL_1_P	E3	MD_DQSL_1_P	B1
MD_DQSL_1_N	E4	MD_DQSL_1_N	B1
MD_DQSL_2_P	J3	MD_DQSL_2_P	B1
MD_DQSL_2_N	J4	MD_DQSL_2_N	B1
MD_DQSL_3_P	R1	MD_DQSL_3_P	B1
MD_DQSL_3_N	T2	MD_DQSL_3_N	B1
MD_DQSU_0_P	E6	MD_DQSU_0_P	B1
MD_DQSU_0_N	F7	MD_DQSU_0_N	B1
MD_DQSU_1_P	J8	MD_DQSU_1_P	B1
MD_DQSU_1_N	K7	MD_DQSU_1_N	B1
MD_DQSU_2_P	P4	MD_DQSU_2_P	B1
MD_DQSU_2_N	P5	MD_DQSU_2_N	B1
MD_DQSU_3_P	U1	MD_DQSU_3_P	B1
MD_DQSU_3_N	V1	MD_DQSU_3_N	B1
MD_DML_0	A4	MD_DML_0	B1
MD_DMU_0	D7	MD_DMU_0	B1
MD_DML_1	F2	MD_DML_1	B1
MD_DMU_1	J6	MD_DMU_1	B1
MD_DML_2	K2	MD_DML_2	B1
MD_DMU_2	N8	MD_DMU_2	B1
MD_DML_3	U4	MD_DML_3	B1
MD_DMU_3	R3	MD_DMU_3	B1

V_MEMIOCD EMPTY
R3R14 40.2 OHM 1% 2 EMPTY 402
R3R13 40.2 OHM 1% 2 EMPTY 402

MD_TEST	AG11	MD_TEST	
MD_CLK0_DP	K9	MD_CLK0_DP	OUT 12 13
MD_CLK0_DN	J10	MD_CLK0_DN	OUT 12 13
MD_A15	V7	MD_A<15>	B1 12 13 14
MD_A14	W10	MD_A<14>	B1 12 13 14
MD_A13	U10	MD_A<13>	B1 12 13 14
MD_A12	W6	MD_A<12>	B1 12 13 14
MD_A11	Y5	MD_A<11>	B1 12 13 14
MD_A10	N10	MD_A<10>	B1 12 13 14
MD_A9	V9	MD_A<9>	B1 12 13 14
MD_A8	W7	MD_A<8>	B1 12 13 14
MD_A7	U7	MD_A<7>	B1 12 13 14
MD_A6	Y7	MD_A<6>	B1 12 13 14
MD_A5	V4	MD_A<5>	B1 12 13 14
MD_A4	R10	MD_A<4>	B1 12 13 14
MD_A3	U8	MD_A<3>	B1 12 13 14
MD_A2	Y4	MD_A<2>	B1 12 13 14
MD_A1	W8	MD_A<1>	B1 12 13 14
MD_A0	T9	MD_A<0>	B1 12 13 14
MD_BA0	T7	MD_BA<0>	B1 12 13 14
MD_BA1	V5	MD_BA<1>	B1 12 13 14
MD_BA2	U6	MD_BA<2>	B1 12 13 14
MD_CKE0	N7	MD_CKE0	OUT 12 13 14
MD_CAS_N	L10	MD_CAS_N	OUT
MD_RAS_N	M9	MD_RAS_N	OUT
MD_CS0_N	R7	MD_CS0_N	OUT 12 13 14
MD_ODT0	T5	MD_ODT0	OUT 12 13 14
MD_WE_N	P9	MD_WE_N	OUT 12 13 14
MD_RESET	Y9	MD_RESET_N	OUT 12 13
MD_CKE1	H9		
MD_CS1_N	F9		
MD_ODT1	G10		
MD_VREFDQ	G7	MD_VREFDQ	1 2 MD_VREFDQ_MM OUT
MD_ZVDDIO	J7	MD_ZVDDIO	0 OHM 5% 402 EMPTY

V_MEMIOCD EMPTY
R2R5 0 OHM 5% 402 EMPTY
V_MEMIOCD EMPTY
R2R6 40.2 OHM 1% 2 CH 402

U7E1 — IC — MEM CHANNEL C 5 of 17 — X862035-001 — BGA1443

MC_DQ63	AN8	MC_DQ63	B1
MC_DQ62	AM11	MC_DQ62	B1
MC_DQ61	AM8	MC_DQ61	B1
MC_DQ60	AN10	MC_DQ60	B1
MC_DQ59	AR7	MC_DQ59	B1
MC_DQ58	AP11	MC_DQ58	B1
MC_DQ57	AP9	MC_DQ57	B1
MC_DQ56	AR11	MC_DQ56	B1
MC_DQ55	AV11	MC_DQ55	B1
MC_DQ54	AR8	MC_DQ54	B1
MC_DQ53	AU11	MC_DQ53	B1
MC_DQ52	AV7	MC_DQ52	B1
MC_DQ51	AV10	MC_DQ51	B1
MC_DQ50	AT8	MC_DQ50	B1
MC_DQ49	AR10	MC_DQ49	B1
MC_DQ48	AV8	MC_DQ48	B1
MC_DQ47	AT2	MC_DQ47	B1
MC_DQ46	AN6	MC_DQ46	B1
MC_DQ45	AP2	MC_DQ45	B1
MC_DQ44	AN4	MC_DQ44	B1
MC_DQ43	AT1	MC_DQ43	B1
MC_DQ42	AM7	MC_DQ42	B1
MC_DQ41	AN3	MC_DQ41	B1
MC_DQ40	AP7	MC_DQ40	B1
MC_DQ39	AU7	MC_DQ39	B1
MC_DQ38	AV3	MC_DQ38	B1
MC_DQ37	AV6	MC_DQ37	B1
MC_DQ36	AU3	MC_DQ36	B1
MC_DQ35	AT6	MC_DQ35	B1
MC_DQ34	AR4	MC_DQ34	B1
MC_DQ33	AV5	MC_DQ33	B1
MC_DQ32	AT4	MC_DQ32	B1
MC_DQ31	AJ8	MC_DQ31	B1
MC_DQ30	AL6	MC_DQ30	B1
MC_DQ29	AH4	MC_DQ29	B1
MC_DQ28	AL8	MC_DQ28	B1
MC_DQ27	AH5	MC_DQ27	B1
MC_DQ26	AM5	MC_DQ26	B1
MC_DQ25	AJ6	MC_DQ25	B1
MC_DQ24	AM4	MC_DQ24	B1
MC_DQ23	AR1	MC_DQ23	B1
MC_DQ22	AK1	MC_DQ22	B1
MC_DQ21	AP1	MC_DQ21	B1
MC_DQ20	AJ4	MC_DQ20	B1
MC_DQ19	AN1	MC_DQ19	B1
MC_DQ18	AK2	MC_DQ18	B1
MC_DQ17	AM1	MC_DQ17	B1
MC_DQ16	AL1	MC_DQ16	B1
MC_DQ15	AC1	MC_DQ15	B1
MC_DQ14	AG4	MC_DQ14	B1
MC_DQ13	AB2	MC_DQ13	B1
MC_DQ12	AG3	MC_DQ12	B1
MC_DQ11	AB1	MC_DQ11	B1
MC_DQ10	AH1	MC_DQ10	B1
MC_DQ9	AC3	MC_DQ9	B1
MC_DQ8	AH2	MC_DQ8	B1
MC_DQ7	AJ3	MC_DQ7	B1
MC_DQ6	AA3	MC_DQ6	B1
MC_DQ5	AJ1	MC_DQ5	B1
MC_DQ4	AA1	MC_DQ4	B1
MC_DQ3	AE3	MC_DQ3	B1
MC_DQ2	AA4	MC_DQ2	B1
MC_DQ1	AE4	MC_DQ1	B1
MC_DQ0	AD1	MC_DQ0	B1
MC_DQSL_0_P	AD2	MC_DQSL_0_P	B1
MC_DQSL_0_N	AE1	MC_DQSL_0_N	B1
MC_DQSL_1_P	AL3	MC_DQSL_1_P	B1
MC_DQSL_1_N	AL4	MC_DQSL_1_N	B1
MC_DQSL_2_P	AV4	MC_DQSL_2_P	B1
MC_DQSL_2_N	AU5	MC_DQSL_2_N	B1
MC_DQSL_3_P	AU9	MC_DQSL_3_P	B1
MC_DQSL_3_N	AV9	MC_DQSL_3_N	B1
MC_DQSU_0_P	AF2	MC_DQSU_0_P	B1
MC_DQSU_0_N	AG1	MC_DQSU_0_N	B1
MC_DQSU_1_P	AK4	MC_DQSU_1_P	B1
MC_DQSU_1_N	AK5	MC_DQSU_1_N	B1
MC_DQSU_2_P	AP5	MC_DQSU_2_P	B1
MC_DQSU_2_N	AR5	MC_DQSU_2_N	B1
MC_DQSU_3_P	AM9	MC_DQSU_3_P	B1
MC_DQSU_3_N	AM10	MC_DQSU_3_N	B1
MC_DML_0	AF1	MC_DML_0	B1
MC_DMU_0	AC4	MC_DMU_0	B1
MC_DML_1	AM2	MC_DML_1	B1
MC_DMU_1	AK7	MC_DMU_1	B1
MC_DML_2	AR6	MC_DML_2	B1
MC_DMU_2	AR3	MC_DMU_2	B1
MC_DML_3	AT10	MC_DML_3	B1
MC_DMU_3	AR9	MC_DMU_3	B1

V_MEMIOCD EMPTY
R3R16 40.2 OHM 1% 2 EMPTY 402
R3R15 40.2 OHM 1% 2 EMPTY 402

MC_TEST	AJ11	MC_TEST	
MC_CLK0_DP	AK9	MC_CLK0_DP	OUT 15 16
MC_CLK0_DN	AJ10	MC_CLK0_DN	OUT 15 16
MC_A15	AC8	MC_A<15>	B1 15 16 17
MC_A14	AA7	MC_A<14>	B1 15 16 17
MC_A13	AA10	MC_A<13>	B1 15 16 17
MC_A12	AC6	MC_A<12>	B1 15 16 17
MC_A11	AB4	MC_A<11>	B1 15 16 17
MC_A10	AE10	MC_A<10>	B1 15 16 17
MC_A9	AC7	MC_A<9>	B1 15 16 17
MC_A8	AB9	MC_A<8>	B1 15 16 17
MC_A7	AD9	MC_A<7>	B1 15 16 17
MC_A6	AA8	MC_A<6>	B1 15 16 17
MC_A5	AD5	MC_A<5>	B1 15 16 17
MC_A4	AC10	MC_A<4>	B1 15 16 17
MC_A3	AD7	MC_A<3>	B1 15 16 17
MC_A2	AB5	MC_A<2>	B1 15 16 17
MC_A1	AB7	MC_A<1>	B1 15 16 17
MC_A0	AE7	MC_A<0>	B1 15 16 17
MC_BA0	AE8	MC_BA<0>	B1 15 16 17
MC_BA1	AD4	MC_BA<1>	B1 15 16 17
MC_BA2	AE6	MC_BA<2>	B1 15 16 17
MC_CKE0	AH9	MC_CKE0	OUT
MC_CAS_N	AG10	MC_CAS_N	OUT 15 16 17
MC_RAS_N	AJ7	MC_RAS_N	OUT 15 16 17
MC_CS0_N	AF9	MC_CS0_N	OUT 15 16 17
MC_ODT0	AF4	MC_ODT0	OUT 15 16 17
MC_WE_N	AG7	MC_WE_N	OUT 15 16 17
MC_RESET	AA6	MC_RESET_N	OUT 15 16
MC_CKE1	AH7		
MC_CS1_N	AF7		
MC_ODT1	AG8		
MC_VREFDQ	AL7	MC_VREFDQ	1 2 MC_VREFDQ_MM OUT
MC_ZVDDIO	AL10	MC_ZVDDIO	0 OHM 5% 402 EMPTY

V_MEMIOCD EMPTY
R3R5 40.2 OHM 1% 2 CH 402
R3R4 0 OHM 5% 402 EMPTY

[PAGE_TITLE=SOC MEMORY PARTITION A & B]

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CONFIDENTIAL

MICROSOFT	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	5/72	5/72	M	1.0

SOC, MEMORY PARTITION A + B

U7E1 IC — MEM CHANNEL B 7 of 17

19	BI	MB_DQ63	V37
19	BI	MB_DQ62	N38
19	BI	MB_DQ61	W38
19	BI	MB_DQ60	P37
19	BI	MB_DQ59	W36
19	BI	MB_DQ58	N35
19	BI	MB_DQ57	V38
19	BI	MB_DQ56	N36
19	BI	MB_DQ55	M37
19	BI	MB_DQ54	Y38
19	BI	MB_DQ53	M38
19	BI	MB_DQ52	Y37
19	BI	MB_DQ51	T38
19	BI	MB_DQ50	W35
19	BI	MB_DQ49	R35
19	BI	MB_DQ48	U36
19	BI	MB_DQ47	R33
19	BI	MB_DQ46	M32
19	BI	MB_DQ45	R31
19	BI	MB_DQ44	N33
19	BI	MB_DQ43	T35
19	BI	MB_DQ42	M34
19	BI	MB_DQ41	P32
19	BI	MB_DQ40	M35
19	BI	MB_DQ39	G35
19	BI	MB_DQ38	L38
19	BI	MB_DQ37	H37
19	BI	MB_DQ36	L35
19	BI	MB_DQ35	H38
19	BI	MB_DQ34	L36
19	BI	MB_DQ33	J35
19	BI	MB_DQ32	K37
18	BI	MB_DQ31	K34
18	BI	MB_DQ30	H34
18	BI	MB_DQ29	L33
18	BI	MB_DQ28	H32
18	BI	MB_DQ27	L31
18	BI	MB_DQ26	H35
18	BI	MB_DQ25	K35
18	BI	MB_DQ24	G31
18	BI	MB_DQ23	C36
18	BI	MB_DQ22	G36
18	BI	MB_DQ21	C38
18	BI	MB_DQ20	G38
18	BI	MB_DQ19	E35
18	BI	MB_DQ18	F38
18	BI	MB_DQ17	D38
18	BI	MB_DQ16	E38
18	BI	MB_DQ15	D34
18	BI	MB_DQ14	C31
18	BI	MB_DQ13	F34
18	BI	MB_DQ12	B32
18	BI	MB_DQ11	F35
18	BI	MB_DQ10	E31
18	BI	MB_DQ9	G33
18	BI	MB_DQ8	D30
18	BI	MB_DQ7	A32
18	BI	MB_DQ6	B36
18	BI	MB_DQ5	D33
18	BI	MB_DQ4	A36
18	BI	MB_DQ3	D35
18	BI	MB_DQ2	A35
18	BI	MB_DQ1	A33
18	BI	MB_DQ0	C35
18	BI	MB_DQSL_0_P	A34
18	BI	MB_DQSL_0_N	B34
18	BI	MB_DQSL_1_P	F37
18	BI	MB_DQSL_1_N	E36
19	BI	MB_DQSL_2_P	K38
19	BI	MB_DQSL_2_N	J38
19	BI	MB_DQSL_3_P	U38
19	BI	MB_DQSL_3_N	T37
18	BI	MB_DQSU_0_P	E33
18	BI	MB_DQSU_0_N	D32
18	BI	MB_DQSU_1_P	J31
18	BI	MB_DQSU_1_N	J33
19	BI	MB_DQSU_2_P	P35
19	BI	MB_DQSU_2_N	P34
19	BI	MB_DQSU_3_P	R38
19	BI	MB_DQSU_3_N	P38
18	BI	MB_DML_0	C33
18	BI	MB_DMU_0	F32
18	BI	MB_DML_1	D37
18	BI	MB_DMU_1	K32
19	BI	MB_DML_2	J36
19	BI	MB_DMU_2	N31
19	BI	MB_DML_3	R36
19	BI	MB_DMU_3	U35

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V_MEMIOAB

EMPTY

1 R3T10 40.2 OHM 1% 2 EMPTY 402

MB_TEST AJ28 MB_TEST

1 R3T13 40.2 OHM 1% 2 EMPTY 402

MB_CLK0_DP	K30	MB_CLK0_DP	OUT	18 19
MB_CLK0_DN	J29	MB_CLK0_DN	OUT	18 19
MB_A15	V32	MB_A<15>	BI	18 19 20
MB_A14	W29	MB_A<14>	BI	18 19 20
MB_A13	U29	MB_A<13>	BI	18 19 20
MB_A12	W33	MB_A<12>	BI	18 19 20
MB_A11	Y34	MB_A<11>	BI	18 19 20
MB_A10	N29	MB_A<10>	BI	18 19 20
MB_A9	V30	MB_A<9>	BI	18 19 20
MB_A8	W32	MB_A<8>	BI	18 19 20
MB_A7	U32	MB_A<7>	BI	18 19 20
MB_A6	Y32	MB_A<6>	BI	18 19 20
MB_A5	V35	MB_A<5>	BI	18 19 20
MB_A4	R29	MB_A<4>	BI	18 19 20
MB_A3	U31	MB_A<3>	BI	18 19 20
MB_A2	Y35	MB_A<2>	BI	18 19 20
MB_A1	W31	MB_A<1>	BI	18 19 20
MB_A0	T30	MB_A<0>	BI	18 19 20
MB_BA0	T32	MB_BA<0>	BI	18 19 20
MB_BA1	V34	MB_BA<1>	BI	18 19 20
MB_BA2	U33	MB_BA<2>	BI	18 19 20
MB_CKE0	N32	MB_CKE0	OUT	20
MB_CAS_N	L29	MB_CAS_N	OUT	19
MB_RAS_N	M30	MB_RAS_N	OUT	18
MB_CS0_N	R32	MB_CS0_N	OUT	18 19 20
MB_ODT0	T34	MB_ODT0	OUT	18 19 20
MB_WE_N	P30	MB_WE_N	OUT	18 19 20
MB_RESET	Y30	MB_RESET_N	OUT	18 19
MB_CKE1	H30			
MB_CS1_N	F30			
MB_ODT1	G29			

V_MEMIOAB

EMPTY

1 R2T1 40.2 OHM 1% 2 CH 402

R2T2

MB_VREFDQ G32 MB_VREFDQ 1 2 MB_VREFDQ_MM OUT

MB_ZVDDIO J32 0 OHM 5% MB_ZVDDIO

402 EMPTY

U7E1 IC — MEM CHANNEL A 8 of 17

22	BI	MA_DQ63	AM28
22	BI	MA_DQ62	AN31
22	BI	MA_DQ61	AP28
22	BI	MA_DQ60	AM30
22	BI	MA_DQ59	AR28
22	BI	MA_DQ58	AM31
22	BI	MA_DQ57	AN29
22	BI	MA_DQ56	AR32
22	BI	MA_DQ55	AV32
22	BI	MA_DQ54	AU28
22	BI	MA_DQ53	AR31
22	BI	MA_DQ52	AV28
22	BI	MA_DQ51	AT31
22	BI	MA_DQ50	AV29
22	BI	MA_DQ49	AR29
22	BI	MA_DQ48	AT29
22	BI	MA_DQ47	AN33
22	BI	MA_DQ46	AT37
22	BI	MA_DQ45	AM32
22	BI	MA_DQ44	AR36
22	BI	MA_DQ43	AP32
22	BI	MA_DQ42	AP37
22	BI	MA_DQ41	AR34
22	BI	MA_DQ40	AT38
22	BI	MA_DQ39	AU36
22	BI	MA_DQ38	AV33
22	BI	MA_DQ37	AV36
22	BI	MA_DQ36	AU32
22	BI	MA_DQ35	AR35
22	BI	MA_DQ34	AT33
22	BI	MA_DQ33	AV35
22	BI	MA_DQ32	AR33
21	BI	MA_DQ31	AL33
21	BI	MA_DQ30	AJ31
21	BI	MA_DQ29	AM34
21	BI	MA_DQ28	AJ33
21	BI	MA_DQ27	AM35
21	BI	MA_DQ26	AH35
21	BI	MA_DQ25	AL31
21	BI	MA_DQ24	AH34
21	BI	MA_DQ23	AJ35
21	BI	MA_DQ22	AP38
21	BI	MA_DQ21	AK38
21	BI	MA_DQ20	AR38
21	BI	MA_DQ19	AK37
21	BI	MA_DQ18	AM37
21	BI	MA_DQ17	AM38
21	BI	MA_DQ16	AN38
21	BI	MA_DQ15	AG36
21	BI	MA_DQ14	AB37
21	BI	MA_DQ13	AG35
21	BI	MA_DQ12	AC38
21	BI	MA_DQ11	AH38
21	BI	MA_DQ10	AB38
21	BI	MA_DQ9	AG38
21	BI	MA_DQ8	AA35
21	BI	MA_DQ7	AA38
21	BI	MA_DQ6	AJ38
21	BI	MA_DQ5	AA36
21	BI	MA_DQ4	AJ36
21	BI	MA_DQ3	AE38
21	BI	MA_DQ2	AH37
21	BI	MA_DQ1	AD37
21	BI	MA_DQ0	AF38
21	BI	MA_DQSL_0_P	AE35
21	BI	MA_DQSL_0_N	AE36
21	BI	MA_DQSL_1_P	AL35
21	BI	MA_DQSL_1_N	AL36
22	BI	MA_DQSL_2_P	AV34
22	BI	MA_DQSL_2_N	AU34
22	BI	MA_DQSL_3_P	AU30
22	BI	MA_DQSL_3_N	AV30
21	BI	MA_DQSU_0_P	AC35
21	BI	MA_DQSU_0_N	AC36
21	BI	MA_DQSU_1_P	AK34
21	BI	MA_DQSU_1_N	AK35
22	BI	MA_DQSU_2_P	AN35
22	BI	MA_DQSU_2_N	AN36
22	BI	MA_DQSU_3_P	AR30
22	BI	MA_DQSU_3_N	AP30
21	BI	MA_DML_0	AD38
21	BI	MA_DMU_0	AF37
21	BI	MA_DML_1	AL38
21	BI	MA_DMU_1	AK32
22	BI	MA_DML_2	AT35
22	BI	MA_DMU_2	AP34
22	BI	MA_DML_3	AV31
22	BI	MA_DMU_3	AM29

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V_MEMIOAB

EMPTY

1 R3T12 40.2 OHM 1% 2 EMPTY 402

MA_TEST AK28 MA_TEST

1 R3T14 40.2 OHM 1% 2 EMPTY 402

MA_CLK0_DP	AK30	MA_CLK0_DP	OUT	21 22
MA_CLK0_DN	AJ29	MA_CLK0_DN	OUT	21 22
MA_A15	AC31	MA_A<15>	BI	21 22 23
MA_A14	AA32	MA_A<14>	BI	21 22 23
MA_A13	AA29	MA_A<13>	BI	21 22 23
MA_A12	AC33	MA_A<12>	BI	21 22 23
MA_A11	AB35	MA_A<11>	BI	21 22 23
MA_A10	AE29	MA_A<10>	BI	21 22 23
MA_A9	AC32	MA_A<9>	BI	21 22 23
MA_A8	AB30	MA_A<8>	BI	21 22 23
MA_A7	AD30	MA_A<7>	BI	21 22 23
MA_A6	AA31	MA_A<6>	BI	21 22 23
MA_A5	AD34	MA_A<5>	BI	21 22 23
MA_A4	AC29	MA_A<4>	BI	21 22 23
MA_A3	AD32	MA_A<3>	BI	21 22 23
MA_A2	AB34	MA_A<2>	BI	21 22 23
MA_A1	AB32	MA_A<1>	BI	21 22 23
MA_A0	AE32	MA_A<0>	BI	21 22 23
MA_BA0	AE31	MA_BA<0>	BI	21 22 23
MA_BA1	AD35	MA_BA<1>	BI	21 22 23
MA_BA2	AE33	MA_BA<2>	BI	21 22 23
MA_CKE0	AH30	MA_CKE0	OUT	23 23
MA_CAS_N	AG29	MA_CAS_N	OUT	22 21
MA_RAS_N	AJ32	MA_RAS_N	OUT	21 22
MA_CS0_N	AF30	MA_CS0_N	OUT	21 22 23
MA_ODT0	AF35	MA_ODT0	OUT	21 22 23
MA_WE_N	AG32	MA_WE_N	OUT	21 22 23
MA_RESET	AA33	MA_RESET_N	OUT	21 22
MA_CKE1	AH32			
MA_CS1_N	AF32			
MA_ODT1	AG31			

V_MEMIOAB

EMPTY

1 R3T17 40.2 OHM 1% 2 CH 402

R3T18

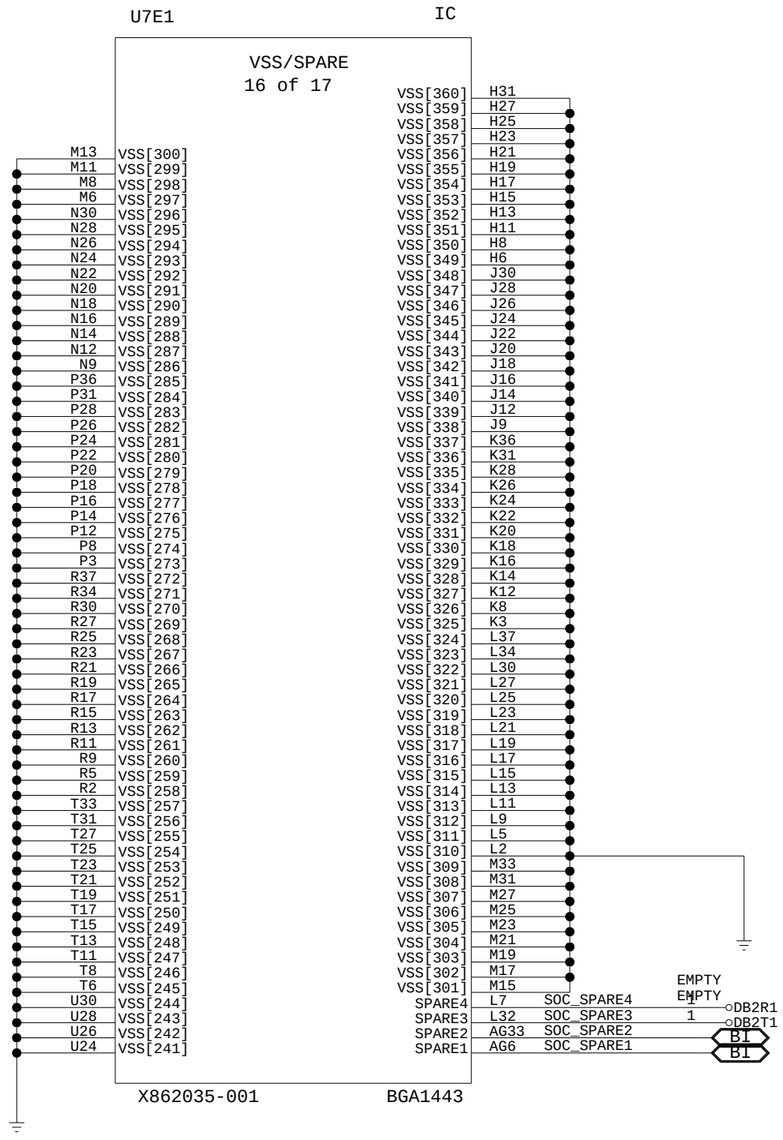
MA_VREFDQ AL32 MA_VREFDQ 1 2 MA_VREFDQ_MM OUT

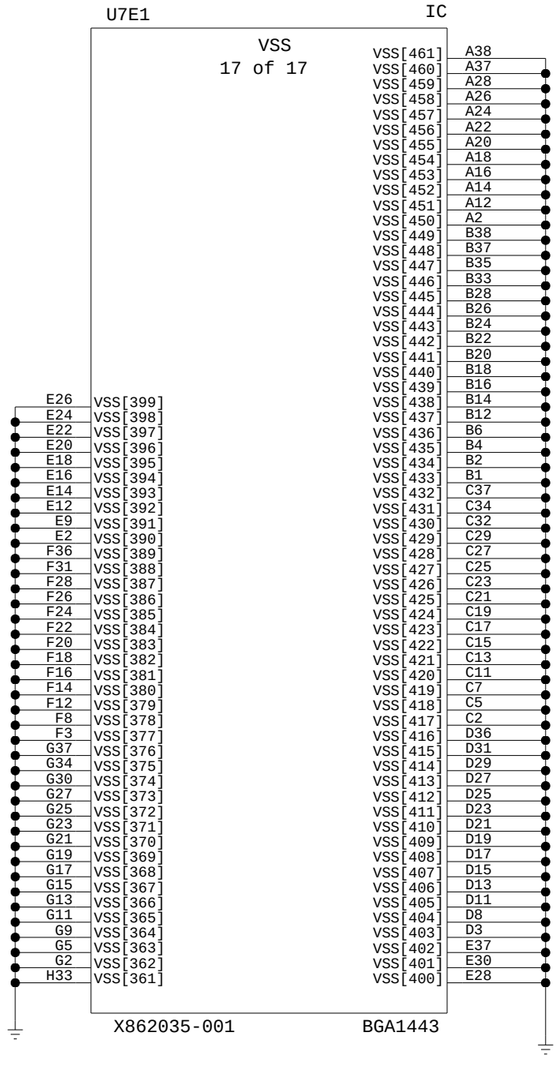
MA_ZVDDIO AL29 0 OHM 5% MA_ZVDDIO

402 EMPTY

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	6/72	6/72	M	1.0

SOC VSS + SPARE



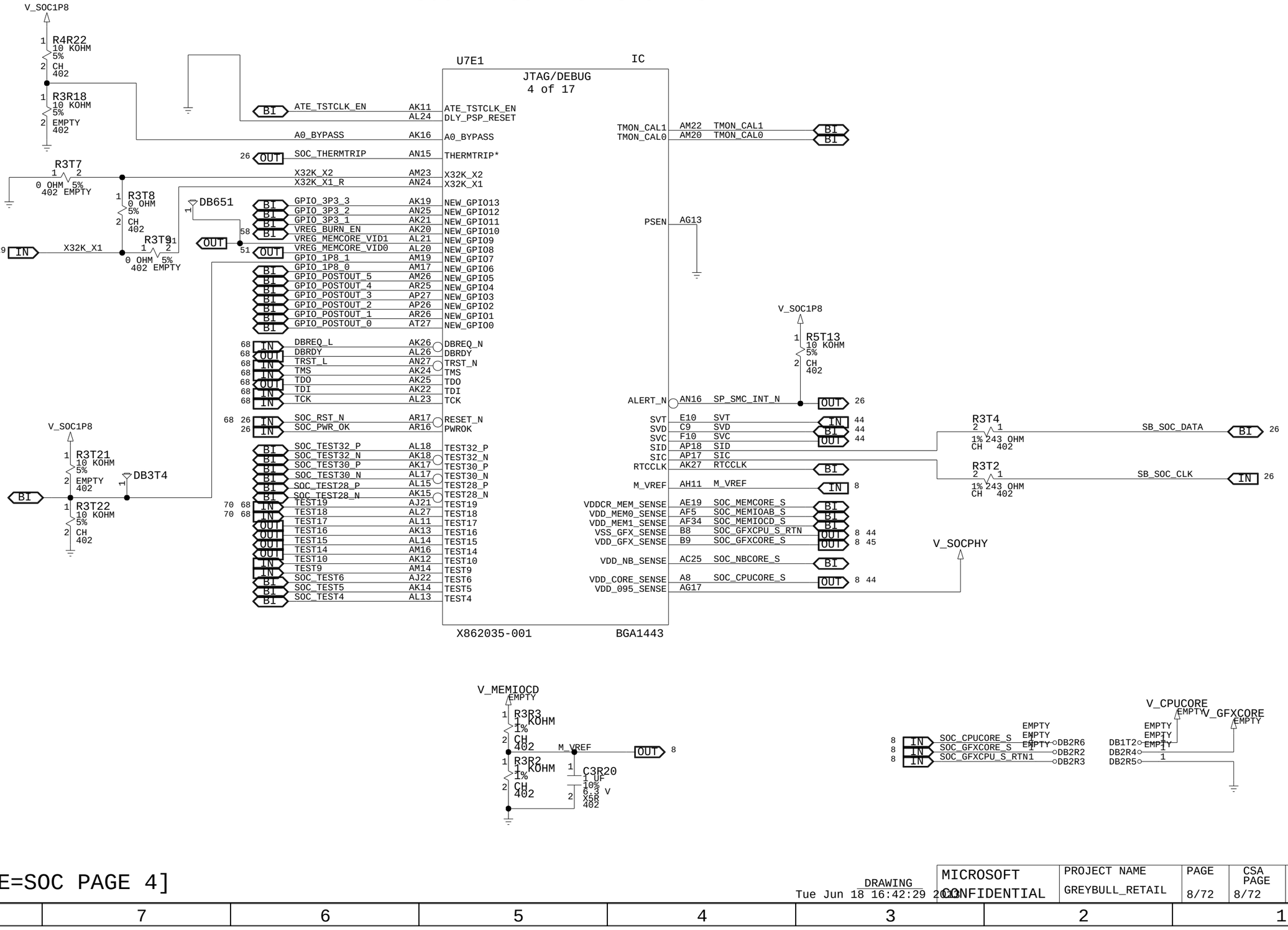


[PAGE_TITLE=SOC PAGE 4]

DRAWING
Tue Jun 18 16:42:24 2013

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	7/72	7/72	M	1.0

SOC DEBUG + SB SIGNALS

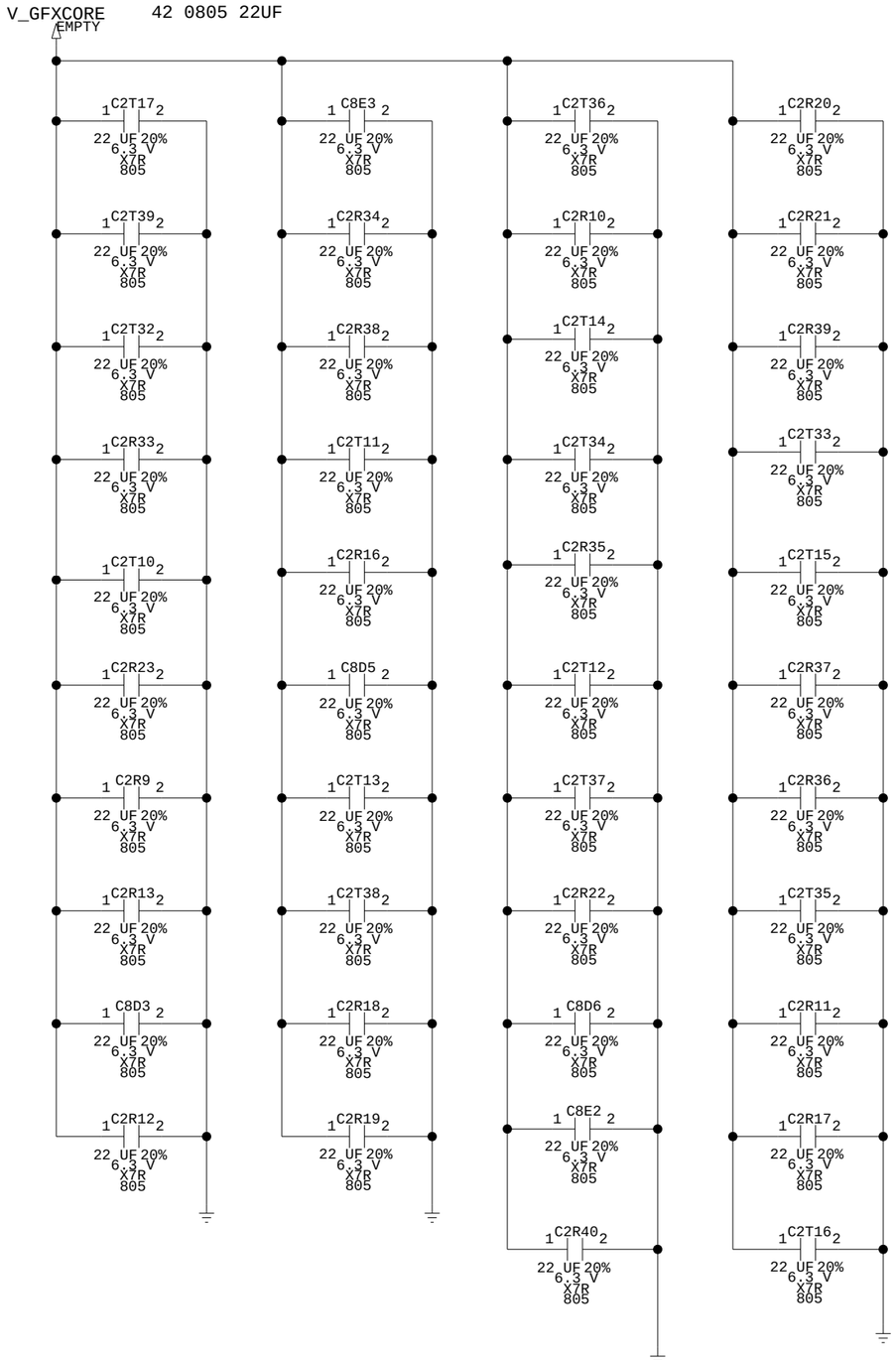


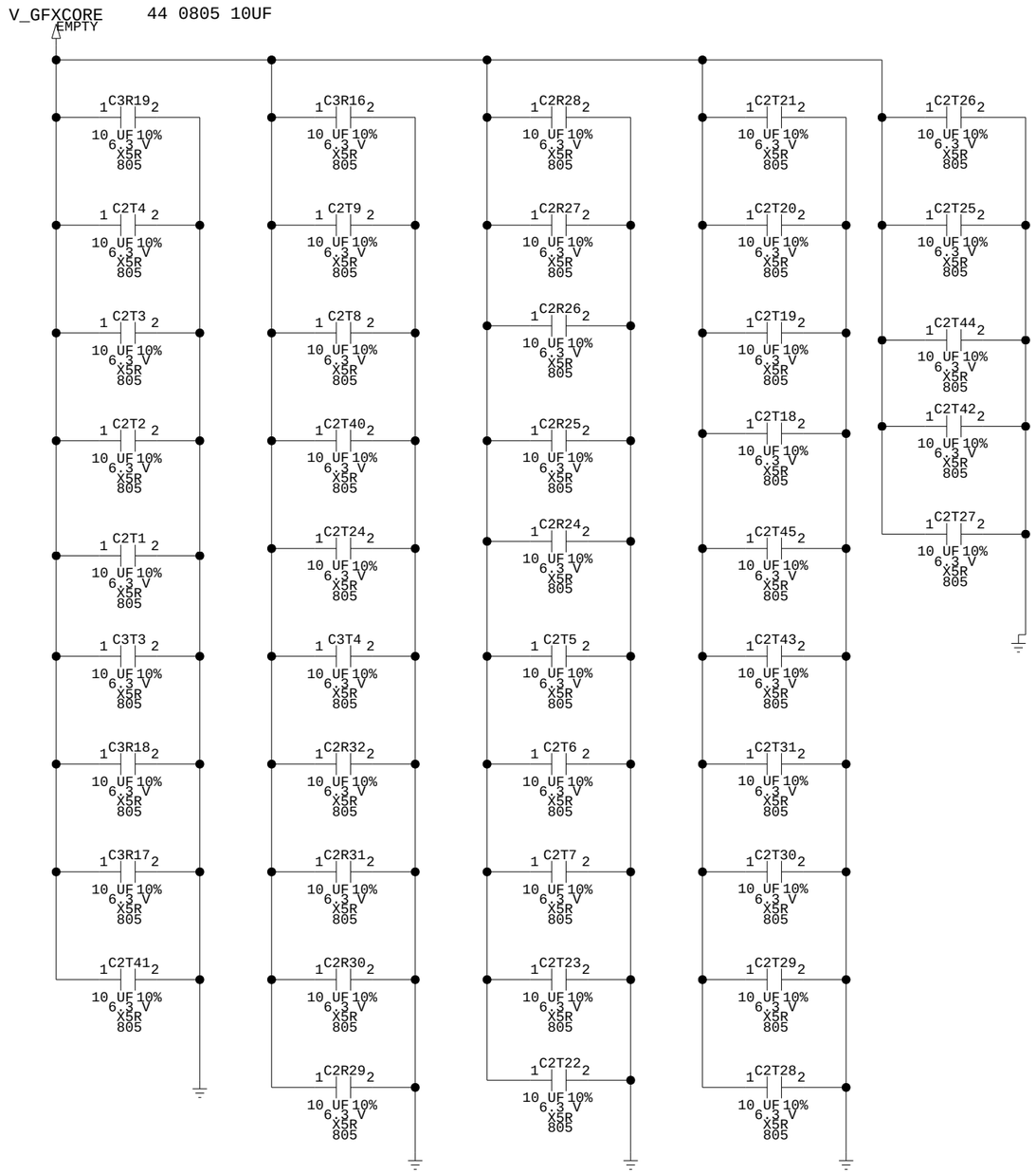
[PAGE_TITLE=SOC PAGE 4]

DRAWING
Tue Jun 18 16:42:29 2013

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	8/72	8/72	M	1.0

SOC, DECOUPLING



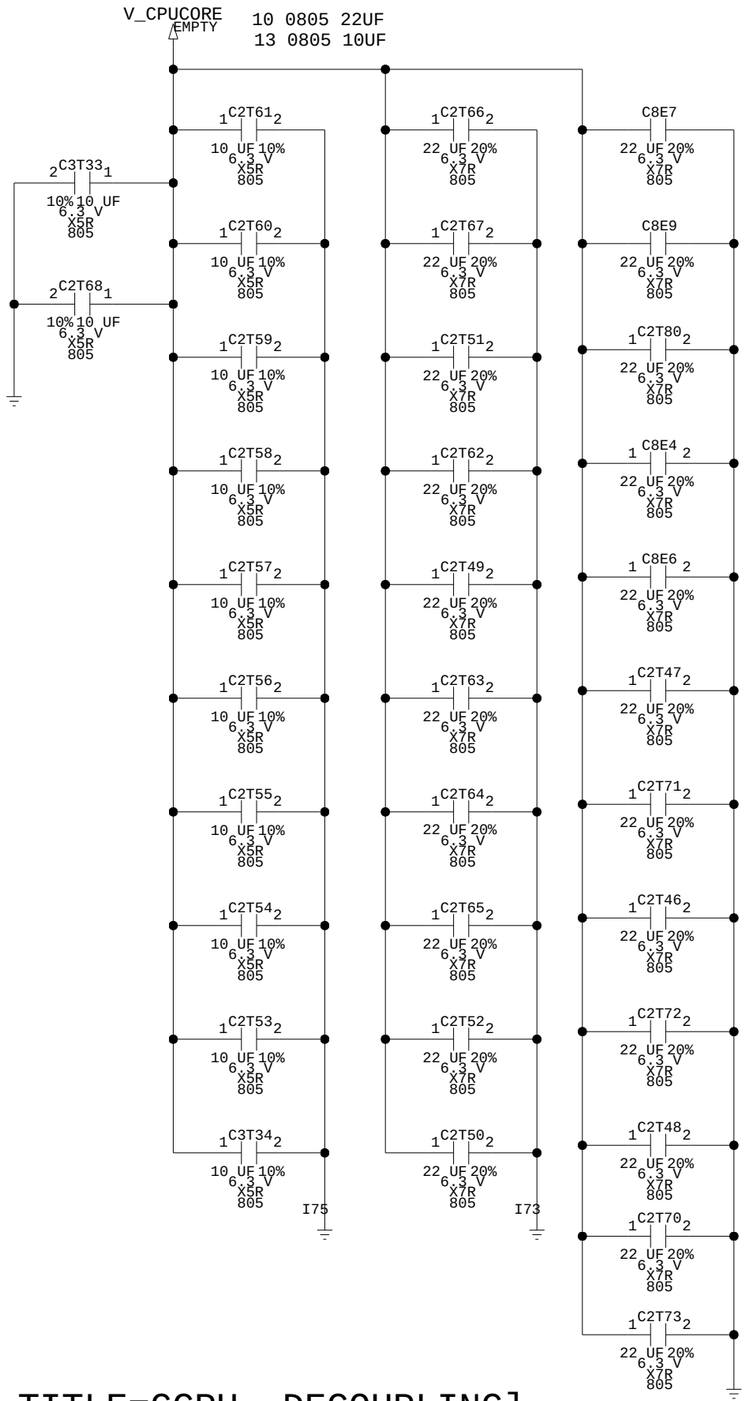


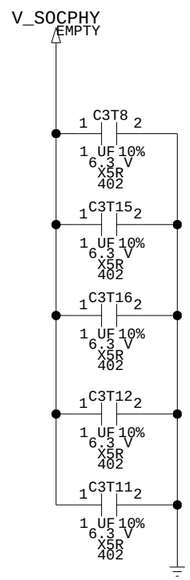
[PAGE_TITLE=GCPU, DECOUPLING]

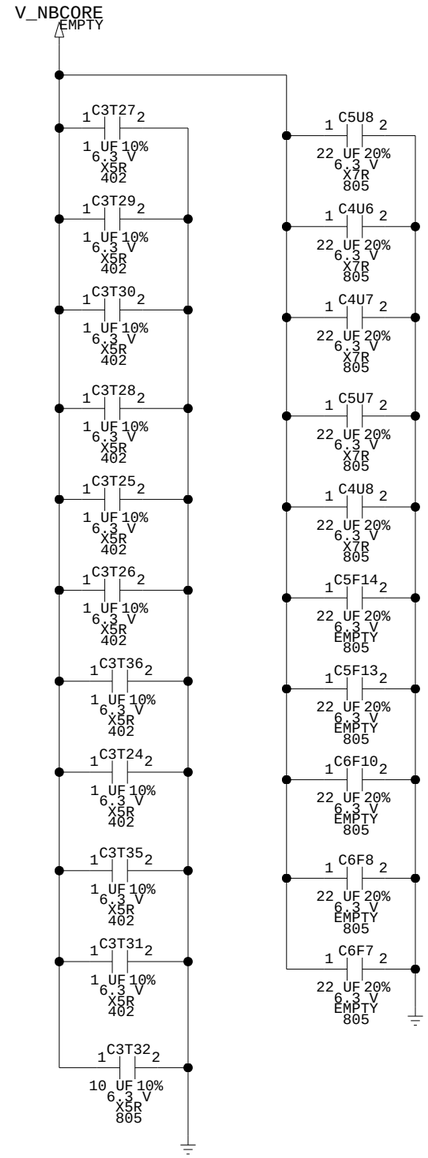
DRAWING
Tue Jun 18 16:42:26 2008

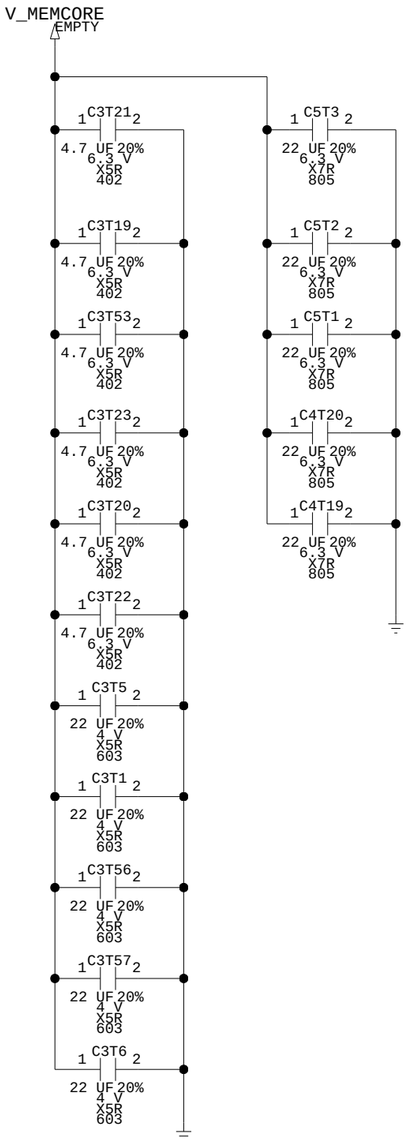
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	9/72	9/72	M	1.0

SOC, DECOUPLING





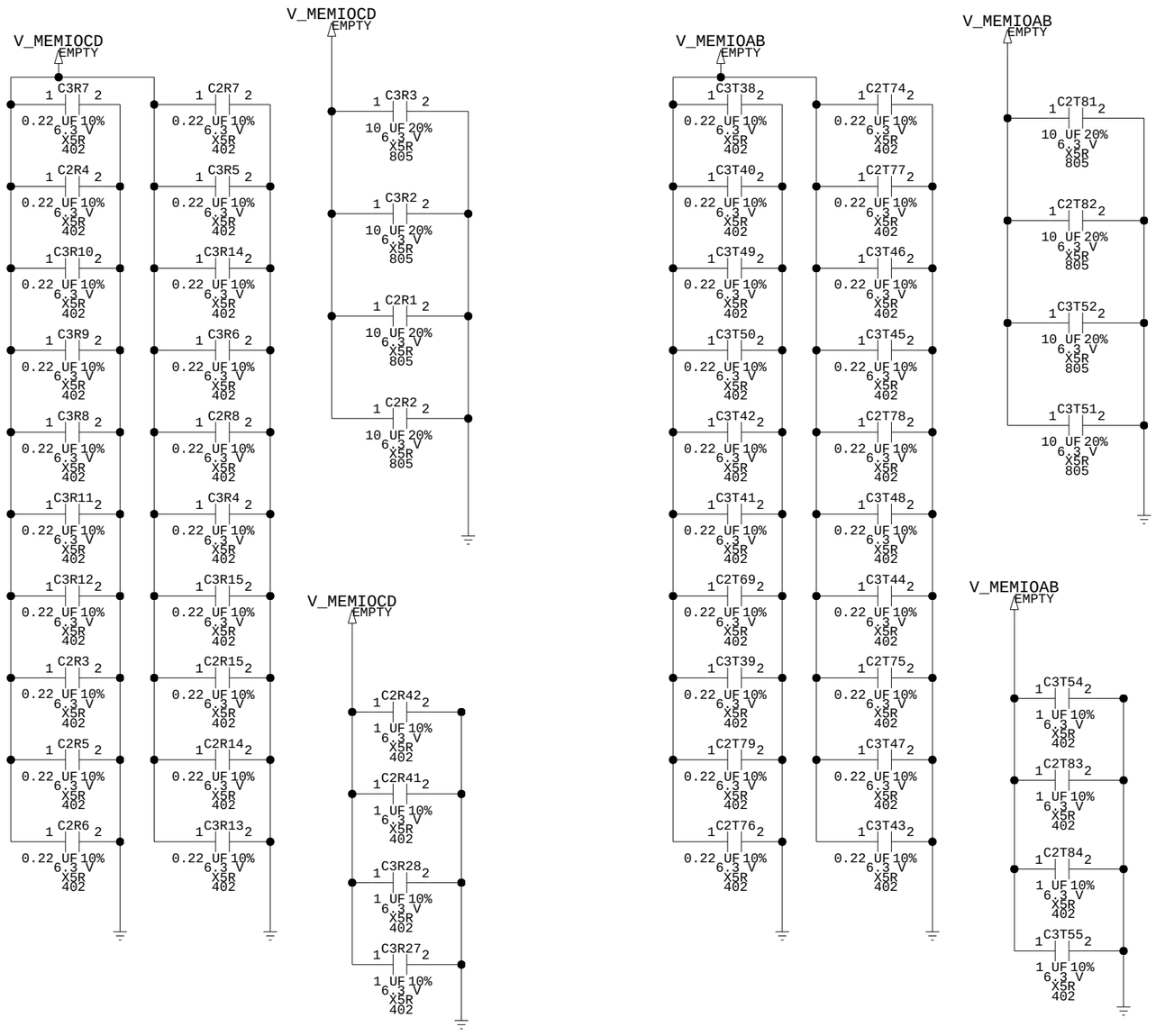




[PAGE_TITLE=GCPU, DECOUPLING]

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	10/72	10/72	M	1.0

SOC, DECOUPLING

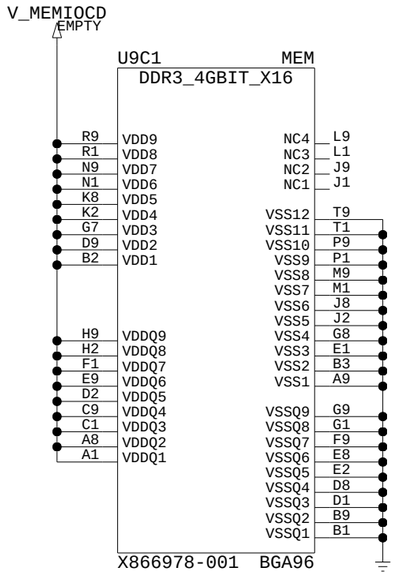


[PAGE_TITLE=GCPU, DECOUPLING]

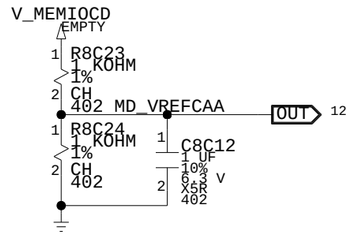
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	11/72	11/72	M	1.0

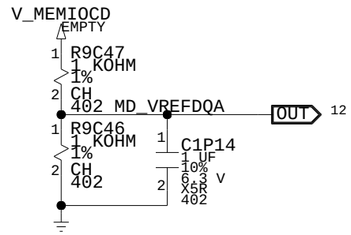
MEMORY CHANNEL D

BYTE LANE 0
BYTE LANE 1

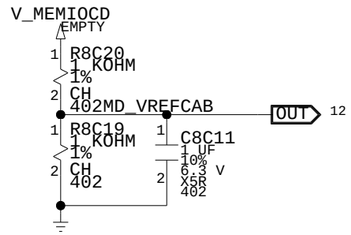


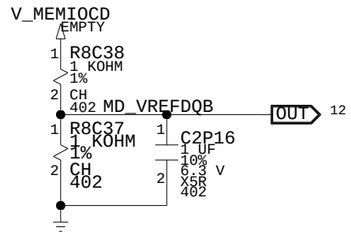
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
U9C1,U8C5	EMPTY	EMPTY	TRUE DUMMY_BOM	

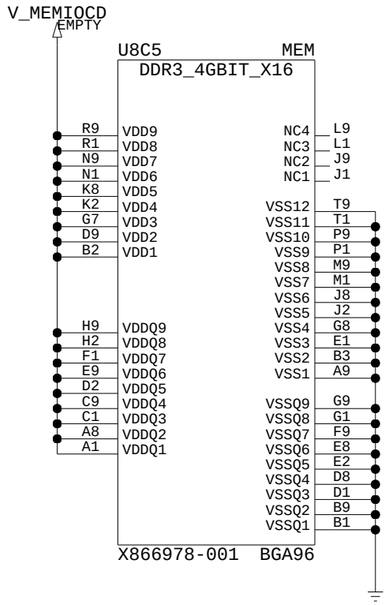


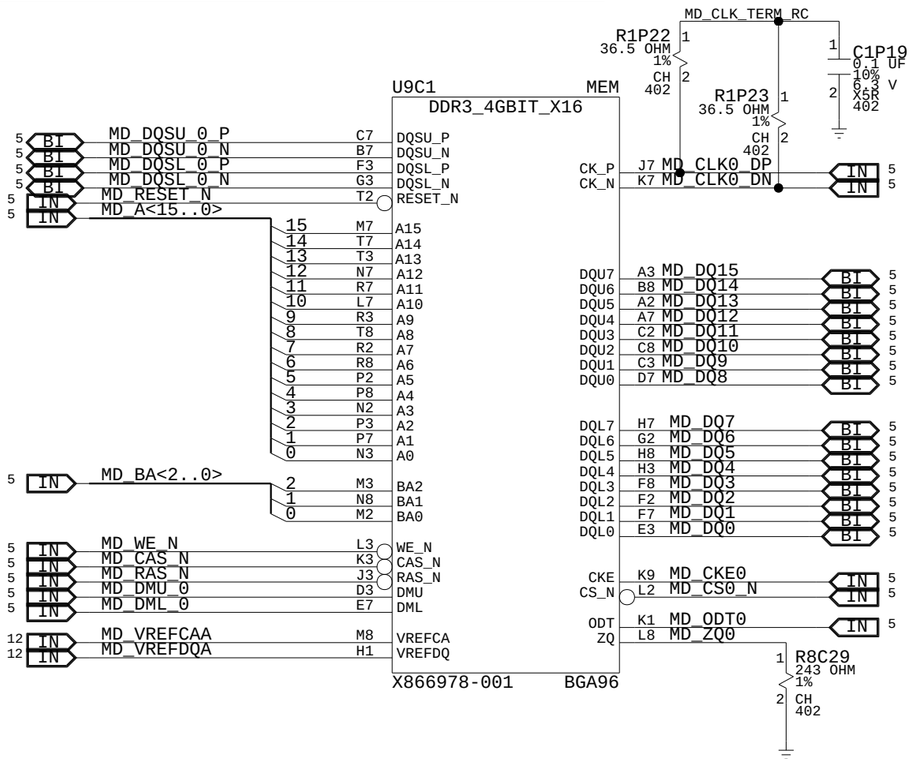


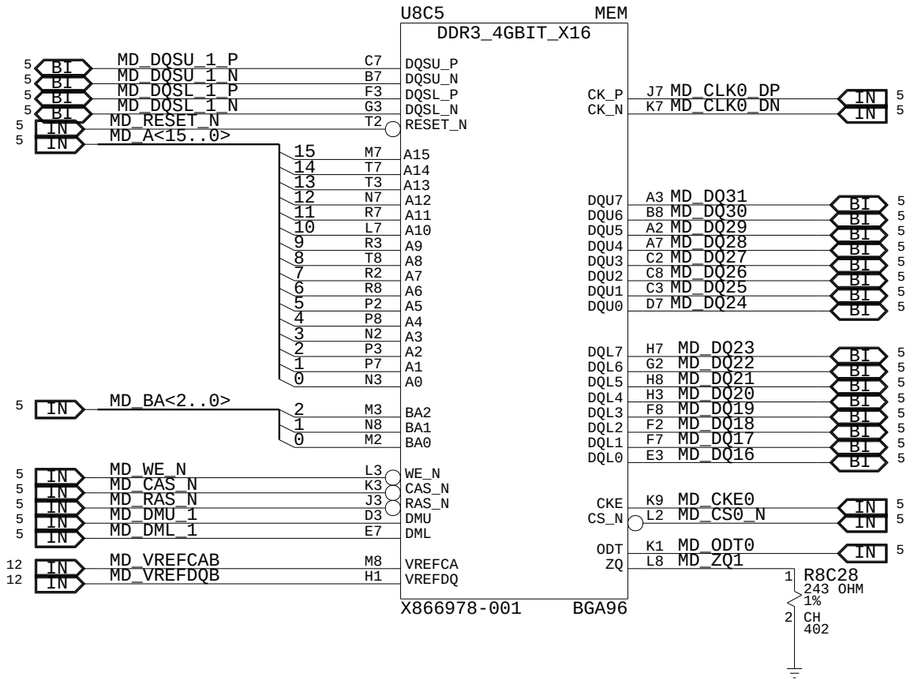
BYTE LANE 2
BYTE LANE 3











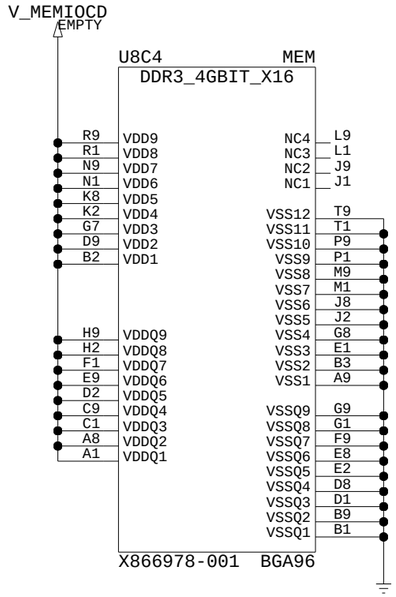
[PAGE_TITLE=MEMORY PARTITION A, LOW]

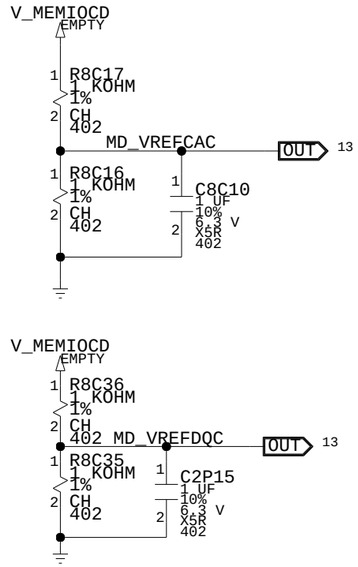
DRAWING
Tue Jun 18 16:42:22 2008

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	12/72	12/72	M	1.0

MEMORY CHANNEL D

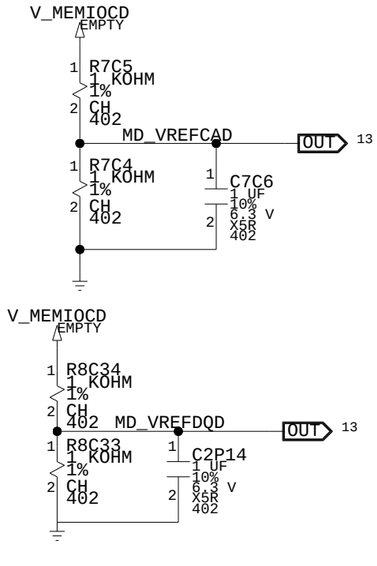
BYTE LANE 4
BYTE LANE 5

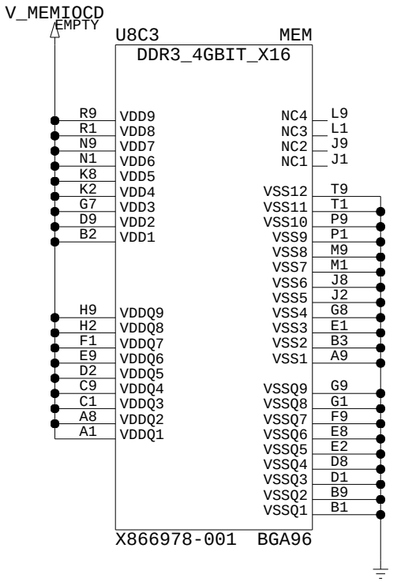


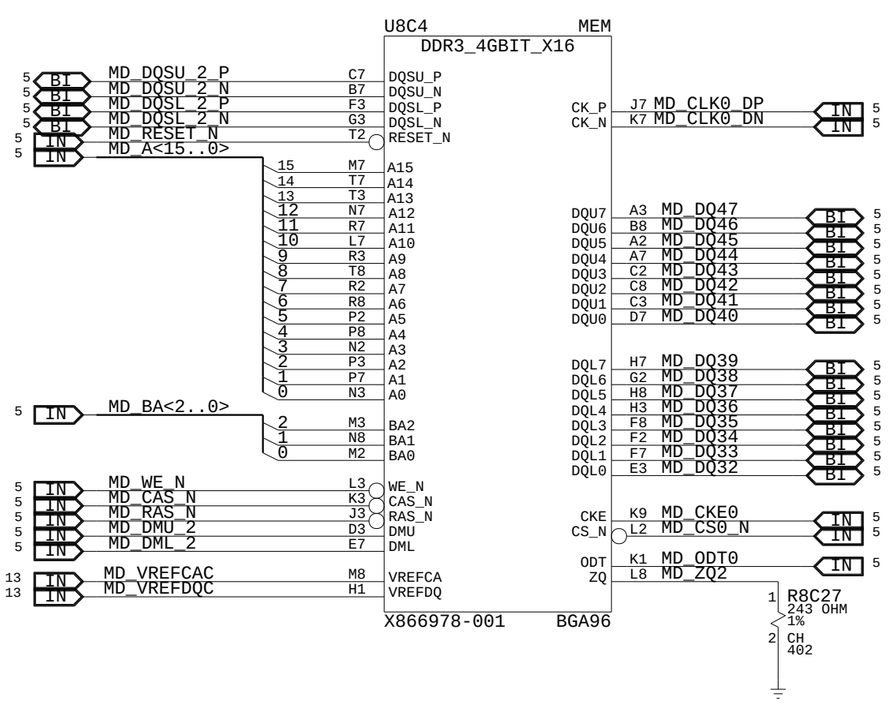


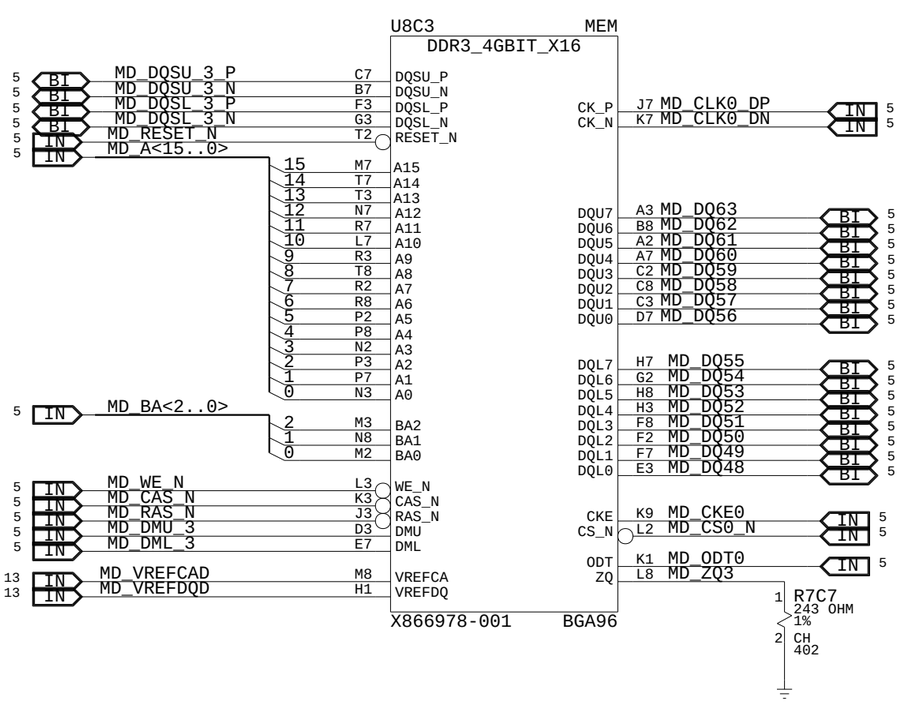
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
U8C4,U8C3	EMPTY	EMPTY	TRUE DUMMY_BOM	

BYTE LANE 6
BYTE LANE 7







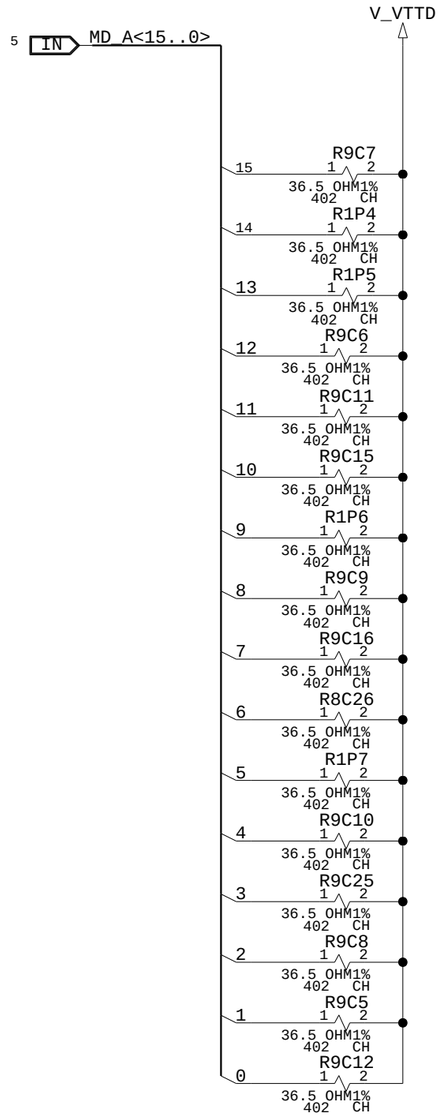


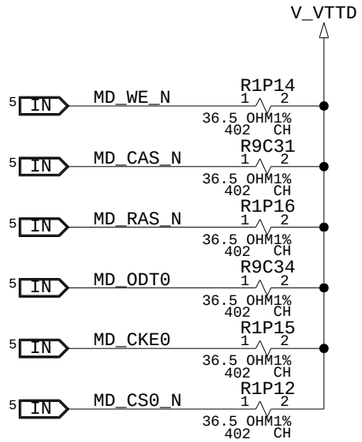
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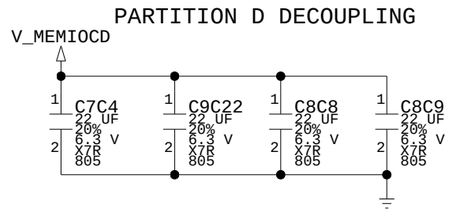
DRAWING
Tue Jun 18 16:42:22 2013

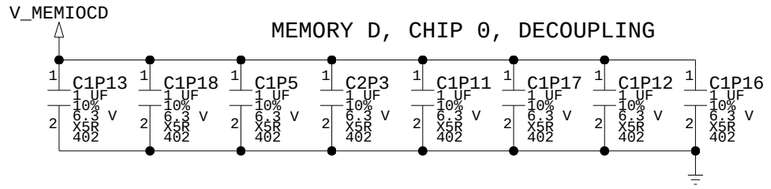
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	13/72	13/72	M	1.0

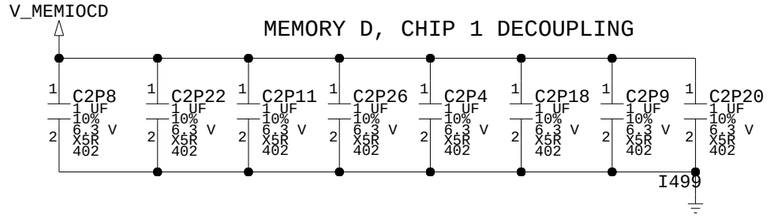
MEMORY CHANNEL D, DECOUPLING + TERMINATION

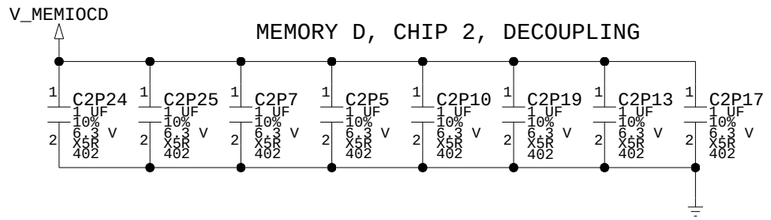


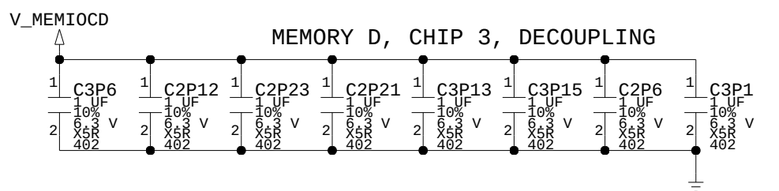












[PAGE_TITLE=MEMORY PARTITION A, DECOUPLING, TERMINATION]

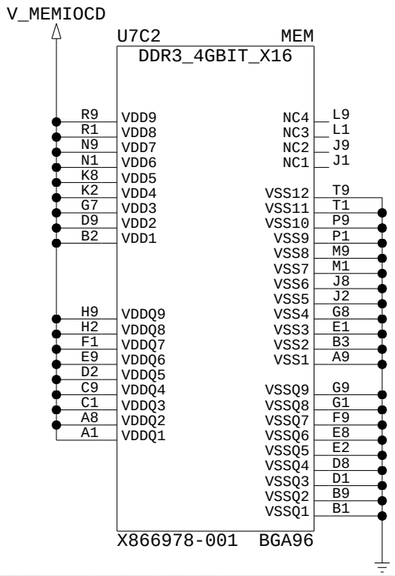
DRAWING
Tue Jun 18 16:42:22 2008

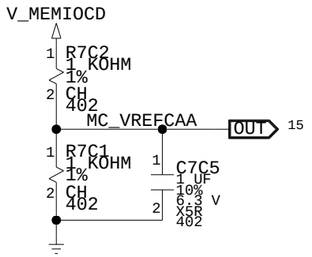
MICROSOFT CONFIDENTIAL

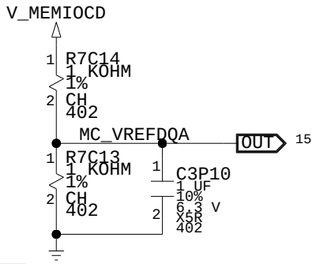
PROJECT NAME	PAGE	CSA PAGE	FAB	REV
GREYBULL_RETAIL	14/72	14/72	M	1.0

MEMORY CHANNEL C

BYTE LANE 0
BYTE LANE 1

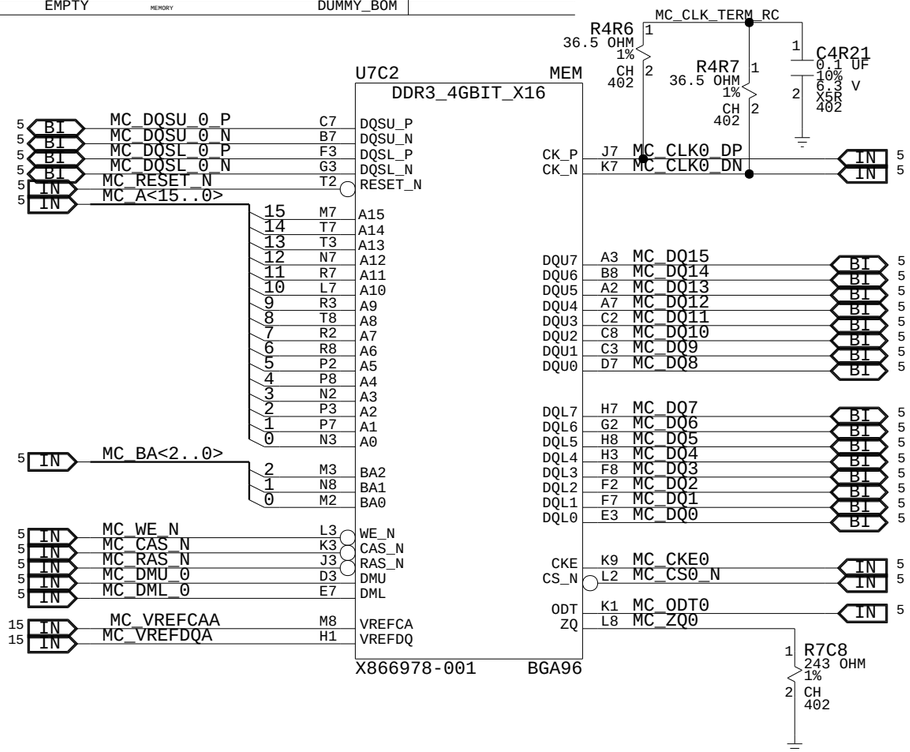




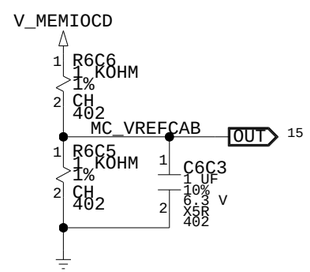


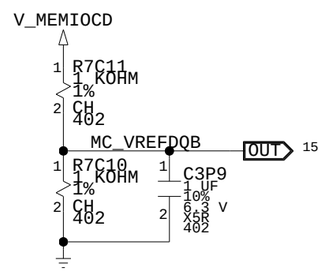
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
U7C2,U7C1	EMPTY	EMPTY	TRUE DUMMY_BOM	

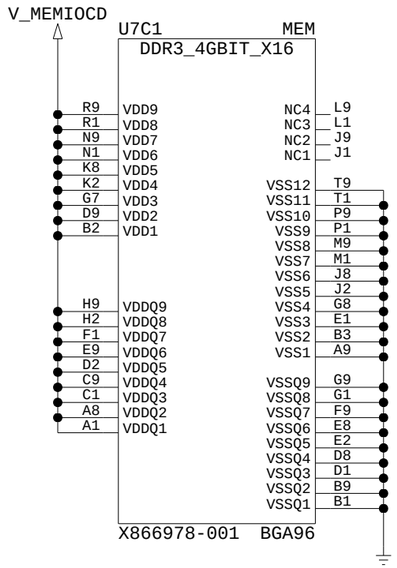
MEMORY

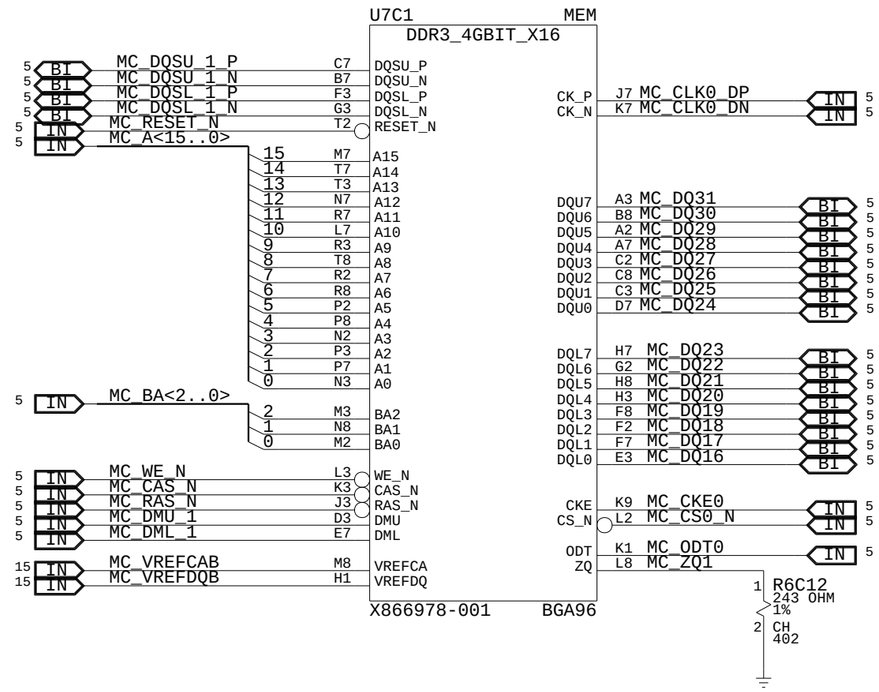


BYTE LANE 2
BYTE LANE 3









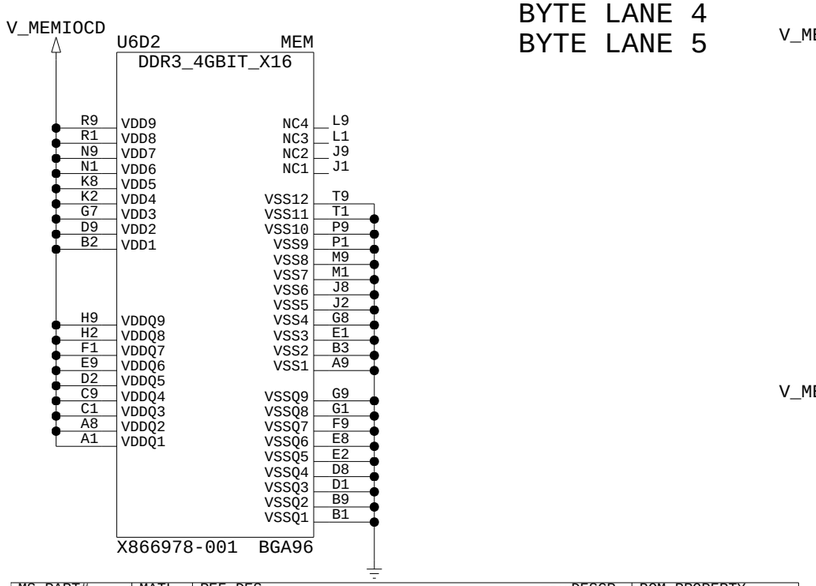
[PAGE_TITLE=MEMORY PARTITION B, LOW]

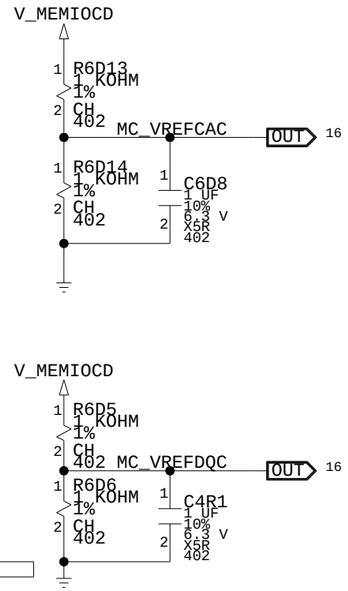
DRAWING CONFIDENTIAL
Tue Jun 18 16:42:34 2013

MICROSOFT	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	15/72	15/72	M	1.0

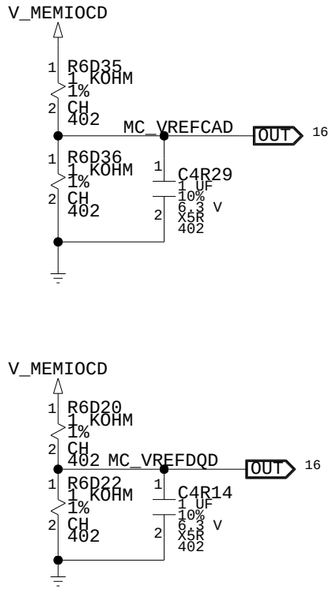
MEMORY CHANNEL C

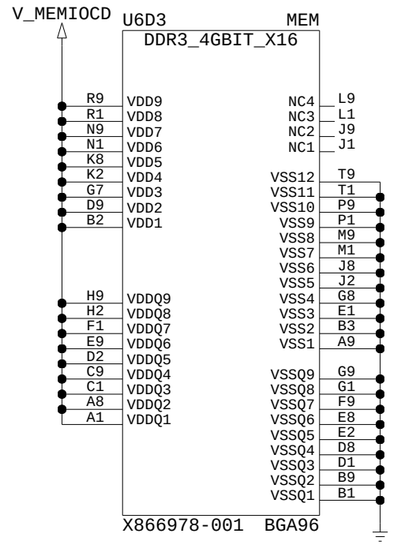
BYTE LANE 4
BYTE LANE 5



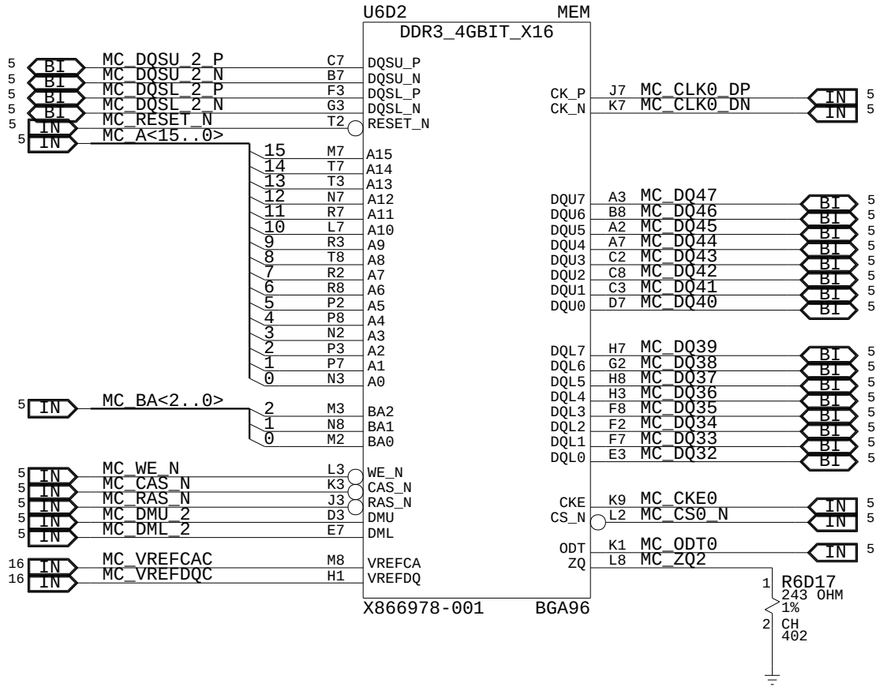


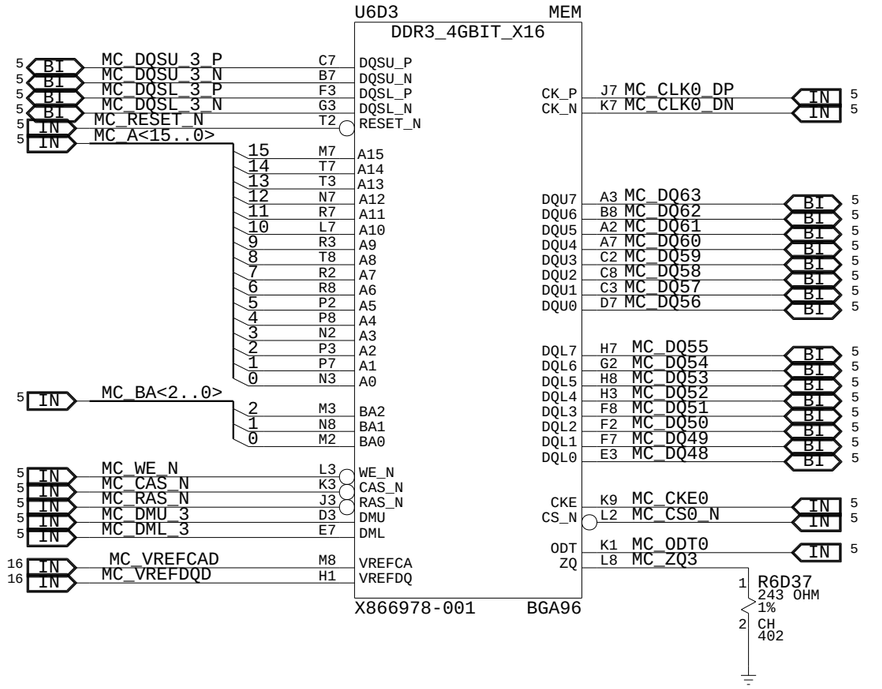
BYTE LANE 6
BYTE LANE 7





MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
U6D2,U6D3	EMPTY	EMPTY	TRUE DUMMY_BOM	





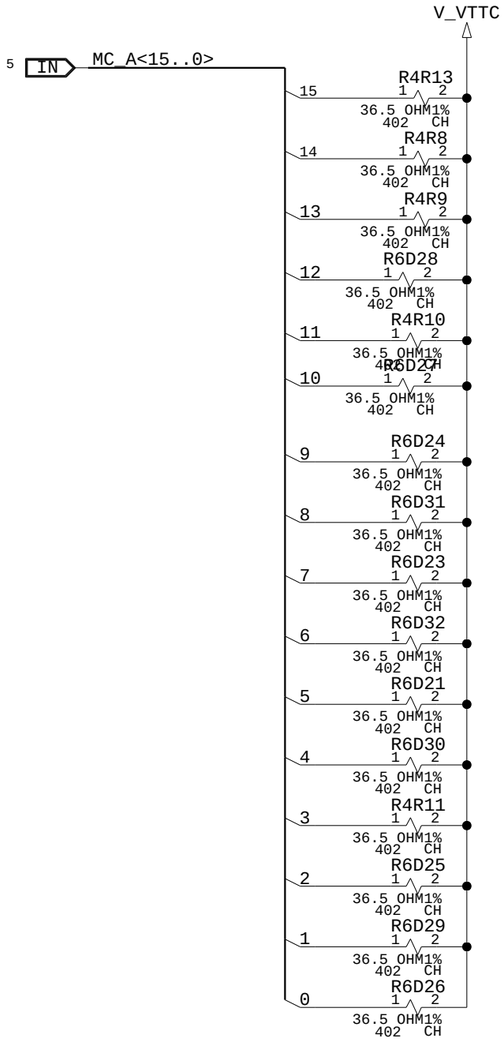
[PAGE_TITLE=MEMORY PARTITION B, HIGH]

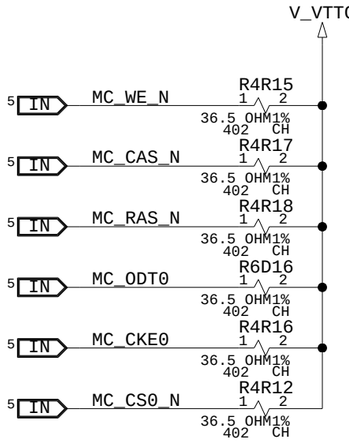
DRAWING

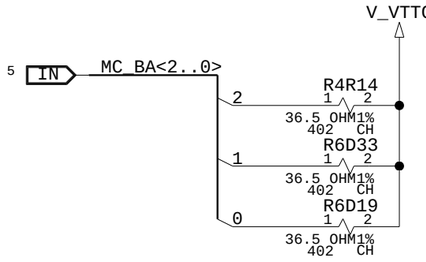
Tue Jun 18 16:42:23 2013

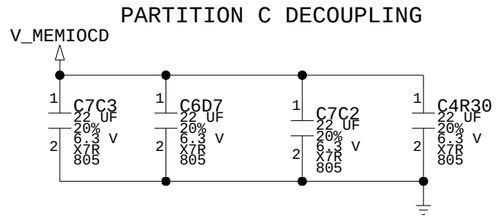
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	16/72	16/72	M	1.0

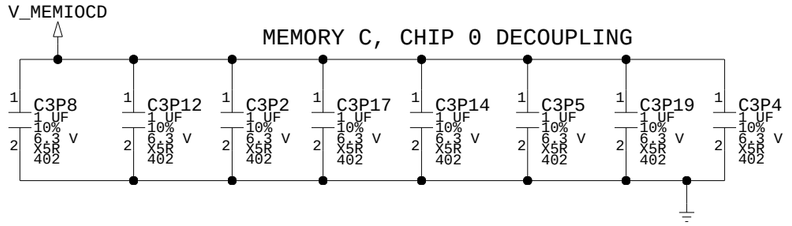
MEMORY CHANNEL C, DECOUPLING + TERMINATION

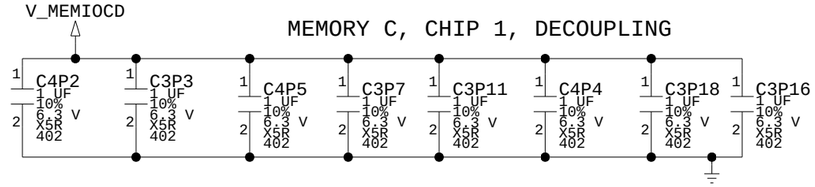


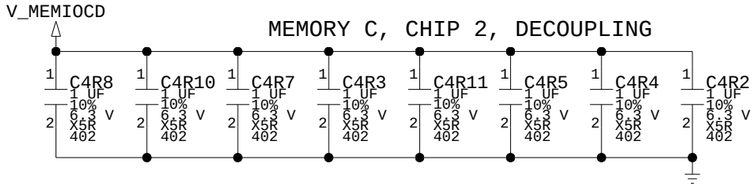


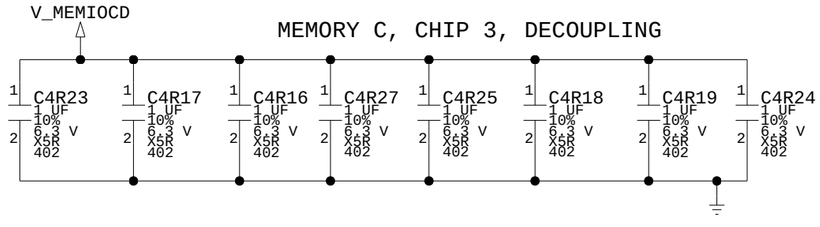












[PAGE_TITLE=MEMORY PARTITION B, DECOUPLING, TERMINATION]

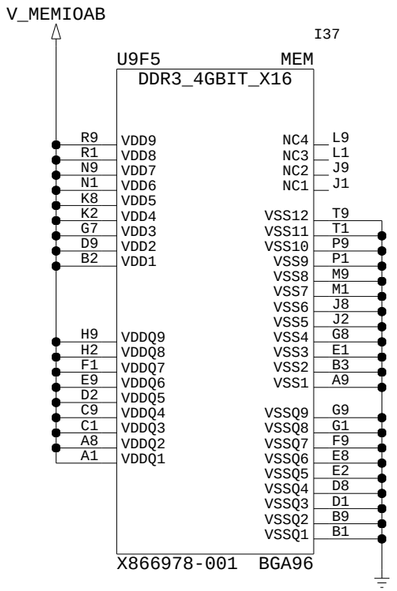
DRAWING
Tue Jun 18 16:42:23 2008

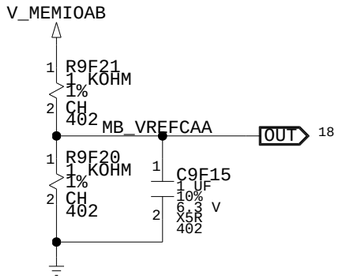
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	17/72	17/72	M	1.0

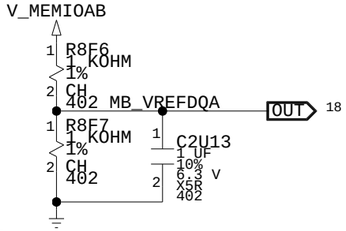
MEMORY CHANNEL B

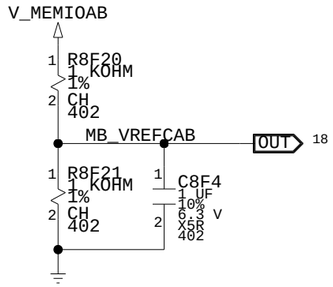
BYTE LANE 0
BYTE LANE 1

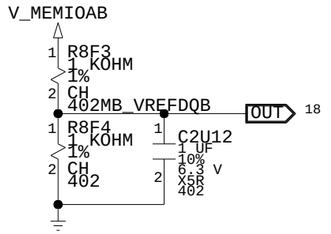
BYTE LANE 2
BYTE LANE 3

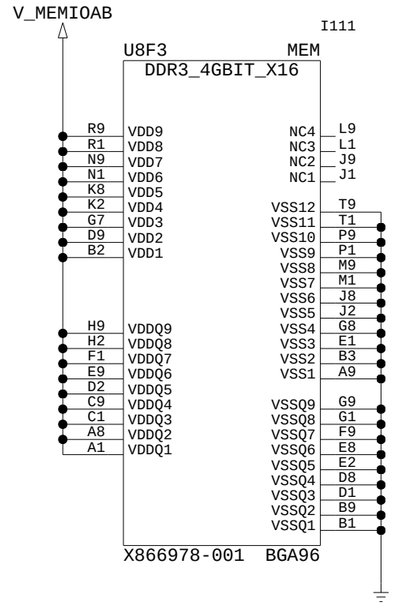




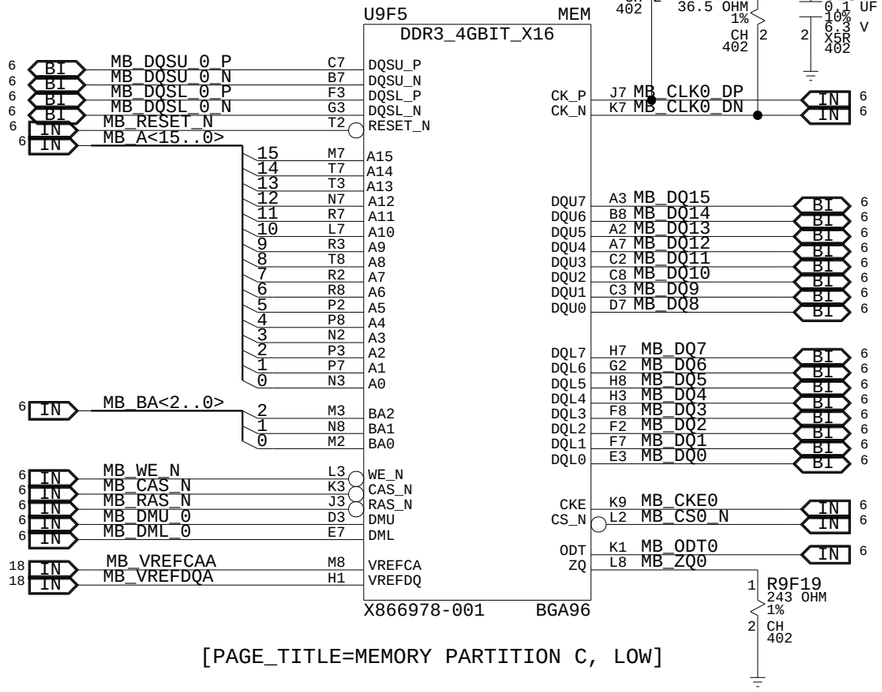




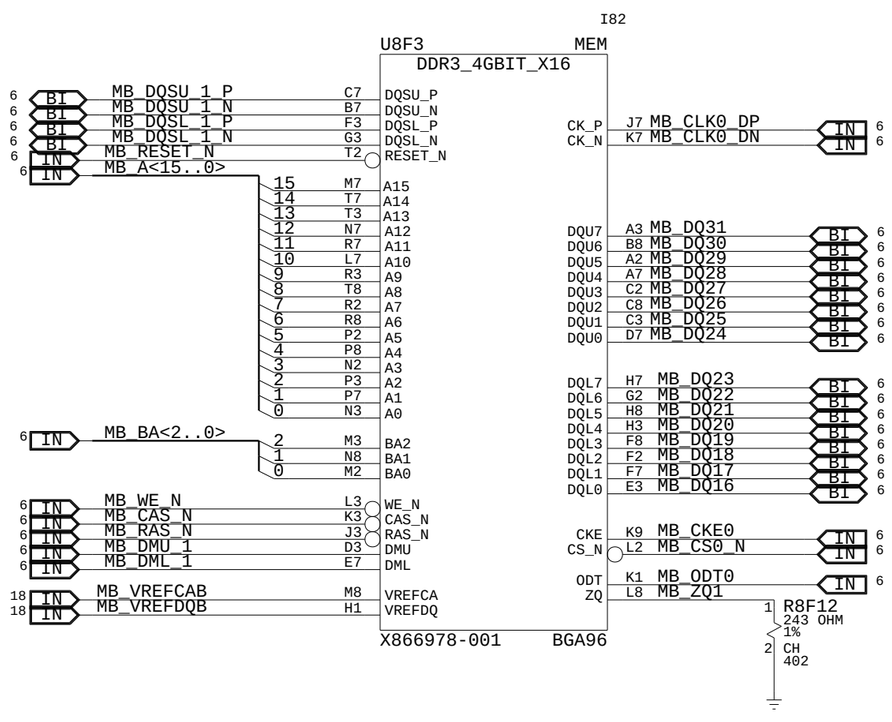




MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
U9F5,U8F3	EMPTY	EMPTY	TRUE DUMMY_BOM	



[PAGE_TITLE=MEMORY PARTITION C, LOW]

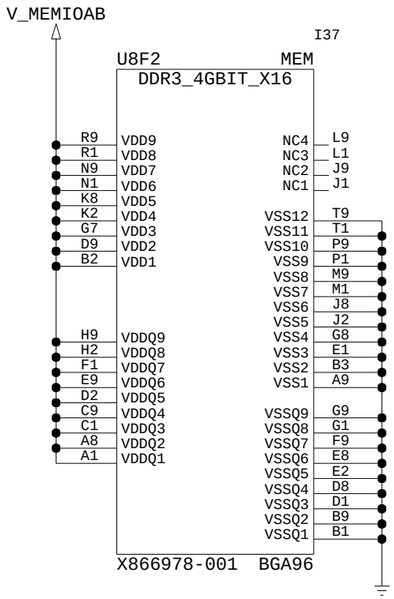


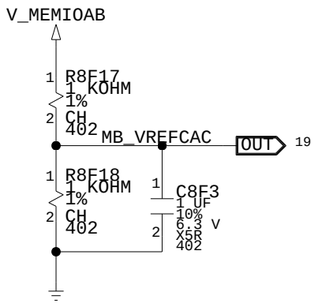
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	18/72	18/72	M	1.0

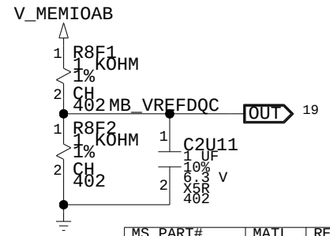
MEMORY CHANNEL B

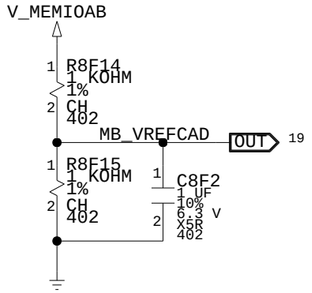
BYTE LANE 4
BYTE LANE 5

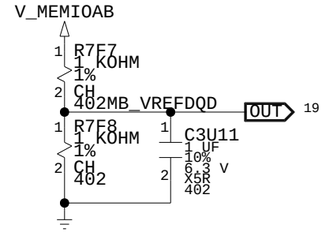
BYTE LANE 6
BYTE LANE 7

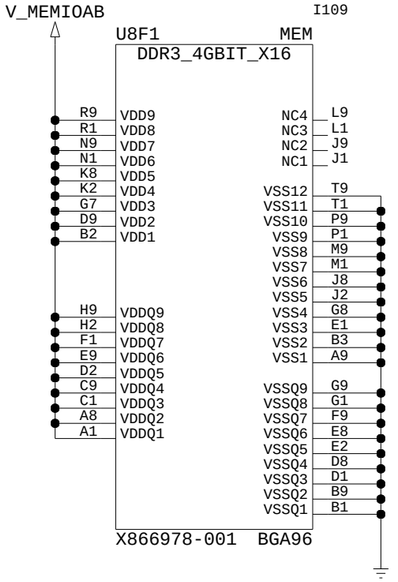




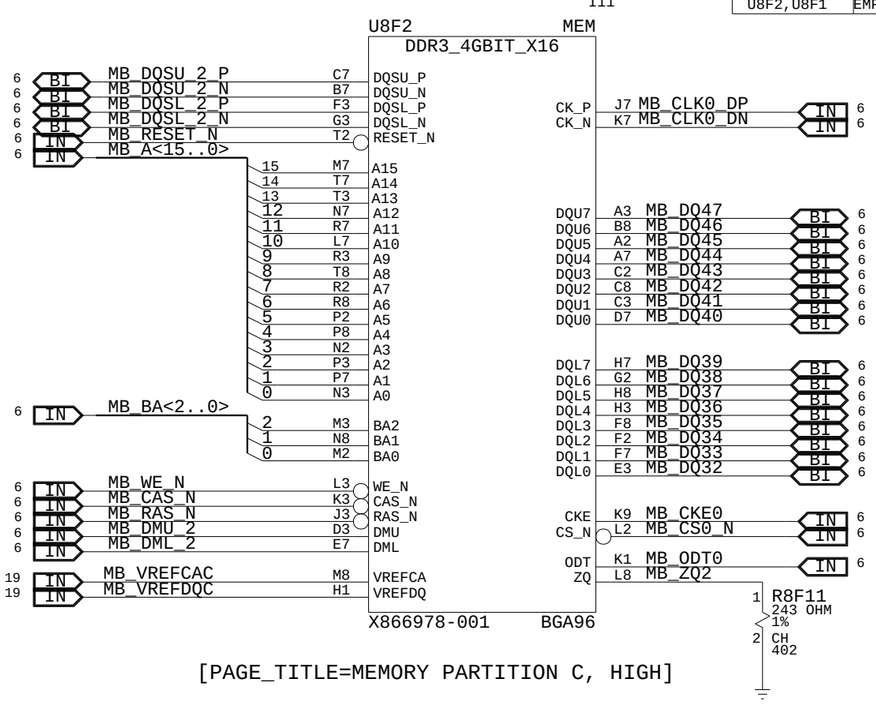


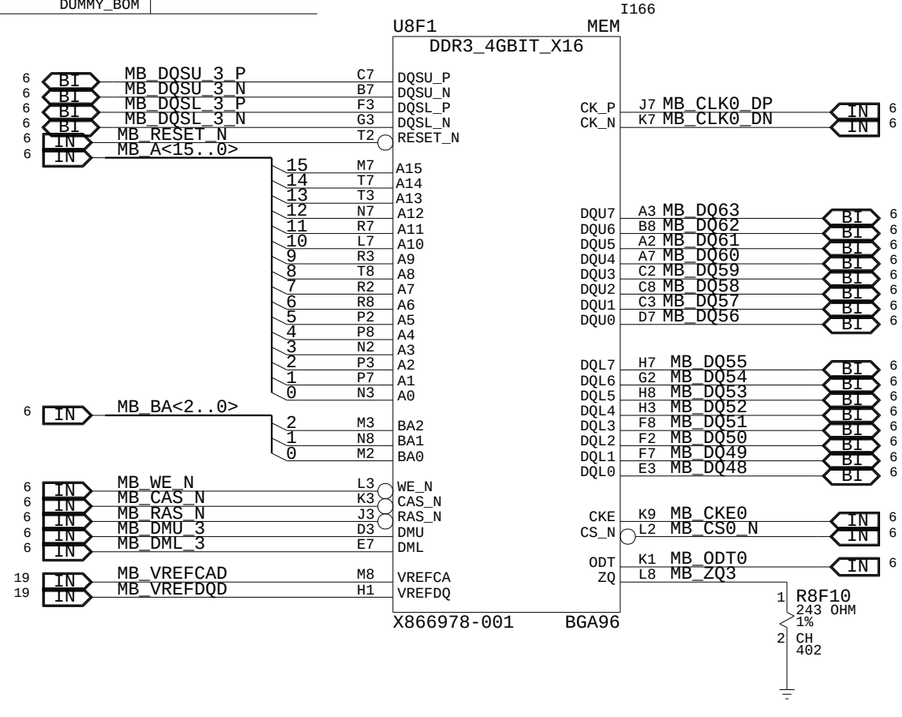






MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
U8F2,U8F1	EMPTY	EMPTY MEMORY	TRUE DUMMY_BOM	

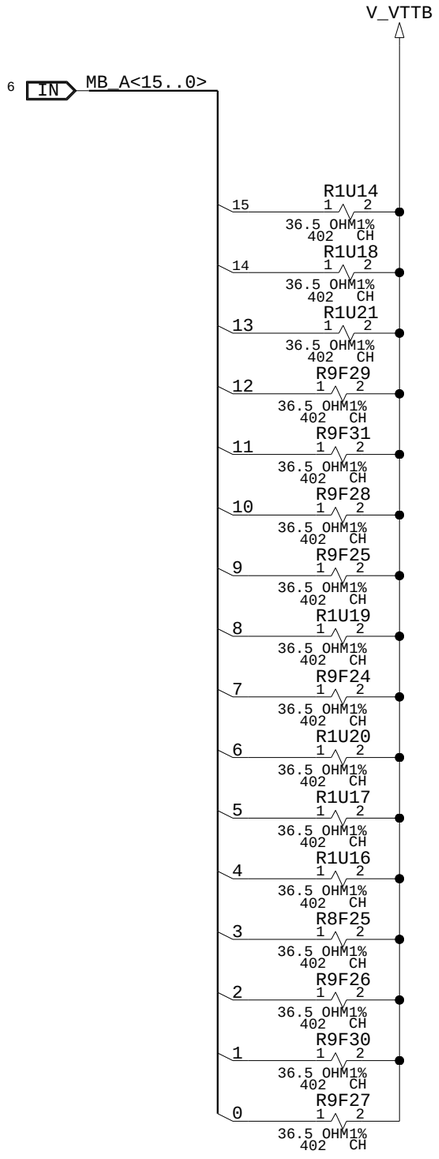


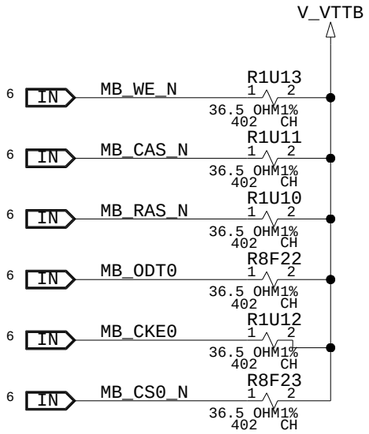


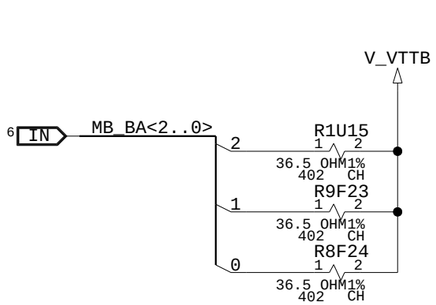
[PAGE_TITLE=MEMORY PARTITION C, HIGH]

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	19/72	19/72	M	1.0

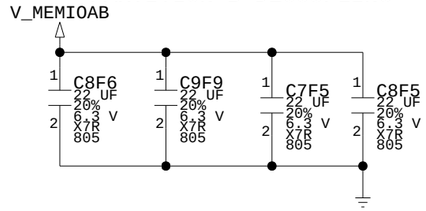
MEMORY CHANNEL B, DECOUPLING + TERMINATION



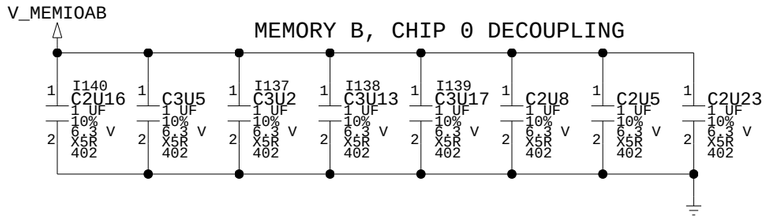




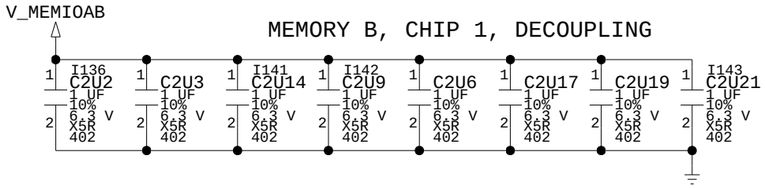
PARTITION B DECOUPLING



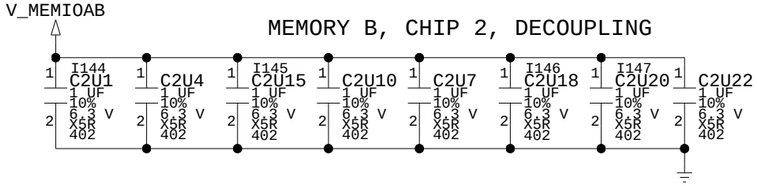
MEMORY B, CHIP 0 DECOUPLING



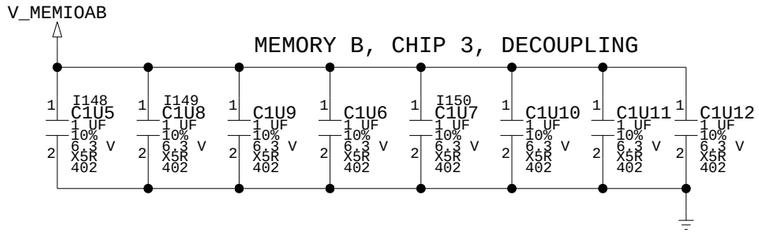
MEMORY B, CHIP 1, DECOUPLING



MEMORY B, CHIP 2, DECOUPLING



MEMORY B, CHIP 3, DECOUPLING



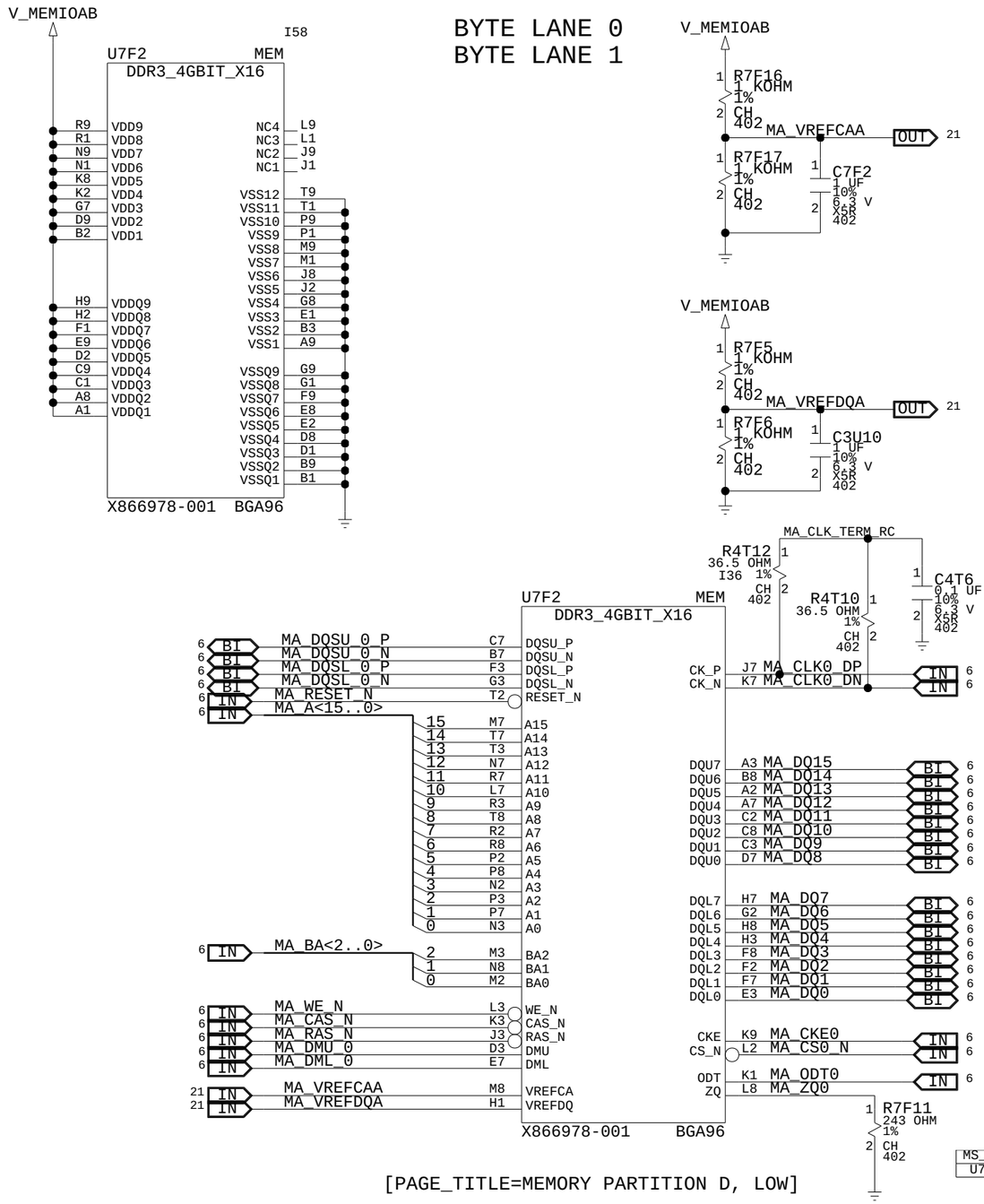
[PAGE_TITLE=MEMORY PARTITION C, DECOUPLING, TERMINATION]

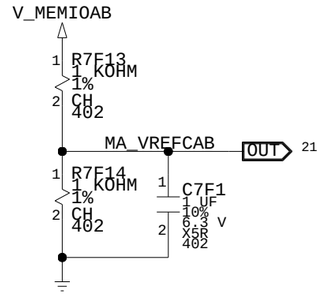
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	20/72	20/72	M	1.0

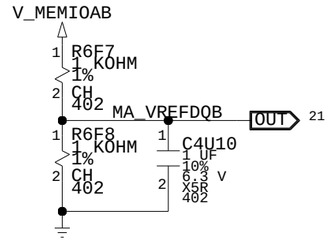
MEMORY CHANNEL A

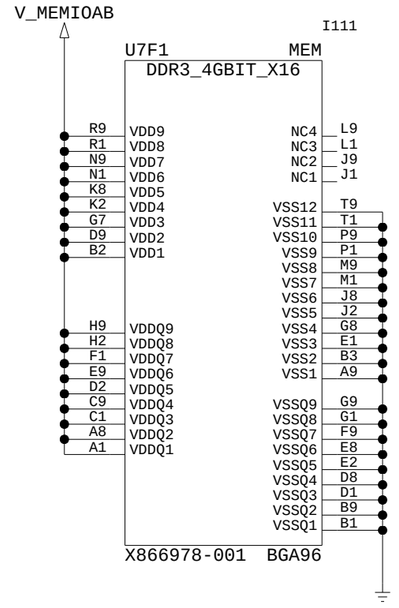
BYTE LANE 0
BYTE LANE 1

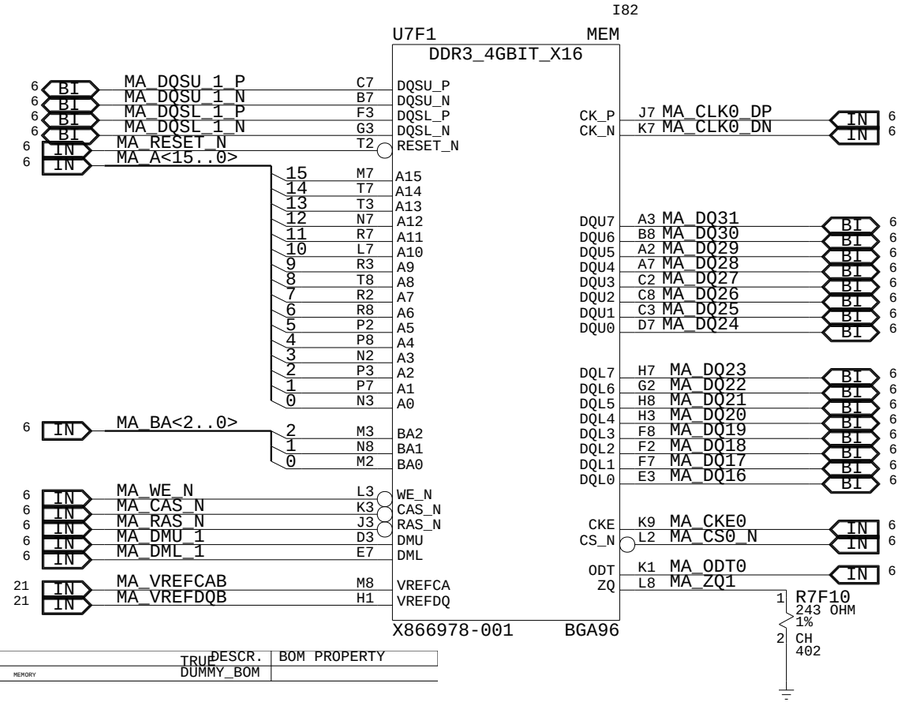
BYTE LANE 2
BYTE LANE 3











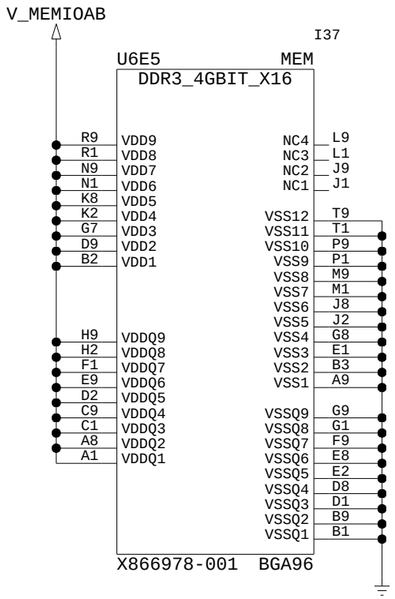
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
U7F2,U7F1	EMPTY	EMPTY	TRUE DUMMY_BOM	

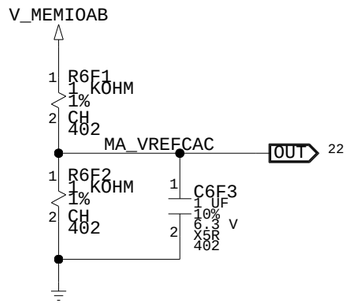
[PAGE_TITLE=MEMORY PARTITION D, LOW]

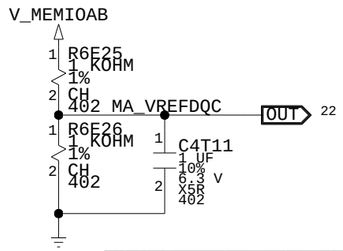
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	21/72	21/72	M	1.0

MEMORY CHANNEL A

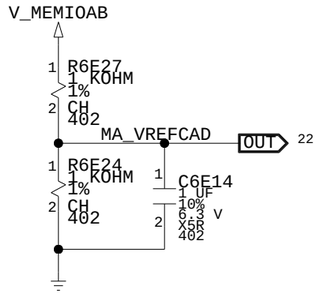
BYTE LANE 4
BYTE LANE 5

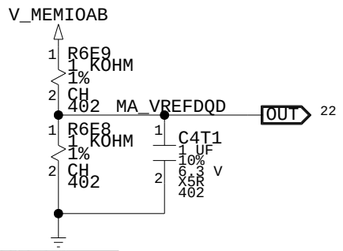


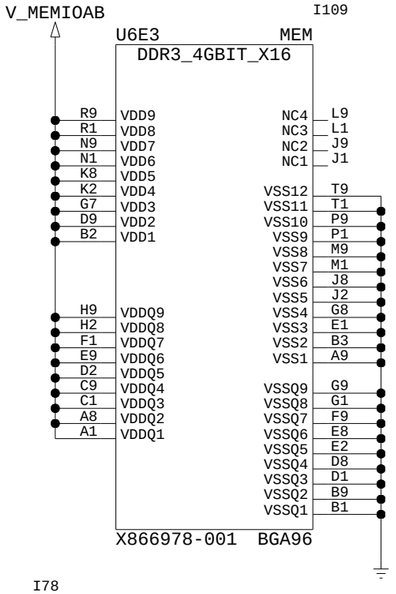




BYTE LANE 6
BYTE LANE 7

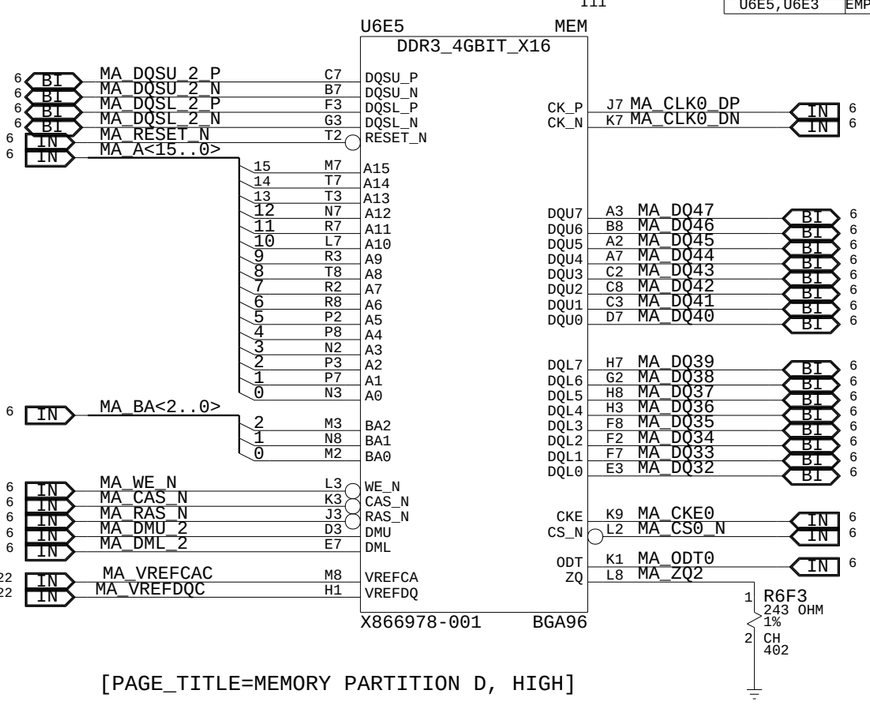


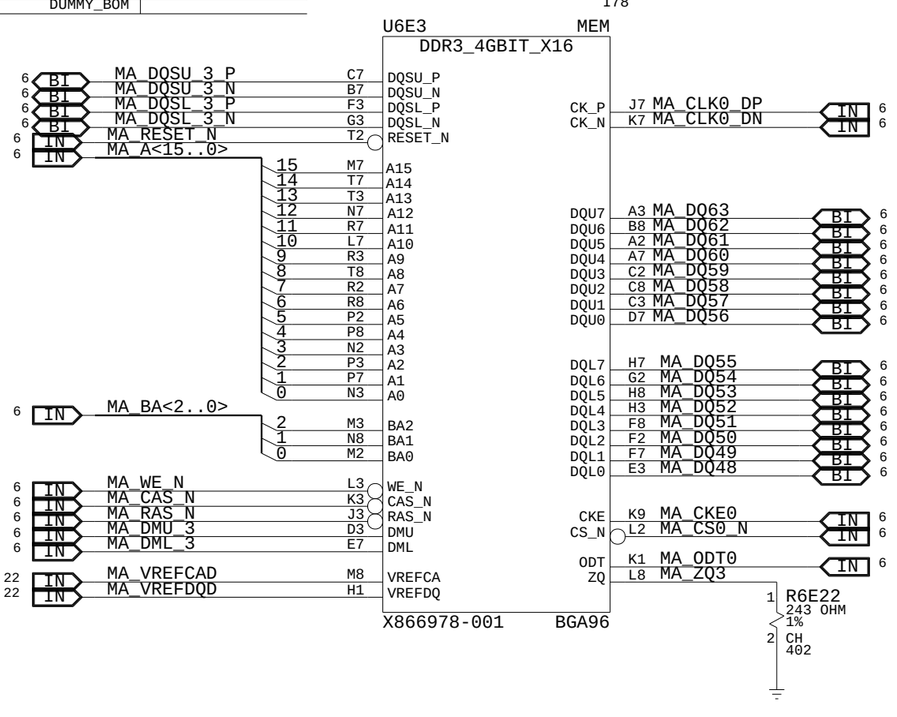




MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
U6E5,U6E3	EMPTY	EMPTY	TRUE DUMMY_BOM	

MEMORY

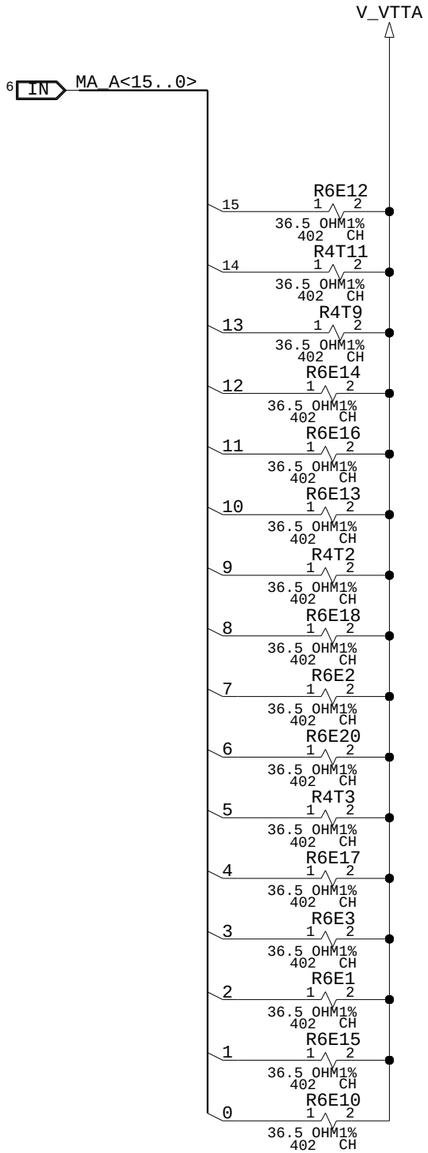


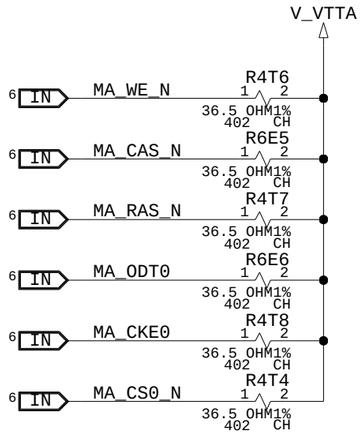


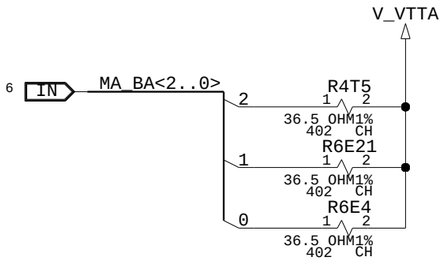
[PAGE_TITLE=MEMORY PARTITION D, HIGH]

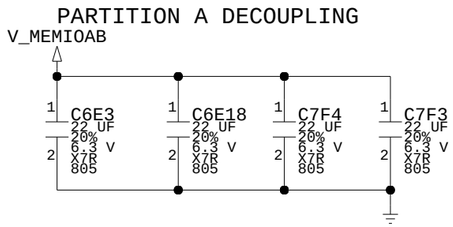
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	22/72	22/72	M	1.0

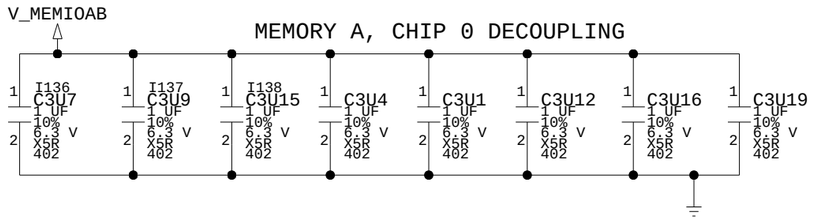
MEMORY CHANNEL A, DECOUPLING + TERMINATION

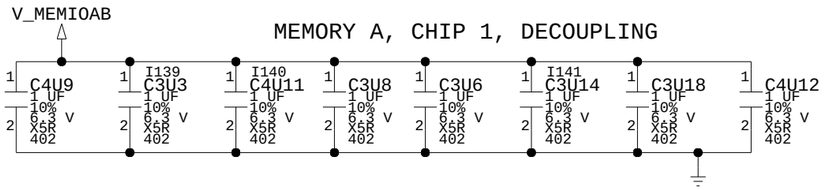


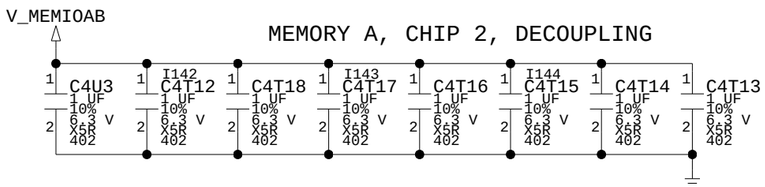


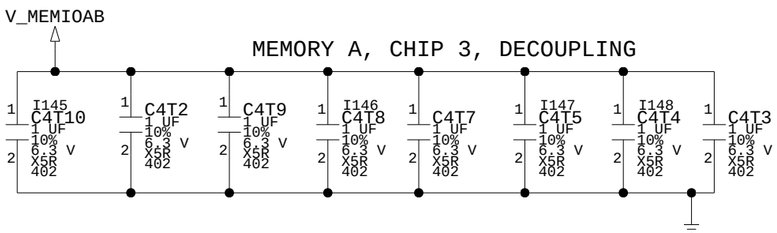












[PAGE_TITLE=MEMORY PARTITION D, DECOUPLING, TERMINATION]

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	23/72	23/72	M	1.0

KIC, USB

U4D2 KIC IC

6 of 11 I84

				USBD_D0_DP	AF8	USB2_WIFI_DP	BI 36
				USBD_D0_DN	AE8	USB2_WIFI_DN	BI 36
				USBC_D0_DP	AC4	USB2_AUX_DP	BI 36
				USBC_D0_DN	AC3	USB2_AUX_DN	BI 36
36	IN	USB3_AUX_RP	Y1	USBC_D1_RXP	USBC_D1_TXP	W3 USB3_AUX_TP_C	C6B9 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_AUX_TP OUT 36
36	IN	USB3_AUX_RN	Y2	USBC_D1_RXN	USBC_D1_TXN	W4 USB3_AUX_TN_C	C6B8 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_AUX_TN OUT 36
35	IN	USB3_BP_P1_RP	V1	USBB_D2_RXP	USBB_D2_TXP	U3 USB3_BP_P1_TP_C	C7B10 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_BP_P1_TP OUT 35
35	IN	USB3_BP_P1_RN	V2	USBB_D2_RXN	USBB_D2_TXN	U4 USB3_BP_P1_TN_C	C7B9 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_BP_P1_TN OUT 35
					USBB_D0_DP	AE4	USB2_BP_P1_DP BI 35
					USBB_D0_DN	AE3	USB2_BP_P1_DN BI 35
					USBB_D1_DP	AD2	USB2_BP_P0_DP BI 35
					USBB_D1_DN	AD1	USB2_BP_P0_DN BI 35
35	IN	USB3_BP_P0_RP	T1	USBB_D3_RXP	USBB_D3_TXP	R3 USB3_BP_P0_TP_C	C7B12 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_BP_P0_TP OUT 35
35	IN	USB3_BP_P0_RN	T2	USBB_D3_RXN	USBB_D3_TXN	R4 USB3_BP_P0_TN_C	C7B11 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_BP_P0_TN OUT 35
35	IN	USB3_FP_P0_RP	AA8	USBA_D2_RXP	USBA_D2_TXP	AB10 USB3_FP_P0_TP_C	C3F4 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_FP_P0_TP OUT 35
35	IN	USB3_FP_P0_RN	AA7	USBA_D2_RXN	USBA_D2_TXN	AB9 USB3_FP_P0_TN_C	C3F5 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_FP_P0_TN OUT 35
					USBA_D0_DP	AC7	USB2_FP_P0_DP BI 35
					USBA_D0_DN	AC8	USB2_FP_P0_DN BI 35
					USBA_D1_DP	AF6	USB2_ACC_DP BI 36
					USBA_D1_DN	AF5	USB2_ACC_DN BI 36
DB4B3	1	USB3_TP_P1_RP	AB1	USBA_D3_RXP	USBA_D3_TXP	AA3 USB3_TP_P1_TP_C	C4B8 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_TP_P1_TP 1 → DB4B5
DB4B2	1	USB3_TP_P1_RN	AB2	USBA_D3_RXN	USBA_D3_TXN	AA4 USB3_TP_P1_TN_C	C4B7 1 2 0.1 UF 10% 6.3 V X5R 402 → USB3_TP_P1_TN 1 → DB4B4
					USBA_D2_AMOUT	AA5 USBA_D2_AMOUNT 1	→ DB6P4
					USBA_D1_AMOUT	AE6 USBA_D1_AMOUNT 1	→ DB4C2
					USBA_D0_AMOUT	AC11 USBA_D0_AMOUNT 1	→ DB4D4
					USBB_D2_AMOUT	R5 USBB_D2_AMOUNT 1	→ DB6P7
					USBB_D1_AMOUT	AD5 USBB_D1_AMOUNT 1	→ DB4C3
					USBB_D0_AMOUT	AF4 USBB_D0_AMOUNT 1	→ DB4D1
					USBC_D1_AMOUT	W5 USBC_D1_AMOUNT 1	→ DB6P5
					USBC_D0_AMOUT	AC5 USBC_D0_AMOUNT 1	→ DB6P6
					USBD_D0_AMOUT	AE9 USBD_D0_AMOUNT 1	→ DB3D8

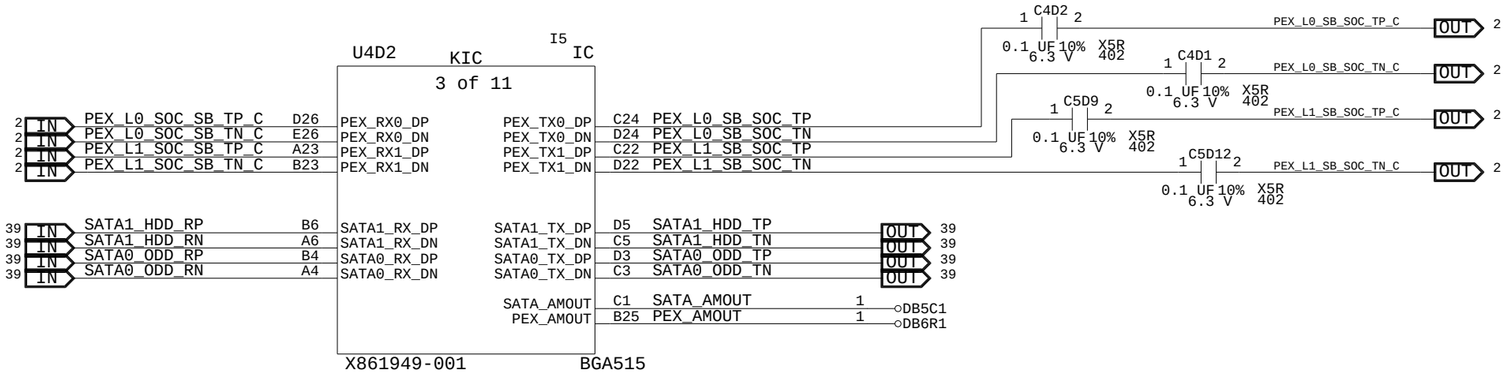
USBA_D1_TXRTUNE AB11 → USBA_D1_TXRTUNE

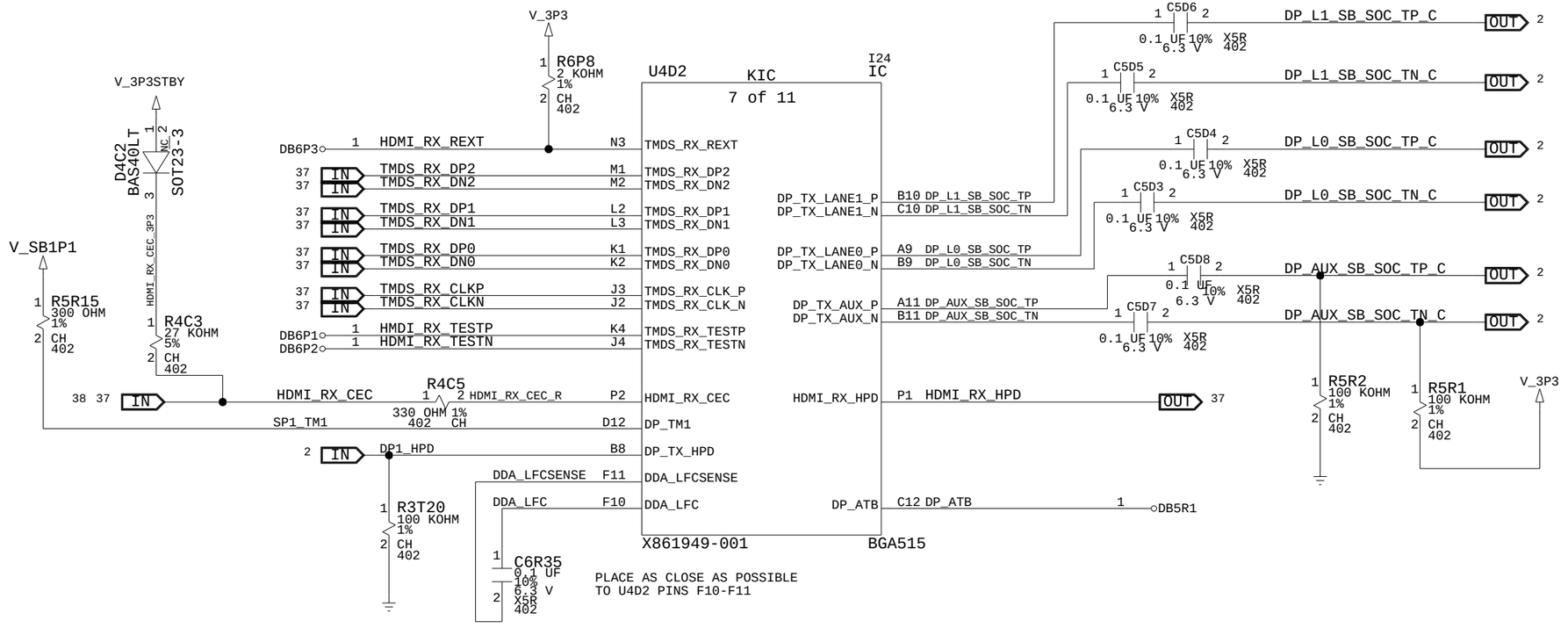
R6R19 1 2 174 OHM 1% CH 402

X861949-001 BGA515

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	24/72	24/72	M	1.0

KIC, PCIEX + SATA + VIDEO

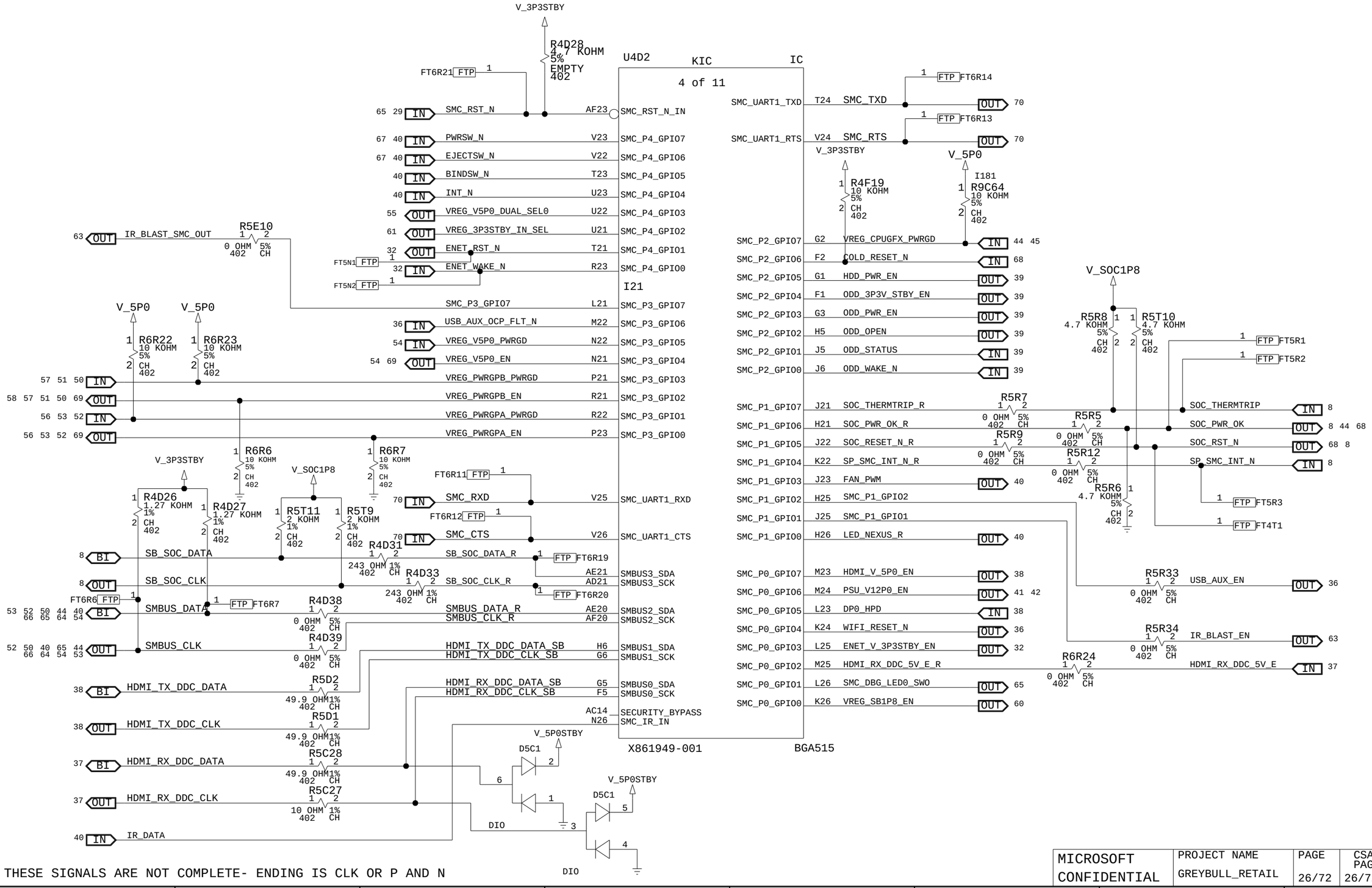




MS_PART#	MATL	REF_DES	TRUE_DESCR.	BOM PROPERTY
U4D2	EMPTY	EMPTY KIC	DUMMY_BOM	

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	25/72	25/72	M	1.0

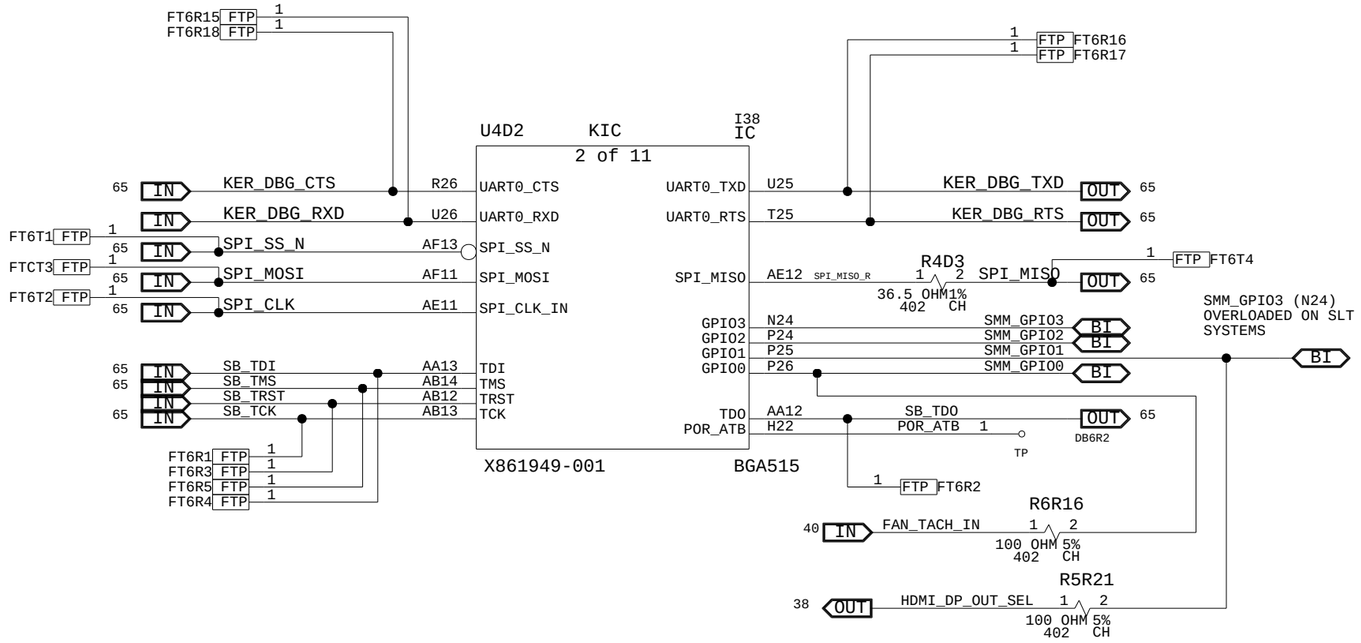
KIC, SMC



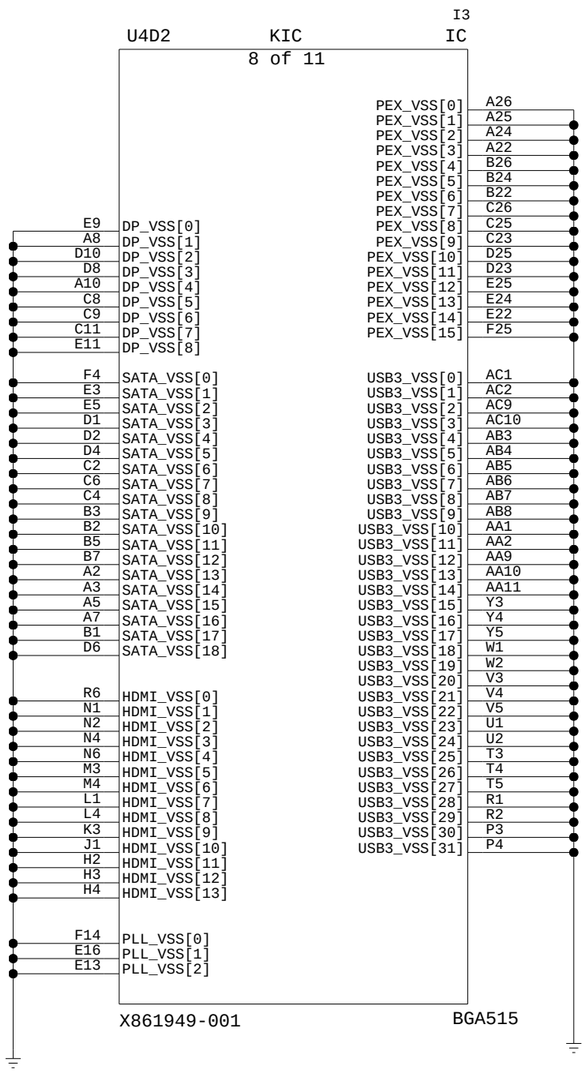
THESE SIGNALS ARE NOT COMPLETE- ENDING IS CLK OR P AND N

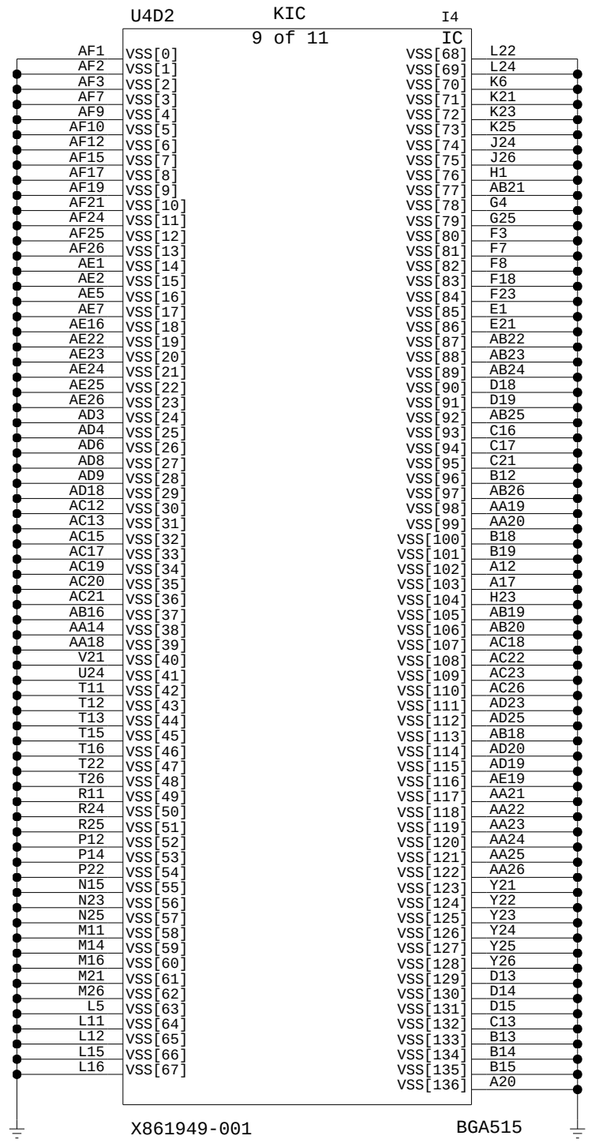
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	26/72	26/72	M	1.0

KIC, FACET



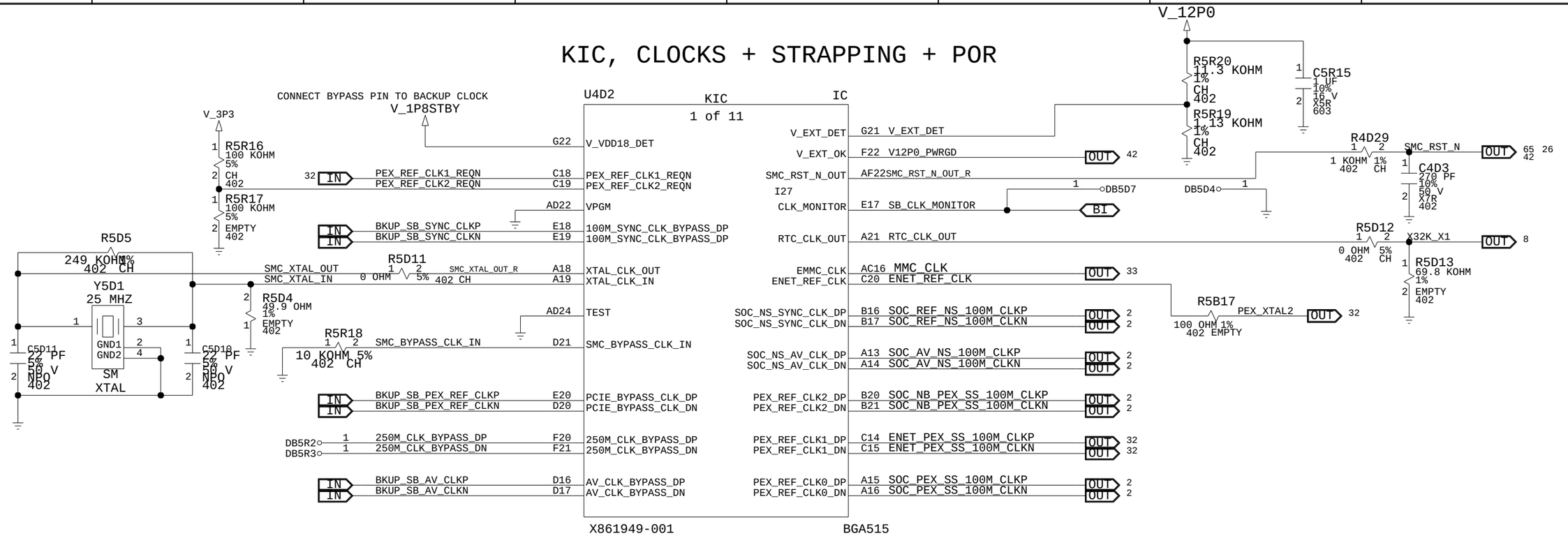
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	27/72	27/72	M	1.0



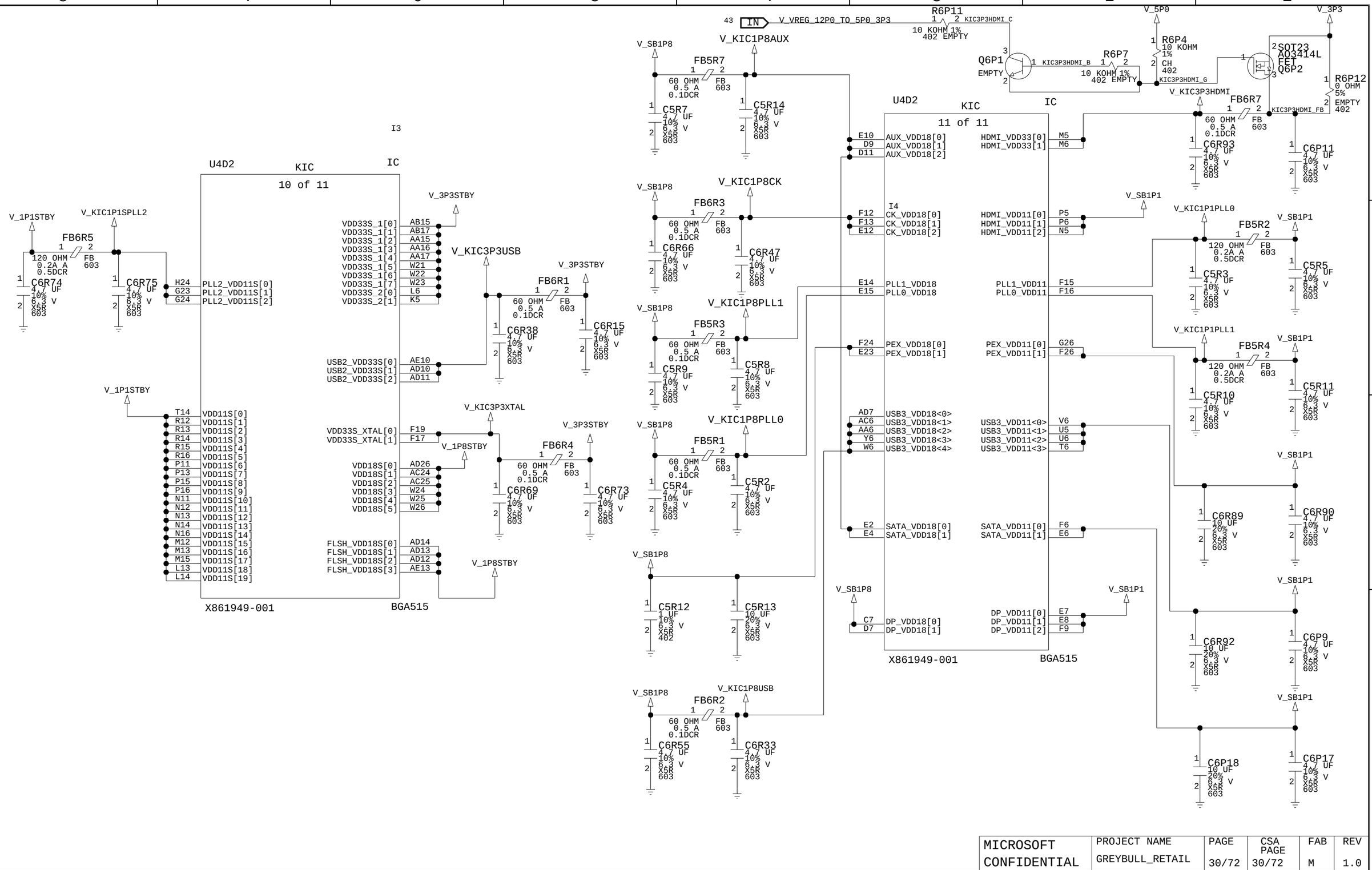


MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	28/72	28/72	M	1.0

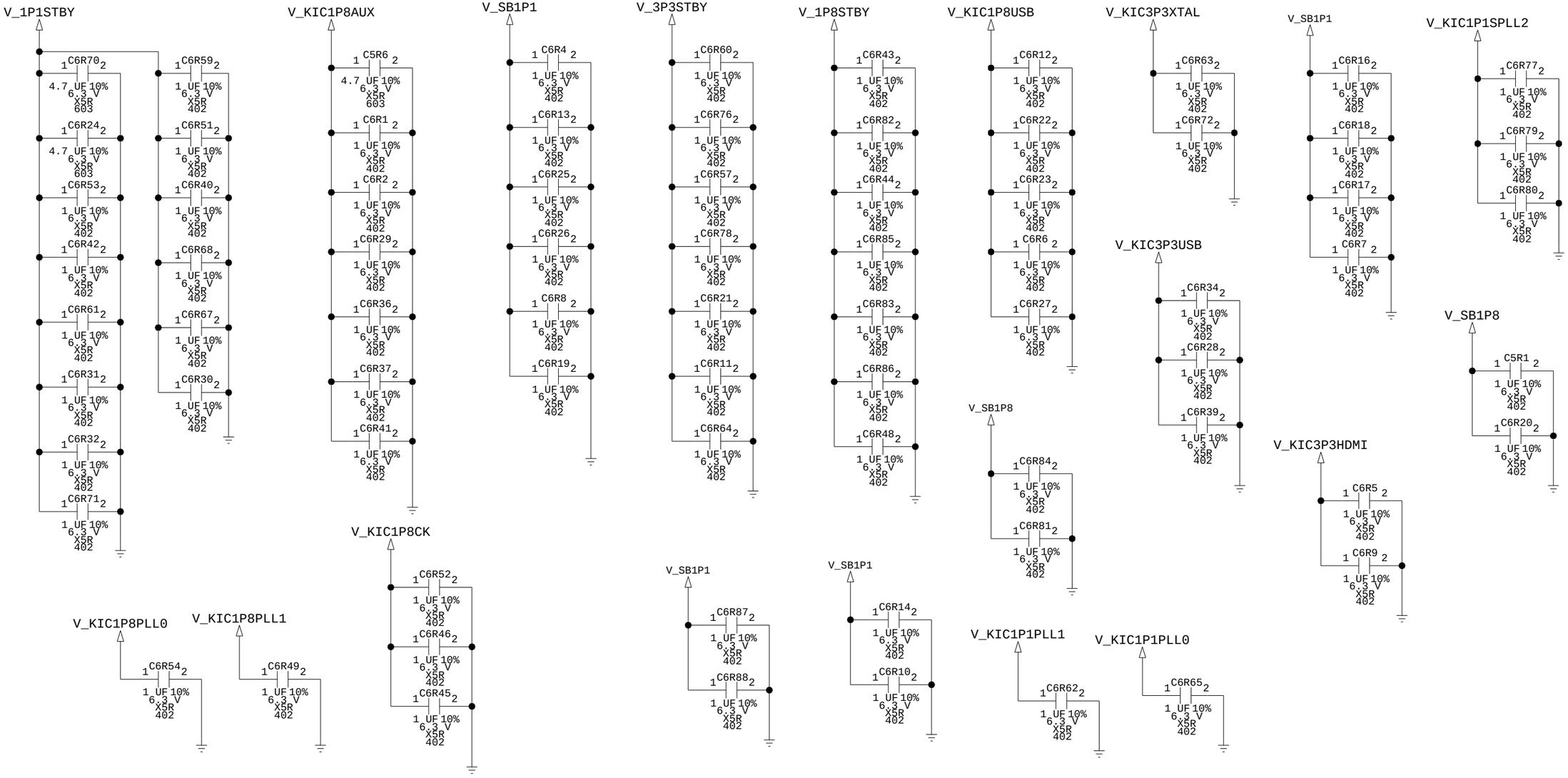
KIC, CLOCKS + STRAPPING + POR



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	29/72	29/72	M	1.0

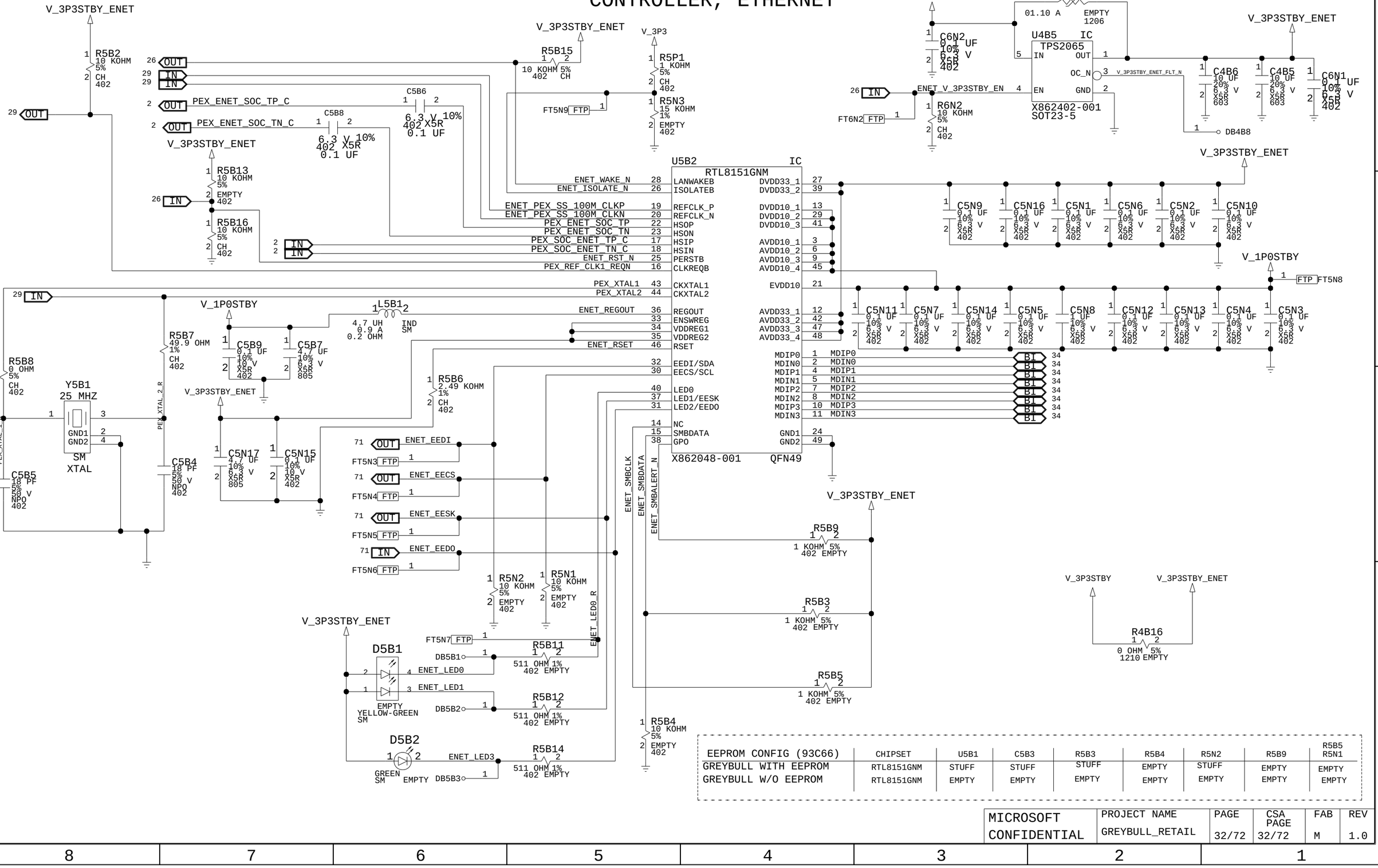

I3
U4D2
KIC
IC
10 of 11
X861949-001
BGA515
V_1P1STBY
V_KIC1P1SPLL2
FB6R5
120 OHM 0.2A A 0.5DCR
C6R74 4.7 UF 10% 6.3 V X5R 603
C6R75 4.7 UF 10% 6.3 V X5R 603
H24 PLL2_VDD11S[0]
G23 PLL2_VDD11S[1]
G24 PLL2_VDD11S[2]
V_3P3STBY
V_KIC3P3USB
FB6R1
60 OHM 0.5 A 0.1DCR FB 603
C6R38 4.7 UF 10% 6.3 V X5R 603
C6R15 4.7 UF 10% 6.3 V X5R 603
VDD33S_1[0] AB15
VDD33S_1[1] AB17
VDD33S_1[2] AA15
VDD33S_1[3] AA16
VDD33S_1[4] AA17
VDD33S_1[5] W21
VDD33S_1[6] W22
VDD33S_1[7] W23
VDD33S_2[0] L6
VDD33S_2[1] K5
USB2_VDD33S[0] AE10
USB2_VDD33S[1] AD10
USB2_VDD33S[2] AD11
T14 VDD11S[0]
R12 VDD11S[1]
R13 VDD11S[2]
R14 VDD11S[3]
R15 VDD11S[4]
R16 VDD11S[5]
P11 VDD11S[6]
P13 VDD11S[7]
P15 VDD11S[8]
P16 VDD11S[9]
N11 VDD11S[10]
N12 VDD11S[11]
N13 VDD11S[12]
N14 VDD11S[13]
N16 VDD11S[14]
M12 VDD11S[15]
M13 VDD11S[16]
M15 VDD11S[17]
L13 VDD11S[18]
L14 VDD11S[19]
V_KIC3P3XTAL
VDD33S_XTAL[0] F19
VDD33S_XTAL[1] F17
V_1P8STBY
FB6R4
60 OHM 0.5 A 0.1DCR FB 603
C6R69 4.7 UF 10% 6.3 V X5R 603
C6R73 4.7 UF 10% 6.3 V X5R 603
VDD18S[0] AD26
VDD18S[1] AC24
VDD18S[2] AC25
VDD18S[3] W24
VDD18S[4] W25
VDD18S[5] W26
FLSH_VDD18S[0] AD14
FLSH_VDD18S[1] AD13
FLSH_VDD18S[2] AD12
FLSH_VDD18S[3] AE13
V_1P8STBY
43 IN V_VREG_12P0_TO_5P0_3P3
R6P11 10 KOHM 1% 402 EMPTY
KIC3P3HDMI_C
Q6P1 EMPTY
KIC3P3HDMI_B
R6P7 10 KOHM 1% 402 EMPTY
KIC3P3HDMI_G
V_5P0
R6P4 10 KOHM 1% CH 402
SOT23 AO3414L FET Q6P2
V_3P3
R6P12 0 OHM 5% EMPTY 402
V_KIC3P3HDMI
FB6R7
60 OHM 0.5 A 0.1DCR FB 603
KIC3P3HDMI_FB
C6R93 4.7 UF 10% 6.3 V X5R 603
C6P11 4.7 UF 10% 6.3 V X5R 603
V_KIC1P8AUX
FB5R7
60 OHM 0.5 A 0.1DCR FB 603
C5R7 4.7 UF 10% 6.3 V X5R 603
C5R14 4.7 UF 10% 6.3 V X5R 603
V_KIC1P8CK
FB6R3
60 OHM 0.5 A 0.1DCR FB 603
C6R66 4.7 UF 10% 6.3 V X5R 603
C6R47 4.7 UF 10% 6.3 V X5R 603
V_KIC1P8PLL1
FB5R3
60 OHM 0.5 A 0.1DCR FB 603
C5R9 4.7 UF 10% 6.3 V X5R 603
C5R8 4.7 UF 10% 6.3 V X5R 603
V_KIC1P8PLL0
FB5R1
60 OHM 0.5 A 0.1DCR FB 603
C5R4 4.7 UF 10% 6.3 V X5R 603
C5R2 4.7 UF 10% 6.3 V X5R 603
C5R12 1 UF 10% 6.3 V X5R 402
C5R13 10 UF 20% 6.3 V X5R 603
V_KIC1P8USB
FB6R2
60 OHM 0.5 A 0.1DCR FB 603
C6R55 4.7 UF 10% 6.3 V X5R 603
C6R33 4.7 UF 10% 6.3 V X5R 603
E10 AUX_VDD18[0]
D9 AUX_VDD18[1]
D11 AUX_VDD18[2]
F12 CK_VDD18[0]
F13 CK_VDD18[1]
E12 CK_VDD18[2]
E14 PLL1_VDD18
E15 PLL0_VDD18
F24 PEX_VDD18[0]
E23 PEX_VDD18[1]
AD7 USB3_VDD18<0>
AC6 USB3_VDD18<1>
AA6 USB3_VDD18<2>
Y6 USB3_VDD18<3>
W6 USB3_VDD18<4>
E2 SATA_VDD18[0]
E4 SATA_VDD18[1]
C7 DP_VDD18[0]
D7 DP_VDD18[1]
HDMI_VDD33[0] M5
HDMI_VDD33[1] M6
HDMI_VDD11[0] P5
HDMI_VDD11[1] P6
HDMI_VDD11[2] N5
PLL1_VDD11 F15
PLL0_VDD11 F16
PEX_VDD11[0] G26
PEX_VDD11[1] F26
USB3_VDD11<0> V6
USB3_VDD11<1> U5
USB3_VDD11<2> U6
USB3_VDD11<3> T6
SATA_VDD11[0] F6
SATA_VDD11[1] E6
DP_VDD11[0] E7
DP_VDD11[1] E8
DP_VDD11[2] F9
V_SB1P1
V_KIC1P1PLL0
FB5R2
120 OHM 0.2A A 0.5DCR FB 603
C5R3 4.7 UF 10% 6.3 V X5R 603
C5R5 4.7 UF 10% 6.3 V X5R 603
V_KIC1P1PLL1
FB5R4
120 OHM 0.2A A 0.5DCR FB 603
C5R10 4.7 UF 10% 6.3 V X5R 603
C5R11 4.7 UF 10% 6.3 V X5R 603
C6R89 10 UF 20% 6.3 V X5R 603
C6R90 4.7 UF 10% 6.3 V X5R 603
C6R92 10 UF 20% 6.3 V X5R 603
C6P9 4.7 UF 10% 6.3 V X5R 603
C6P18 10 UF 20% 6.3 V X5R 603
C6P17 4.7 UF 10% 6.3 V X5R 603
MICROSOFT CONFIDENTIAL
PROJECT NAME GREYBULL_RETAIL
PAGE 30/72
CSA PAGE 30/72
FAB M
REV 1.0

KIC, DECOUPLING



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	31/72	31/72	M	1.0

CONTROLLER, ETHERNET



EEPROM CONFIG (93C66)	CHIPSET	U5B1	C5B3	R5B3	R5B4	R5N2	R5B9	R5B5 R5N1
GREYBULL WITH EEPROM	RTL8151GNM	STUFF	STUFF	STUFF	EMPTY	STUFF	EMPTY	EMPTY
GREYBULL W/O EEPROM	RTL8151GNM	EMPTY	EMPTY	EMPTY	EMPTY	EMPTY	EMPTY	EMPTY

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	32/72	32/72	M	1.0

EMMC

MS_PART#	MATL	REF_DES		TRUE DESCR.	BOM PROPERTY
U3D1	X866677-001	IC	MEM,SM,8GB,FLASH,EMMC	EMMC_TOSH	
U3D1	X875870-001	IC	MEM,SM,8GB,FLASH,EMMC	EMMC_SEC	
U3D1	X867925-001	IC	MEM,SM,8GB,FLASH,EMMC	EMMC_HX	

V_1P8STBY

R4D4 R4D9 R4D13 R4D11 R4D17 R4D15 R4D21 R4D19 R4D23 — 402 — 1/16W — 1% — 10 KOHM

R4D7 10 KOHM 1% 402 CH 1/16W X806693-001

R4D10 R4D14 R4D12 R4D18 R4D16 R4D22 R4D20 R4D24 R4D8 R4D2 — 0 OHM 5% 402 CH

R4D6 1 KOHM 5% 402 CH

R4D36 10 KOHM 1% EMPTY 402

33	BI	MMC_DATA<7>
33	BI	MMC_DATA<6>
33	BI	MMC_DATA<5>
33	BI	MMC_DATA<4>
33	BI	MMC_DATA<3>
33	BI	MMC_DATA<2>
33	BI	MMC_DATA<1>
33	BI	MMC_DATA<0>
33	BI	MMC_CMD
29	BI	MMC_CLK
33	BI	MMC_RST_N

MMC_RST_N_R — DB3D3 FTP — FT7R1 FTP — FT7R2 FTP

U3D1 IC — emmc_169pin 1 of 3 — X866676-001 BGA169B

MMC_DATA_R<7>	J6	DATA7	
MMC_DATA_R<6>	J5	DATA6	
MMC_DATA_R<5>	J4	DATA5	
MMC_DATA_R<4>	J3	DATA4	
MMC_DATA_R<3>	J2	DATA3	
MMC_DATA_R<2>	H5	DATA2	
MMC_DATA_R<1>	H4	DATA1	
MMC_DATA_R<0>	H3	DATA0	
MMC_CMD_R	W5	CMD	
MMC_CLK_R	W6	CLK	
	U5	RSTN	
	AA5	VCCQ4	
	AA3	VCCQ3	
	Y4	VCCQ2	
	W4	VCCQ1	
	K6	VCCQ0	
	U9	VCC3	
	T10	VCC2	
	N5	VCC1	
	M6	VCC0	
	K2	VDDI	
	AA6	VSSQ4	
	AA4	VSSQ3	
	Y5	VSSQ2	
	Y2	VSSQ1	
	K4	VSSQ0	
	U8	VSS3	
	R10	VSS2	
	P5	VSS1	
	M7	VSS0	

V_1P8STBY

C7R4 0.1 UF 10% 6.3 V X5R 402 — C3D8 0.1 UF 10% 6.3 V X5R 402 — C7R5 2.2 UF 20% 6.3 V X5R 402 — C7R9 1 UF 10% 6.3 V X5R 402 — C6R50 0.1 UF 10% 6.3 V X5R 402 — C6R58 0.1 UF 10% 6.3 V X5R 402 — C6R56 2.2 UF 20% 6.3 V X5R 402

V_3P3STBY

C7R6 0.1 UF 10% 6.3 V X5R 402 — C7R7 0.1 UF 10% 6.3 V X5R 402 — C7R8 2.2 UF 20% 6.3 V X5R 402

VDDI_ROUTE

C3D9 1 UF 10% 6.3 V X5R 402

MMC_X NETS ARE DUMMY NETS TO IMPROVE PAD ADHESION

U4D2 KIC IC — 5 of 12 — I446 — X861949-001 BGA515

MMC_DATA<7..0> BI 33

MMC_DATA7	AE18	7
MMC_DATA6	AD15	6
MMC_DATA5	AE15	5
MMC_DATA4	AD16	4
MMC_DATA3	AF16	3
MMC_DATA2	AF18	2
MMC_DATA1	AD17	1
MMC_DATA0	AE17	0
MMC_CMD	AF14	MMC_CMD — OUT 33
MMC_RESET_N	AE14	MMC_RST_N — OUT 33

U3D1 IC — eMMC_169PIN 2 of 3 — X866676-001 BGA169B

A4	NC0	NC35	K14
A6	NC1	NC36	L1
A9	NC2	NC37	L2
A11	NC3	NC38	L3 D5
B2	NC4	NC39	L4
B13	NC5	NC40	L12
D1	NC6	NC41	L13
D14	NC7	NC42	L14
MMC_X7 H1	NC8	NC43	M1 — DB633
D4 H2	NC9	NC44	M2
H8	NC10	NC45	M3
H9	NC11	NC46	M12
H10	NC12	NC47	M13
H11	NC13	NC48	M14 — DB634
H12	NC14	NC49	N1
MMC_X6 H13	NC15	NC50	N2
MMC_X4 H14	NC16	NC51	N3
MMC_X8 J1	NC17	NC52	N12
J7	NC18	NC53	N13
J8	NC19	NC54	N14 — DB637
J9	NC20	NC55	P1
J10	NC21	NC56	P2
J11	NC22	NC57	P12
J12	NC23	NC58	P13
J13	NC24	NC59	P14 — DB635
MMC_X5 J14	NC25	NC60	R1
D3 K1	NC26	NC61	R2
D5 K3	NC27	NC62	R3
D7 K7	NC28	NC63	R12
K8	NC29	NC64	R13
K9	NC30	NC65	R14 — DB636
K10	NC31	NC66	T1
K11	NC32	NC67	T2
K12	NC33	NC68	T3
K13	NC34	NC69	T12

U3D1 IC — eMMC_169PIN 3 of 3 — X866676-001 BGA169B

AA7	RFU0	NC87	W7
AA10	RFU1	NC88	W8
H6	RFU2	NC89	W9
H7	RFU3	NC90	W10
D6 K5	RFU4	NC91	W11
M5	RFU5	NC92	W12
M8	RFU6	NC93	W13
M9	RFU7	NC94	W14
M10	RFU8	NC95	Y1 MMC_X11
N10	RFU9	NC96	Y3
P3	RFU10	NC97	Y6
P10	RFU11	NC98	Y7
R5	RFU12	NC99	Y8
T5	RFU13	NC100	Y9
U6	RFU14	NC101	Y10
U7	RFU15	NC102	Y11
U10	RFU16	NC103	Y12
		NC104	Y13
T13	NC70	NC105	Y14 MMC_X3
T14	NC71	NC106	AA1 MMC_X10
U1	NC72	NC107	AA2 MMC_X12
U2	NC73	NC108	AA8
U3	NC74	NC109	AA9
U12	NC75	NC110	AA11
U13	NC76	NC111	AA12
U14	NC77	NC112	AA13 MMC_X2
V1	NC78	NC113	AA14 MMC_X1
V2	NC79	NC114	AE1
V3	NC80	NC115	AE14
V12	NC81	NC116	AG2
V13	NC82	NC117	AG13
V14	NC83	NC118	AH4
W1	NC84	NC119	AH6
W2	NC85	NC120	AH9
W3	NC86	NC121	AH11

DB638

SIGNALS ROUTED TO NC PINS FOR ESCAPE

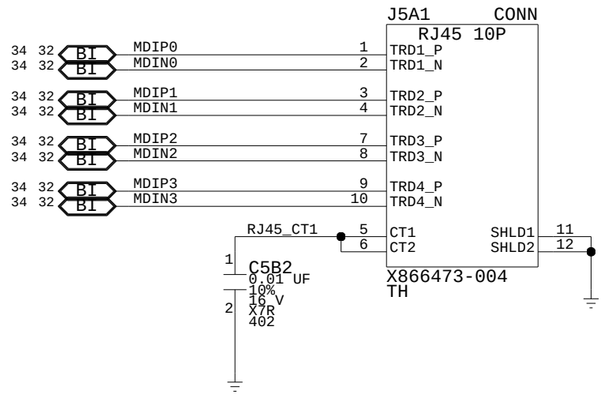
D7	PIN J6 ROUTED TO PIN K7, K8
D6	PIN J5 ROUTED TO PIN K5
D5	PIN J4 ROUTED TO PIN K3, L3
D4	PIN J3 ROUTED TO PIN H2
D3	PIN J2 ROUTED TO PIN K1
VDDI	PIN K2 ROUTED TO PIN L1
GND	PIN Y2 ROUTED TO PIN W1

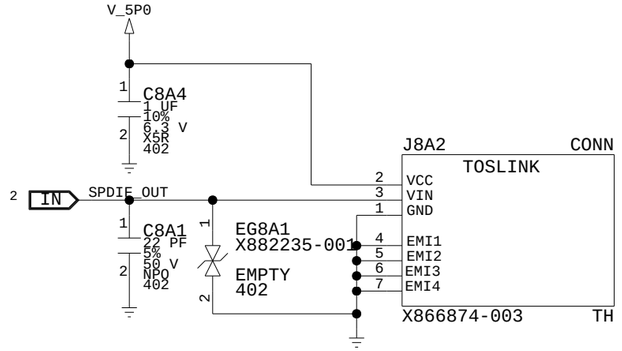
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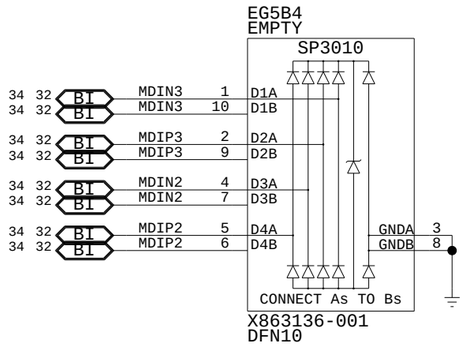
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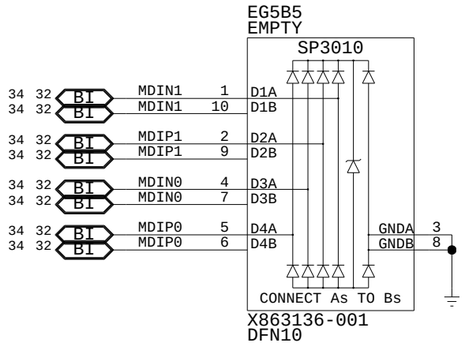
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	33/72	33/72	M	1.0

CONN, RJ45 + TOSLINK



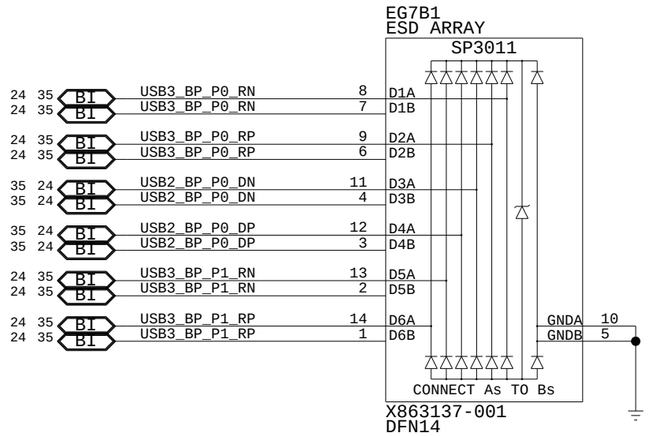


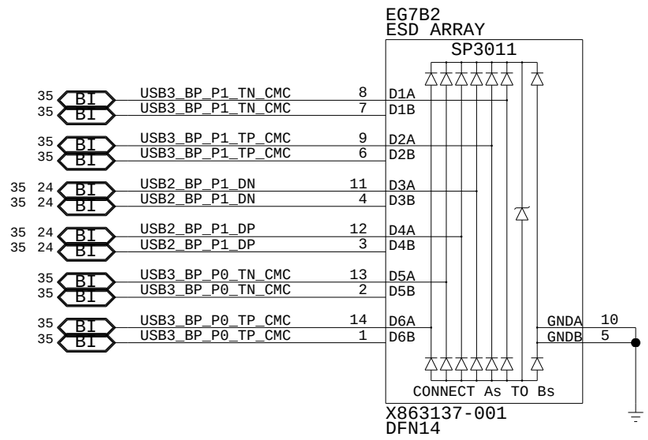


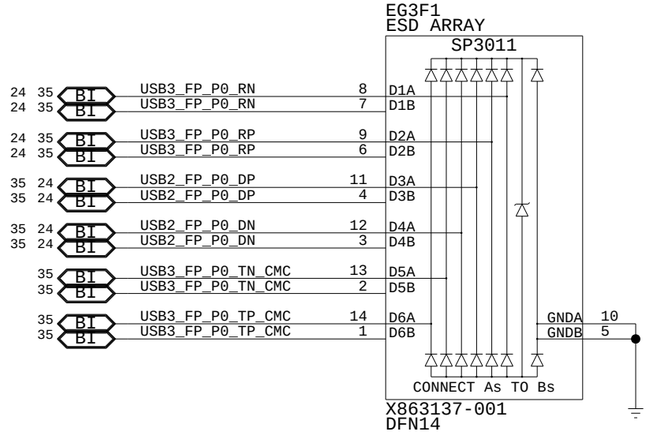


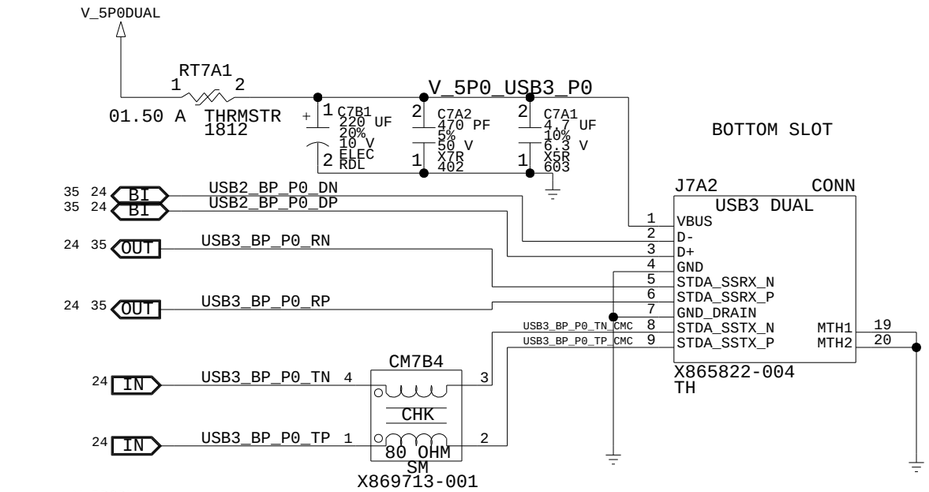
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	34/72	34/72	M	1.0

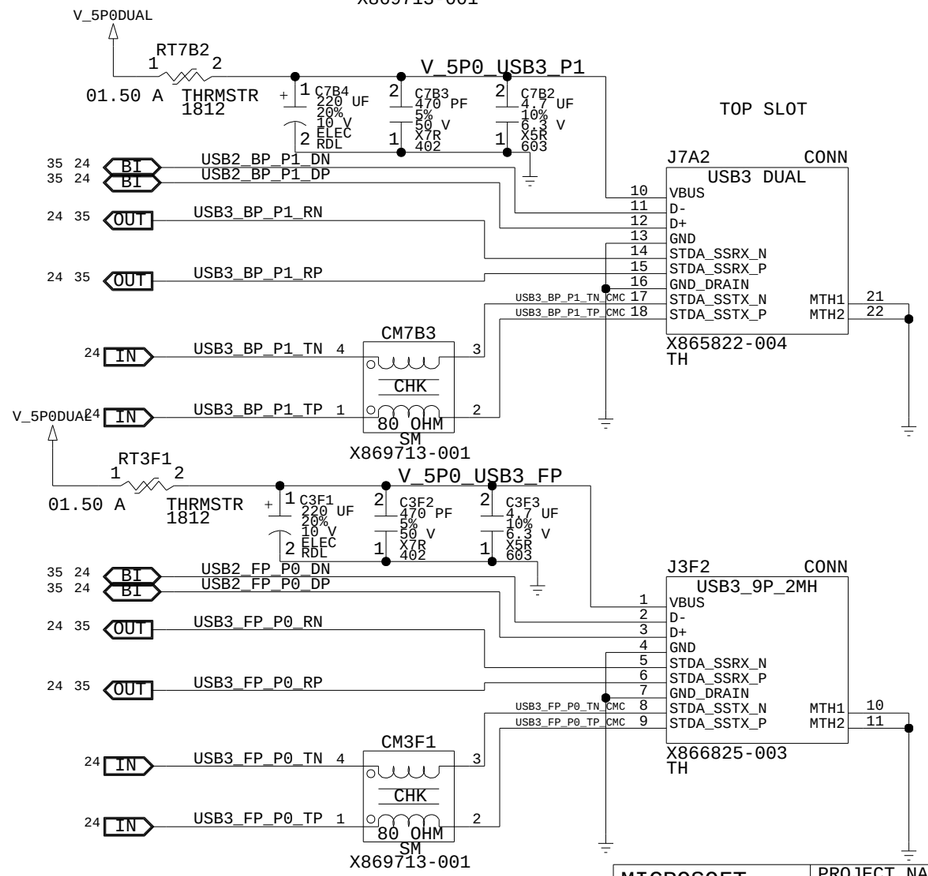
CONN, USB





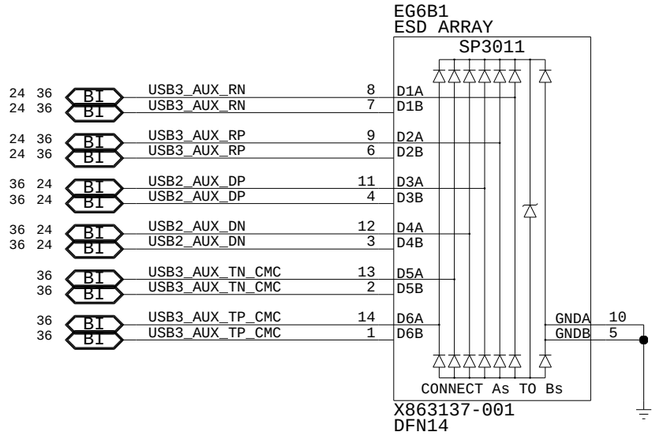


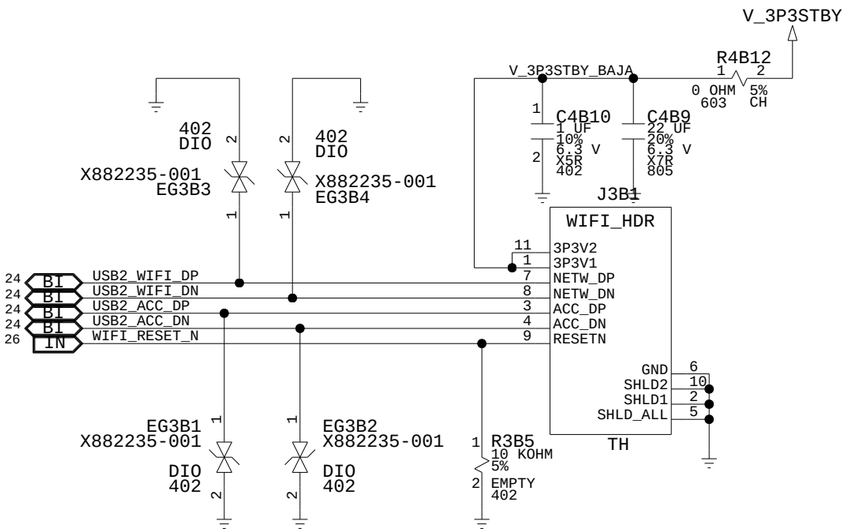


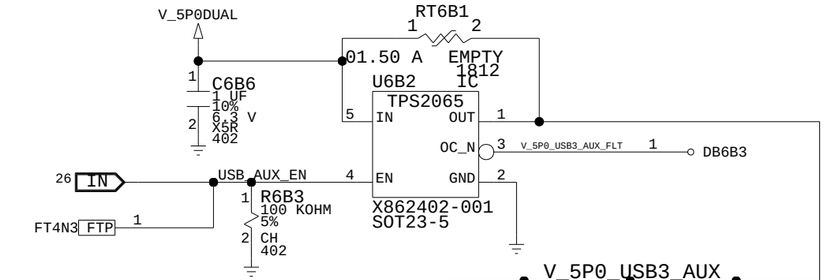


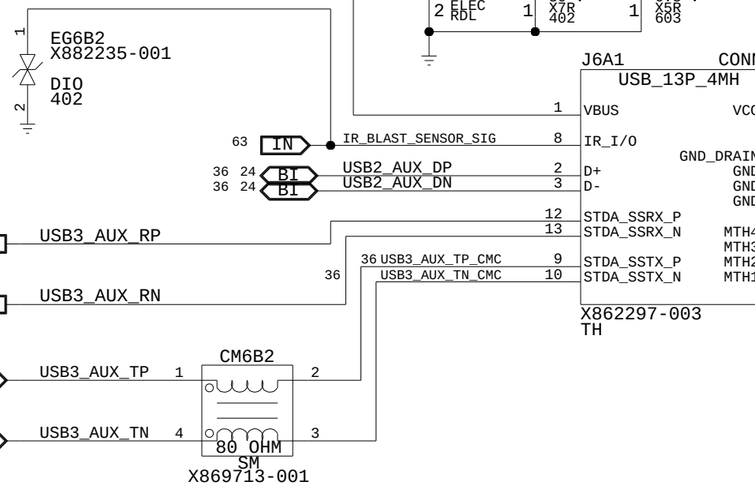
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	35/72	35/72	M	1.0

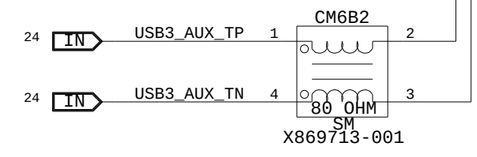
CONN, USB

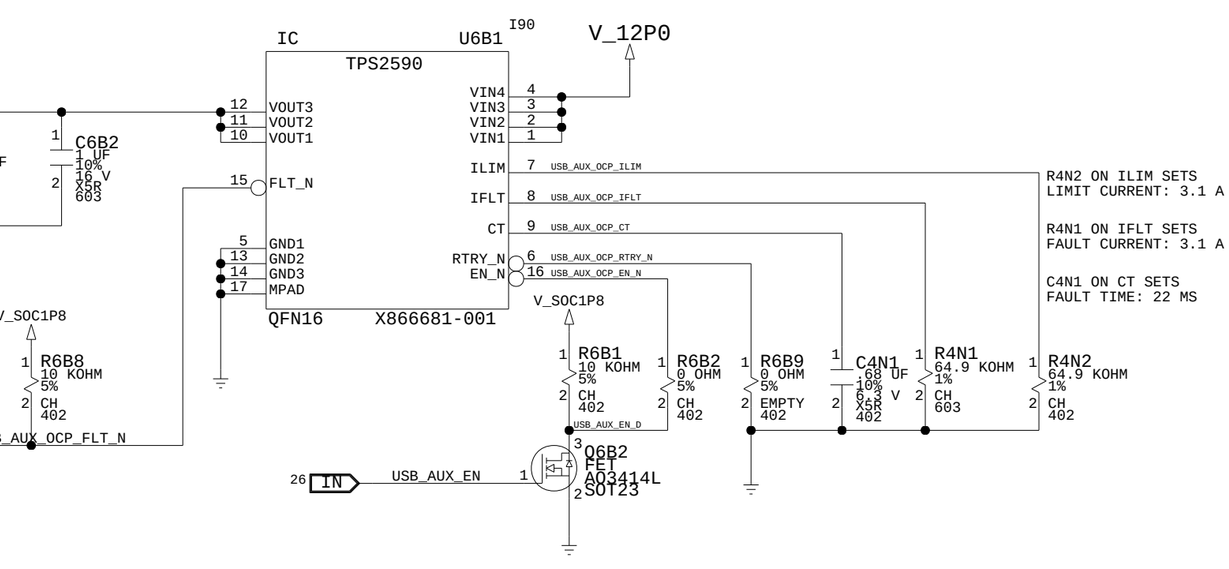












MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	36/72	36/72	M	1.0

CONN, HDMI IN

25 OUT TMDS_RX_DP2
25 OUT TMDS_RX_DN2
25 OUT TMDS_RX_DP1
25 OUT TMDS_RX_DN1
25 OUT TMDS_RX_DP0
25 OUT TMDS_RX_DN0
25 OUT TMDS_RX_CLKP
25 OUT TMDS_RX_CLKN

EG7B3
ESD ARRAY
SP3010
1 D1A
10 D1B
2 D2A
9 D2B
4 D3A
7 D3B
5 D4A
6 D4B
GNDA 3
GNDB 8
CONNECT As TO Bs
X863136-001
DFN10

EG7B4
ESD ARRAY
SP3010
1 D1A
10 D1B
2 D2A
9 D2B
4 D3A
7 D3B
5 D4A
6 D4B
GNDA 3
GNDB 8
CONNECT As TO Bs
X863136-001
DFN10

J7A1 CONN
HDMI
1 TMDS_DATA2_DP
2 TMDS_DATA2_SHD
3 TMDS_DATA2_DN
4 TMDS_DATA1_DP
5 TMDS_DATA1_SHD
6 TMDS_DATA1_DN
7 TMDS_DATA0_DP
8 TMDS_DATA0_SHD
9 TMDS_DATA0_DN
10 TMDS_CLK_DP
11 TMDS_CLK_SHD
12 TMDS_CLK_DN
13 CEC
14 RESERVED
15 SCL
16 SDA
17 DDC_CEC_GND
18 5VCC
19 HOT_PLUG_DET
MT1 20
MT2 21
MT3 22
MT4 23
X864613-003 SM

25 38 BI HDMI_RX_CEC

EG7A9
X882235-001
DIO
402

V_3P3STBY
R8B19
10 KOHM
1%
CH
402

26 OUT HDMI_RX_DDC_5V_E
R8B20
0 OHM
5%
CH
402
D8B3
SOT-523
EMPTY
HDMI_RX_DDC_5V_D
R8B21
2 KOHM
1%
EMPTY
402
U8B2
FET
SOT23
HDMI_RX_DDC_5V_G
R8B9
10 KOHM 1%
603 CH
HDMI_RX_DDC_5V

R8A2
100 KOHM
1%
CH
402
C8B1
0.1 UF
10%
6.3 V
EMPTY
402
EG7A7
X882235-001
DIO
402
R3M2
47 KOHM
1%
CH
402
R3M1
47 KOHM
1%
CH
402

HDMI_HPD_PIN
R8A3
1 KOHM 5%
402 CH
HDMI_RX_HPD
IN 25
EG8A7
X882235-001
EMPTY
402

26 BI HDMI_RX_DDC_CLK
26 BI HDMI_RX_DDC_DATA
EG7A3
X882235-001
DIO
402
EG7A4
X882235-001
DIO
402

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	37/72	37/72	M	1.0

CONN, HDMI OUT

TMDS_TX_DP2
TMDS_TX_DN2
TMDS_TX_DP1
TMDS_TX_DN1
TMDS_TX_DP0
TMDS_TX_DN0
TMDS_TX_CLKP
TMDS_TX_CLKN

C8B2 402 X5R 6.3 V 0.1 UF 10%
C8B3 402 X5R 6.3 V 0.1 UF 10%
C8B4 402 X5R 6.3 V 0.1 UF 10%
C8B5 402 X5R 6.3 V 0.1 UF 10%
C8B6 402 X5R 6.3 V 0.1 UF 10%
C8B7 402 X5R 6.3 V 0.1 UF 10%
C8B8 402 X5R 6.3 V 0.1 UF 10%
C8B9 402 X5R 6.3 V 0.1 UF 10%

TMDS_TX_DP2_C
TMDS_TX_DN2_C
TMDS_TX_DP1_C
TMDS_TX_DN1_C
TMDS_TX_DP0_C
TMDS_TX_DN0_C
TMDS_TX_CLKP_C
TMDS_TX_CLKN_C

R8B1 R8B2 R8B3 R8B4 R8B5 R8B6 R8B7 R8B8 499 OHM 1% CH 402

TMDS_COMMON_FET

V_5P0DUAL
R2N3 47.5 KOHM 1% EMPTY 402

FT2N1 FTP TO SB GPIO
HDMI_DP_OUT_SEL
R2N2 47.5 KOHM 1% CH 402
U2N1 FET SOT23 X801032-001

PLACEHOLDER, ACTUAL FET REQURIEMENTS NEED TO BE ASSESSED.

C8B19 0.1 UF 10% 25 V EMPTY 402

EG8B2 ESD ARRAY SP3010 X863136-001 DFN10
D1A D1B D2A D2B D3A D3B D4A D4B GNDA GNDB CONNECT As TO Bs

EG8B1 ESD ARRAY SP3010 X863136-001 DFN10

ESD DIODE ARRAYS ARE INCLUDED FOR EVALUATION. TO BE CLEANED UP LATER.

J8A1 CONN HDMI X864613-003 SM

1	TMDS_DATA2_DP
2	TMDS_DATA2_SHD
3	TMDS_DATA2_DN
4	TMDS_DATA1_DP
5	TMDS_DATA1_SHD
6	TMDS_DATA1_DN
7	TMDS_DATA0_DP
8	TMDS_DATA0_SHD
9	TMDS_DATA0_DN
10	TMDS_CLK_DP
11	TMDS_CLK_SHD
12	TMDS_CLK_DN
13	CEC
14	RESERVED
15	SCL
16	SDA
17	DDC_CEC_GND
18	5VCC
19	HOT_PLUG_DET
MT1	20
MT2	21
MT3	22
MT4	23

HDMI_RX_CEC
HDMI_TX_CEC
R8A1 0 OHM 5% 402 CH

V_5P0_HDMI_OUT
R1N2 10 KOHM 5% 402 EMPTY
HDMI_OUT_DDC_BUFF_EN

IC U1N1 PCA9507 VCCB VCCA SCLB SCLA SDAB SDAA GND EN
X871021-001 MSOP8

R1N4 2 KOHM 1% CH 402
C1N1 0.1 UF 10% 25 V X5R 402
R1N3 2 KOHM 1% CH 402
R1M4 2 KOHM 1% CH 402
R1M3 2 KOHM 1% CH 402
C9B3 0.1 UF 10% 25 V X5R 402

HDMI_TX_DDC_CLK
HDMI_TX_DDC_DATA
R9B4 0 OHM 5% 402 CH
R9B5 0 OHM 5% 402 CH
R9A7 22 OHM 5% 402 CH
R9A8 22 OHM 5% 402 CH
HDMI_OUT_DDC_BUFF_CLK
HDMI_OUT_DDC_BUFF_DATA

HDMI_OUT_DDC_CLK
HDMI_OUT_DDC_DATA

C9B13 0.1 UF 10% 6.3 V EMPTY 402
C9B15 0.1 UF 10% 6.3 V EMPTY 402
HDMI_OUT_DDC_CLK_R
HDMI_OUT_DDC_DATA_R
V_3P3
R9A6 100 KOHM 5% 2 EMPTY 402
R9A5 100 KOHM 5% 2 EMPTY 402

V_5P0DUAL
C2N1 1 UF 10% 6.3 V X5R 402
RT8A1 01.10 A EMPTY 1206
U8B1 IC TPS2051C IN OUT FLT* EN GND
X867973-001 SOT23-5
V_5P0_HDMI_FLT_N
HDMI_V_5P0_EN
R2N4 10 KOHM 5% CH 402
FT2N2 FTP

V_5P0_HDMI_OUT
DB8B1
C8A2 4.7 UF 10% 10 V X5R 805
C8A3 0.1 UF 10% 6.3 V X5R 402

EG9A2 X882235-001 DIO 402
EG8A8 X882235-001 DIO 402
V_3P3STBY
EG9A1 X882235-001 DIO 402
EG9B1 X882235-001 DIO 402
HDMI_TX_HPD_PIN

R9B1 10 KOHM 1% CH 402
U9B2 FET SOT23 X857156-001 DP0_HPD
HDMI_TX_HPD_D
R9B16 10 KOHM 1% 402 CH
HDMI_TX_HPD_C
R1N1 100 KOHM 1% CH 402
C9B1 0.1 UF 10% 6.3 V EMPTY 402
U9B1 FET SOT23

AUX VS DDX STUFFING

MODE	R9B8/R9B9	R9A5/R9A6	R9A3/R9A4
AUX	0.1 UF CAP	STUFF	EMPTY
DDC	0 OHM RES	EMPTY	STUFF

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	38/72	38/72	M	1.0

CONN, ODD + HDD

HDD SATA

25 IN SATA1_HDD_TP — C5C6 0.01 UF 10% 16 V X7R 402 — SATA1_HDD_TP_C

25 IN SATA1_HDD_TN — C5C7 0.01 UF 10% 16 V X7R 402 — SATA1_HDD_TN_C

25 OUT SATA1_HDD_RN — C5C8 0.01 UF 10% 16 V X7R 402 — SATA1_HDD_RN_C

25 OUT SATA1_HDD_RP — C5C9 0.01 UF 10% 16 V X7R 402 — SATA1_HDD_RP_C

J5C1 1X7SATA (9, 1–8) TH

39 IN V_5P0_HDD — C5C13 4.7 UF 10% 6.3 V X5R 603; C5P4 1 UF 10% 6.3 V X5R 402 — J4B2 1X5HDR2 (1–4) TH

V_3P3STBY — R5C8 0 OHM 5% CH 402

R5C7 10 KOHM 5% CH 402 — ODD_3P3V_STBY_S

26 IN ODD_3P3V_STBY_EN — U5C2 FET SOT23 X857156-001

ODD_3P3V_STBY_PULSE — OUT 39

V_12P0 — C4C3 1 UF 10% 16 V X7R 603

V_5P0 — C4C4 1 UF 10% 6.3 V X5R 402

U4C2 IC NCP45521 X866665-001 DFN8

VCC	3		
VIN1	1	VOUT1	7 V_12P0_ODD
VIN2	9	VOUT2	8
		BLEED	5
SR	6 ODD_PWR_SR		
EN	2	GND	4

26 IN ODD_PWR_EN — R6P3 10 KOHM 5% EMPTY 402

C4C7 0.015 UF 10% 50 V X7R 603

V_12P0_ODD — OUT 39 — C4C9 10 UF 10% 16 V X5R 805; C6P4 10 UF 10% 16 V X5R 805; C6P5 10 UF 10% 16 V EMPTY 805; C4C10 10 UF 10% 16 V EMPTY 805

V_5P0 — C5C1 220 UF 20% 10 V ELEC RDL

U5C4 IC MP5010B X878106-001 DFN11

VCC	11		
SOURCE1	1 V_5P0_HDD		
SOURCE2	2		
SOURCE3	3		
SOURCE4	4		
SOURCE5	5		
DV/DT	9	ILIMIT	7 HDD_PWR_ILIMIT
EN/FLT	8	NC	6
		GND	10

HDD_PWR_DVDT — C5C25 270 PF 10% 50 V X7R 402

26 IN HDD_PWR_EN — R5C2 10 KOHM 5% CH 402; FT5P1 FTP — R5C5 1 KOHM 1% 402 CH — HDD_PWR_EN_R

V_5P0_HDD — OUT 39 — R5C6 22 OHM 5% CH 402; C5C12 10 UF 20% 6.3 V X5R 805

ODD SATA

25 IN SATA0_ODD_TP — C4C17 0.01 UF 10% 16 V X7R 402 — SATA0_ODD_TP_C

25 IN SATA0_ODD_TN — C4C18 0.01 UF 10% 16 V X7R 402 — SATA0_ODD_TN_C

25 OUT SATA0_ODD_RN — C4C19 0.01 UF 10% 16 V X7R 402 — SATA0_ODD_RN_C

25 OUT SATA0_ODD_RP — C4C20 0.01 UF 10% 16 V X7R 402 — SATA0_ODD_RP_C

J4C1 1X7SATA (9, 1–8) TH

V_3P3STBY — R5C14 100 KOHM 5% CH 402

26 OUT

ODD_STATUS_R — R5C13 100 OHM 1% 402 CH — ODD_STATUS — R5C12 100 KOHM 5% CH 402 — OUT 26

26 IN ODD_OPEN — ODD_WAKE_N

J5C3 2X6HDR2 (1, 3, 5, 7, 9, 11 / 4, 6, 8, 10, 12) TH

C5C17 75 PF 5% 50 V NPO 402

ODD_3P3V_STBY_PULSE — IN 39

V_5P0_ODD — IN 39

V_12P0_ODD — IN 39

C5C3 4.7 UF 10% 16 V X5R 805; C4C11 1 UF 10% 16 V X5R 603; C5C14 4.7 UF 10% 6.3 V X5R 603; C5P3 1 UF 10% 6.3 V X5R 402; C5C24 1 UF 10% 6.3 V X5R 402

V_5P0 — C5C2 1 UF 10% 6.3 V X5R 402

U5C1 IC MP5010B X878106-001 DFN11

VCC	11		
SOURCE1	1 V_5P0_ODD		
SOURCE2	2		
SOURCE3	3		
SOURCE4	4		
SOURCE5	5		
DV/DT	9	ILIMIT	7 ODD_PWR_ILIMIT
EN/FLT	8	NC	6
		GND	10

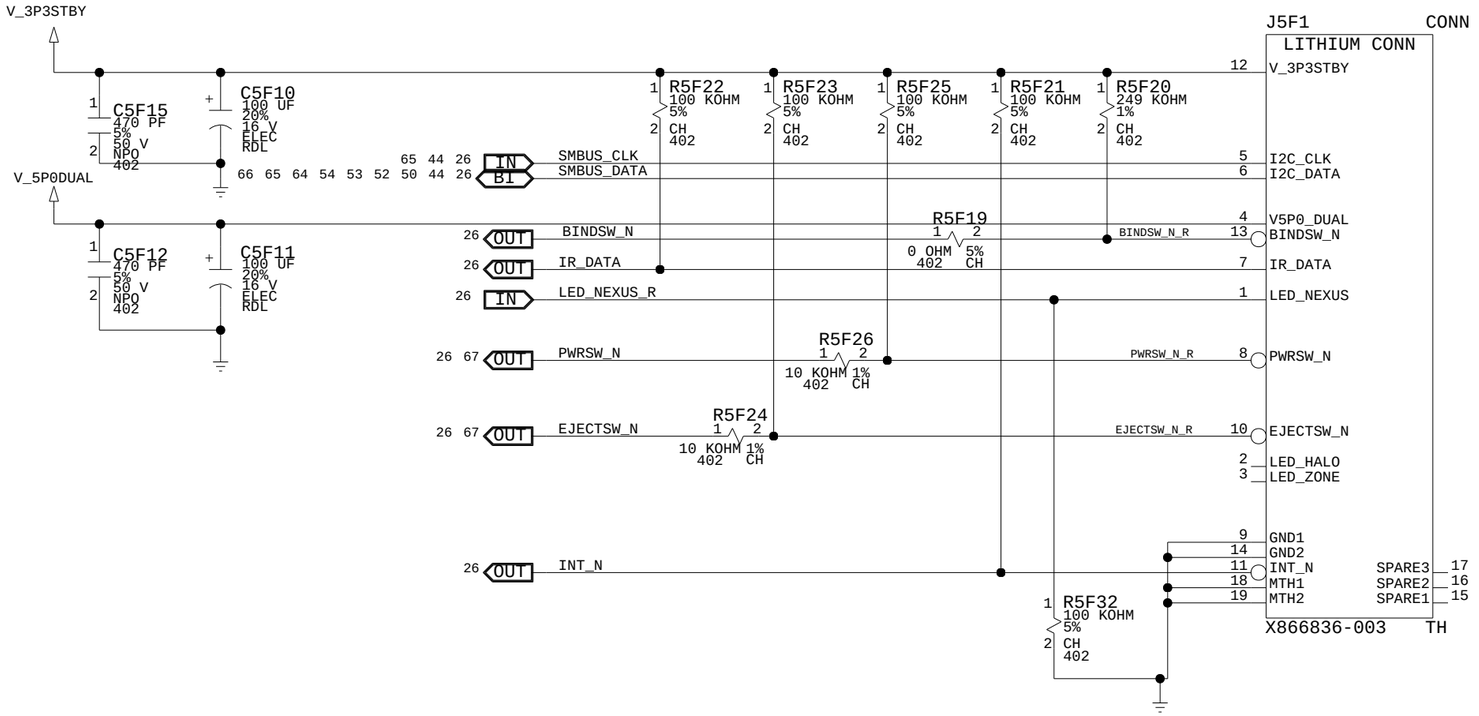
ODD_PWR_DVDT — C5C16 270 PF 10% 50 V X7R 402

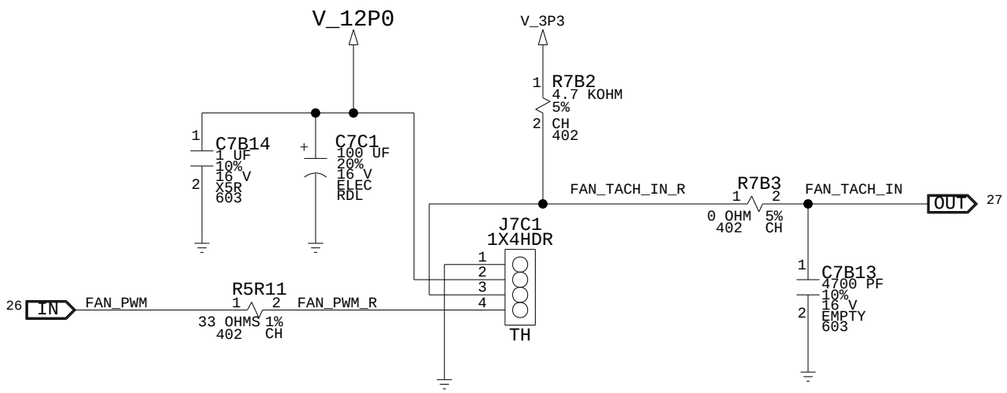
26 IN ODD_PWR_EN — R5C3 10 KOHM 5% CH 402; FT5P2 FTP — R5C1 1 KOHM 1% 402 CH — ODD_PWR_EN_R

V_5P0_ODD — OUT 39 — R5C4 22 OHM 5% CH 402; C5C11 10 UF 20% 6.3 V X5R 805

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	39/72	39/72	M	1.0

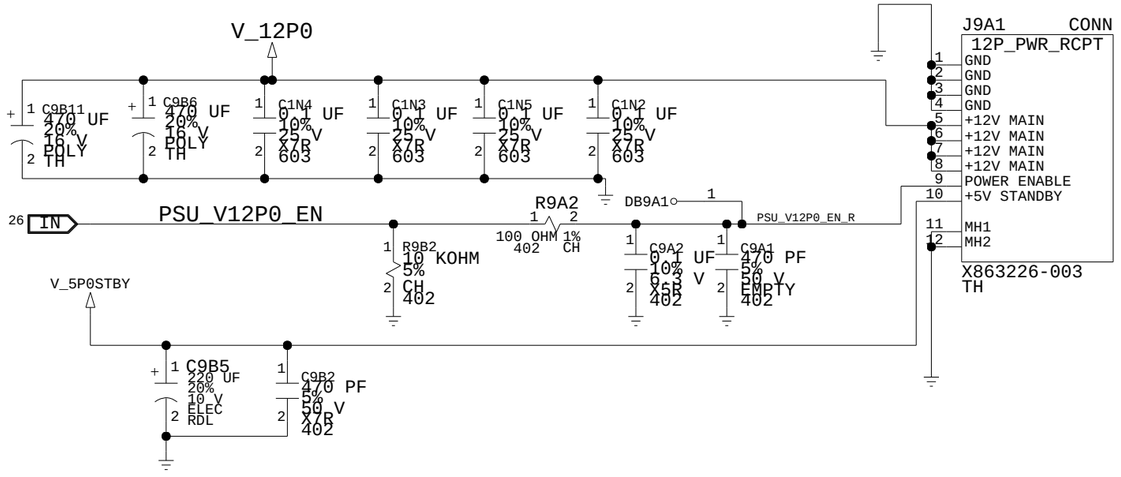
CONN, LITHIUM + FAN





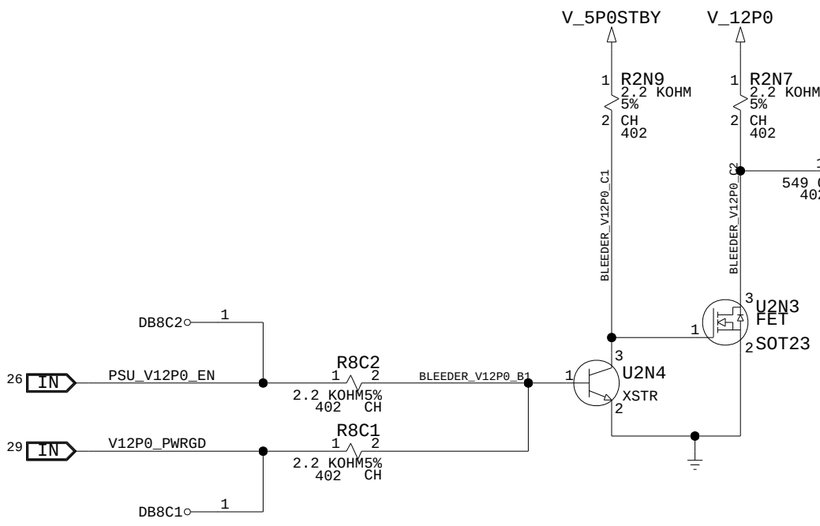
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	40/72	40/72	M	1.0

CONN, PWR

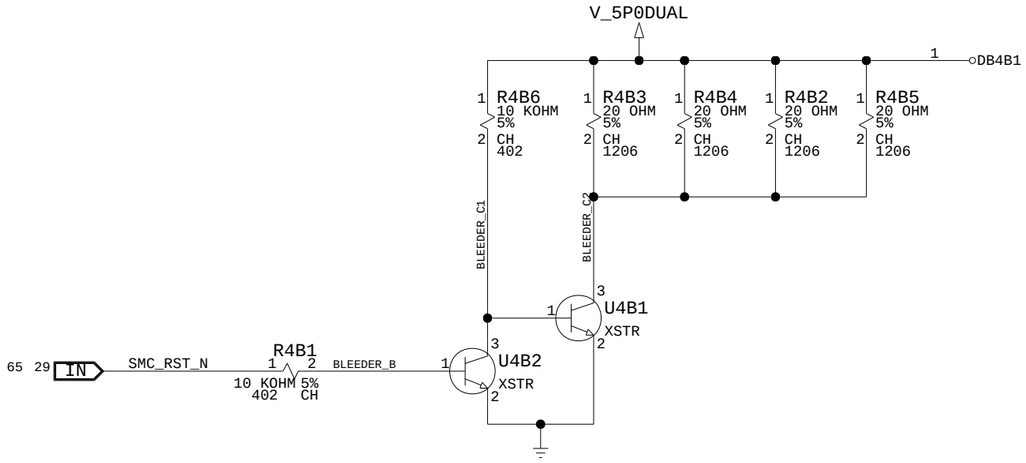


DRAWING	MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
Tue Jun 18 16:42:28 2013		GREYBULL_RETAIL	41/72	41/72	M	1.0

VREGS, BLEEDERS



V_12P0 BLEEDER



V_5P0DUAL BLEEDER

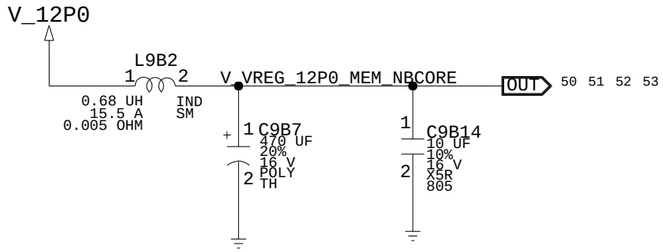
[PAGE_TITLE=VREGS, INPUT + OUTPUT FILTERS]

DRAWING
Tue Jun 18 16:42:20 2008

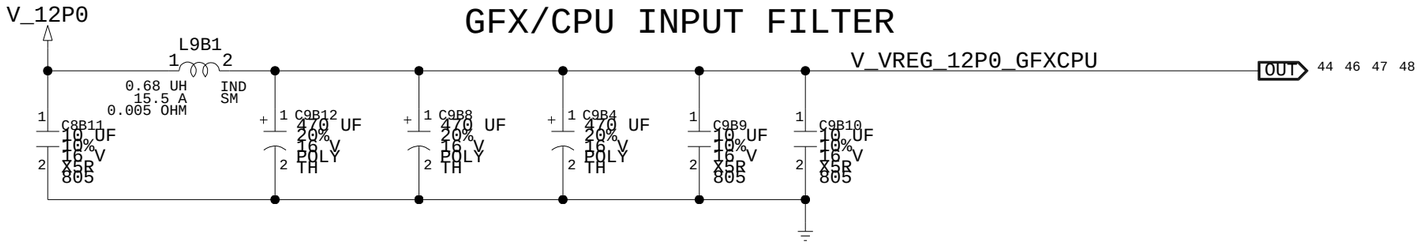
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	42/72	42/72	M	1.0

VREGS, INPUT + OUTPUT FILTERS

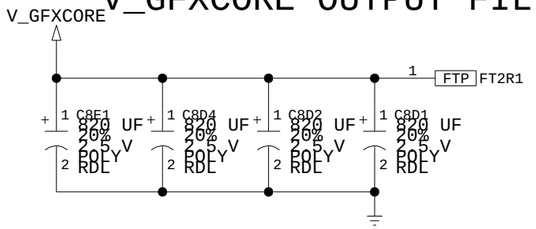
IO/NBCORE/MEMCORE INPUT FILTER



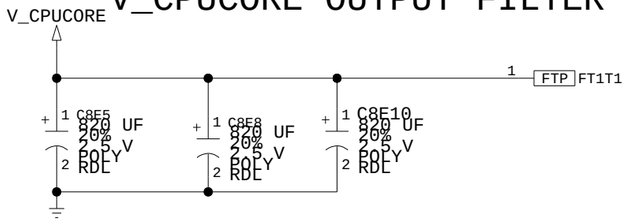
GFX/CPU INPUT FILTER



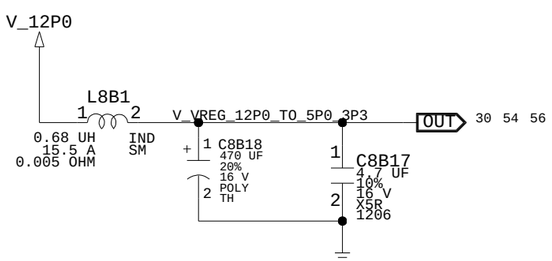
V_GFXCORE OUTPUT FILTER



V_CPUCORE OUTPUT FILTER



5P0/3P3 INPUT FILTER

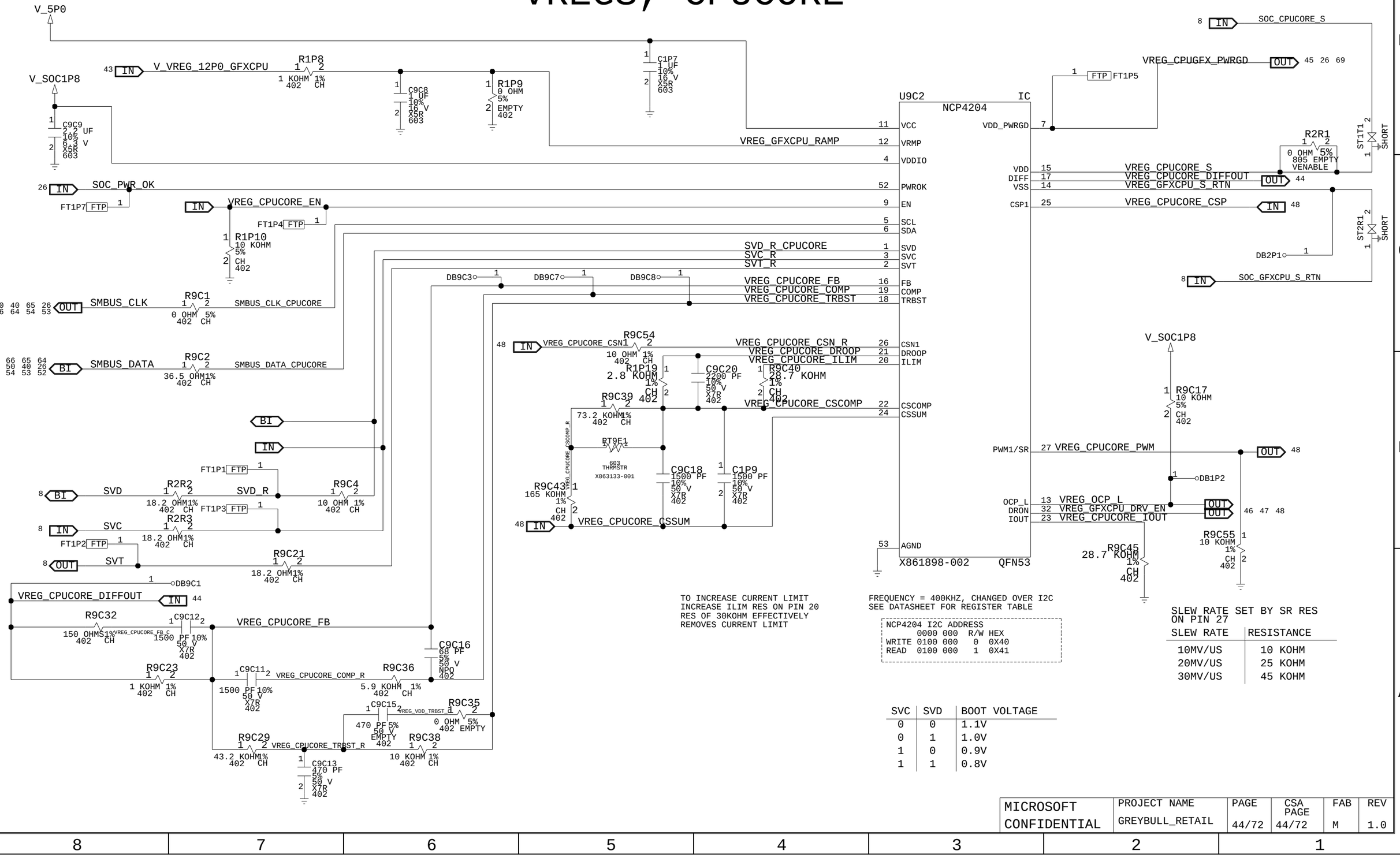


[PAGE_TITLE=VREGS, INPUT + OUTPUT FILTERS]

DRAWING
Tue Jun 18 16:42:25 2008

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	43/72	43/72	M	1.0

VREGS, CPUCORE



TO INCREASE CURRENT LIMIT
INCREASE ILIM RES ON PIN 20
RES OF 30KOHM EFFECTIVELY
REMOVES CURRENT LIMIT

FREQUENCY = 400KHZ, CHANGED OVER I2C
SEE DATASHEET FOR REGISTER TABLE

NCP4204 I2C ADDRESS
0000 000 R/W HEX
WRITE 0100 000 0 0X40
READ 0100 000 1 0X41

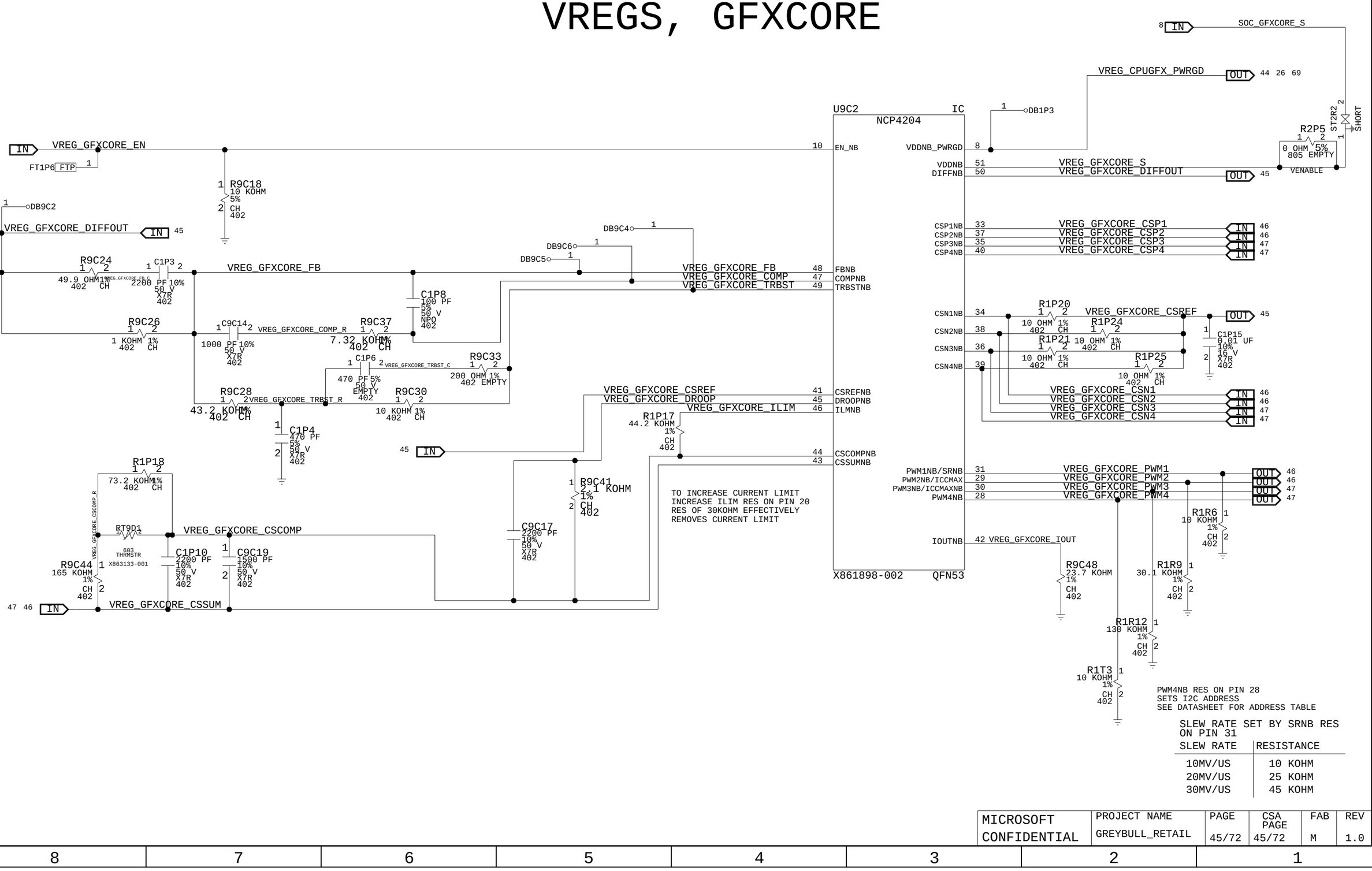
SLEW RATE SET BY SR RES
ON PIN 27

SLEW RATE	RESISTANCE
10MV/US	10 KOHM
20MV/US	25 KOHM
30MV/US	45 KOHM

SVC	SVD	BOOT VOLTAGE
0	0	1.1V
0	1	1.0V
1	0	0.9V
1	1	0.8V

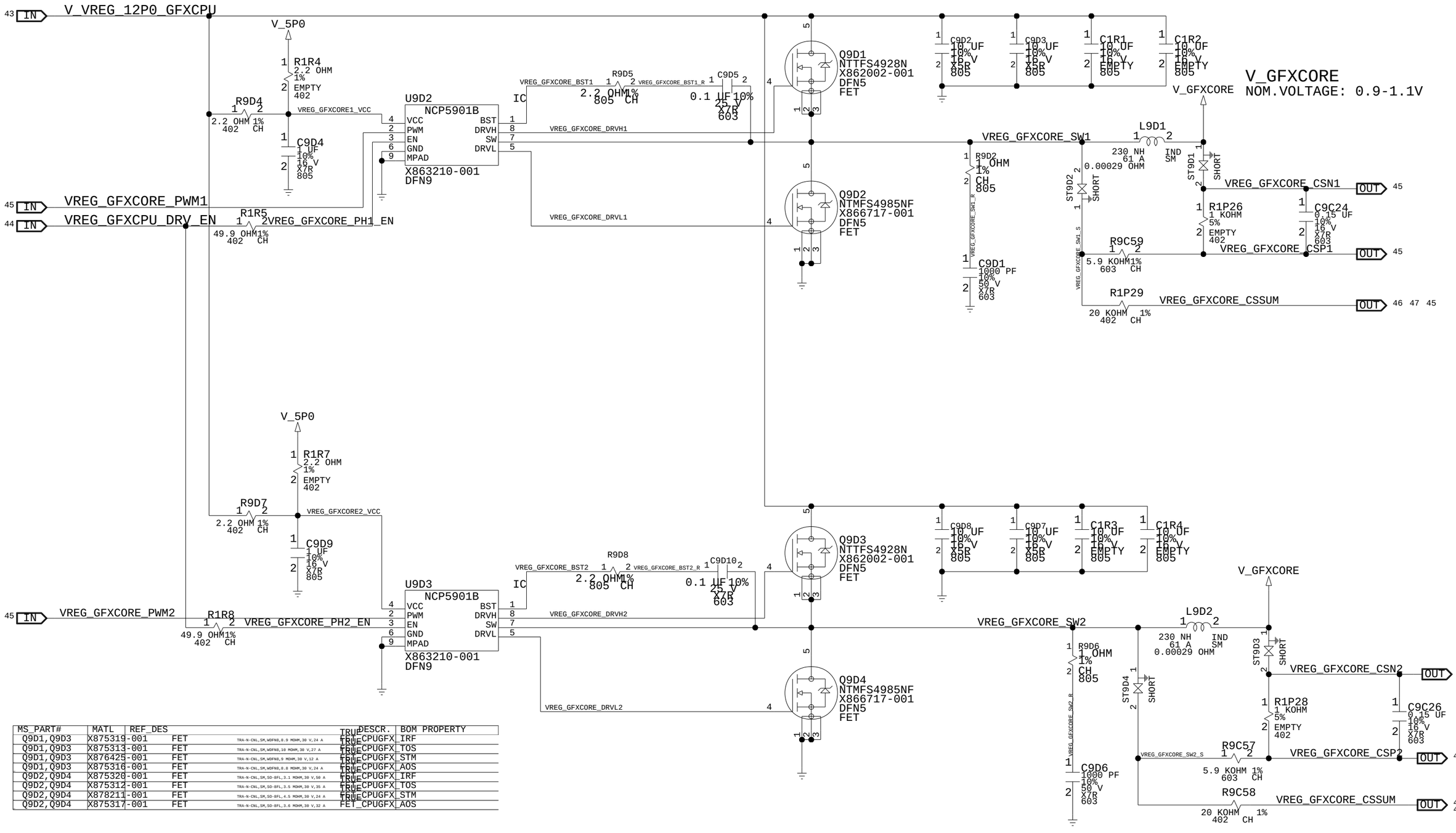
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	44/72	44/72	M	1.0

VREGS, GFXCORE



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	45/72	45/72	M	1.0

VREGS, GFXCORE OUTPUT PHASE 1 & 2

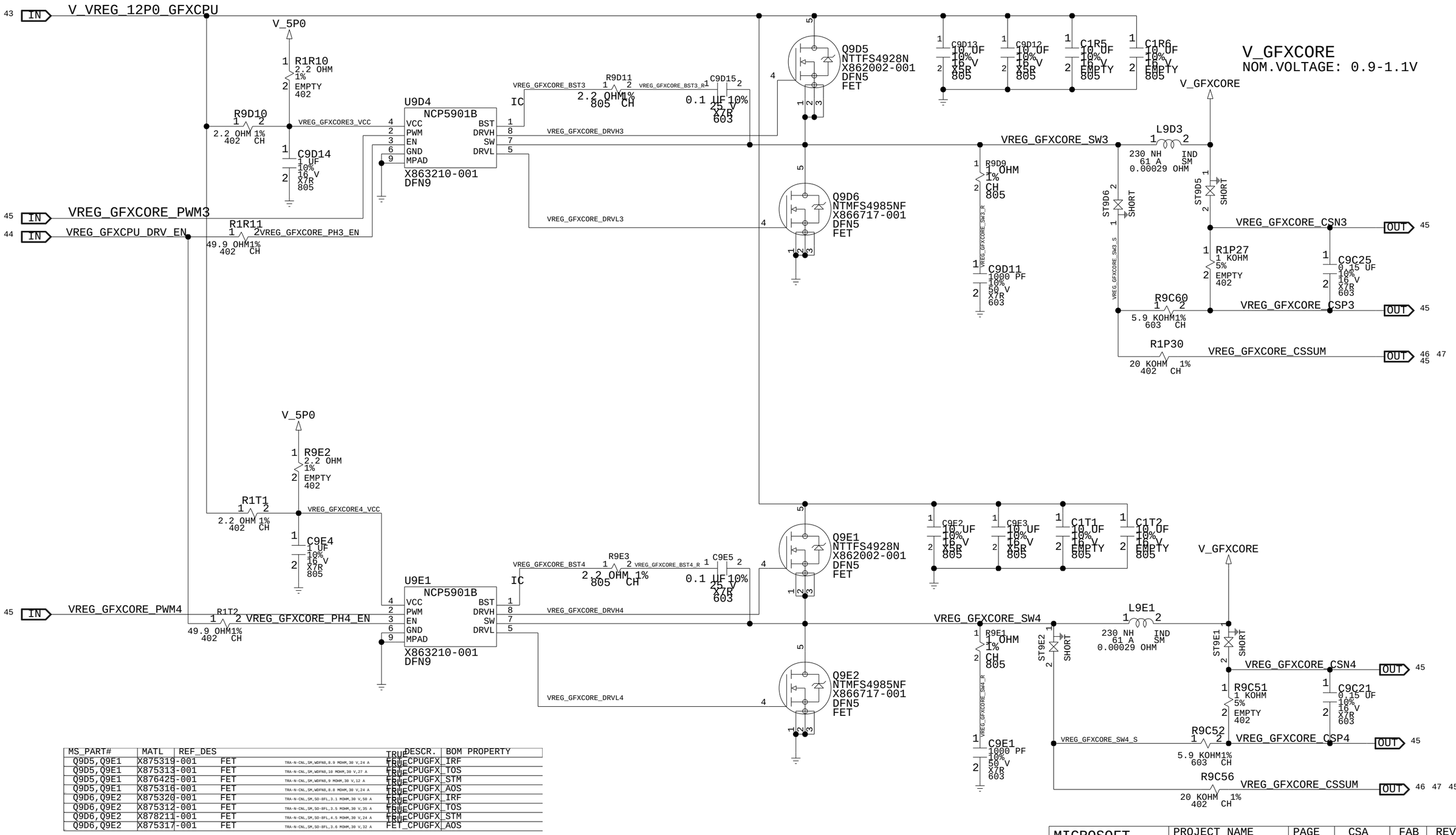


MS_PART#	MATL	REF_DES		DESCR.	BOM PROPERTY
Q9D1,Q9D3	X875319-001	FET	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A	TRUE_CPUGFX_IRF	
Q9D1,Q9D3	X875313-001	FET	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A	TRUE_CPUGFX_TOS	
Q9D1,Q9D3	X876425-001	FET	TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A	TRUE_CPUGFX_STM	
Q9D1,Q9D3	X875316-001	FET	TRA-N-CNL,SM,WDFN8,8.0 MOHM,30 V,24 A	TRUE_CPUGFX_AOS	
Q9D2,Q9D4	X875320-001	FET	TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30 V,50 A	TRUE_CPUGFX_IRF	
Q9D2,Q9D4	X875312-001	FET	TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30 V,35 A	TRUE_CPUGFX_TOS	
Q9D2,Q9D4	X878211-001	FET	TRA-N-CNL,SM,SO-8FL,4.5 MOHM,30 V,24 A	TRUE_CPUGFX_STM	
Q9D2,Q9D4	X875317-001	FET	TRA-N-CNL,SM,SO-8FL,3.6 MOHM,30 V,32 A	FET_CPUGFX_AOS	

DRAWING
Tue Jun 18 16:42:25 2019

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	46/72	46/72	M	1.0

VREGS, GFXCORE OUTPUT PHASE 3 & 4

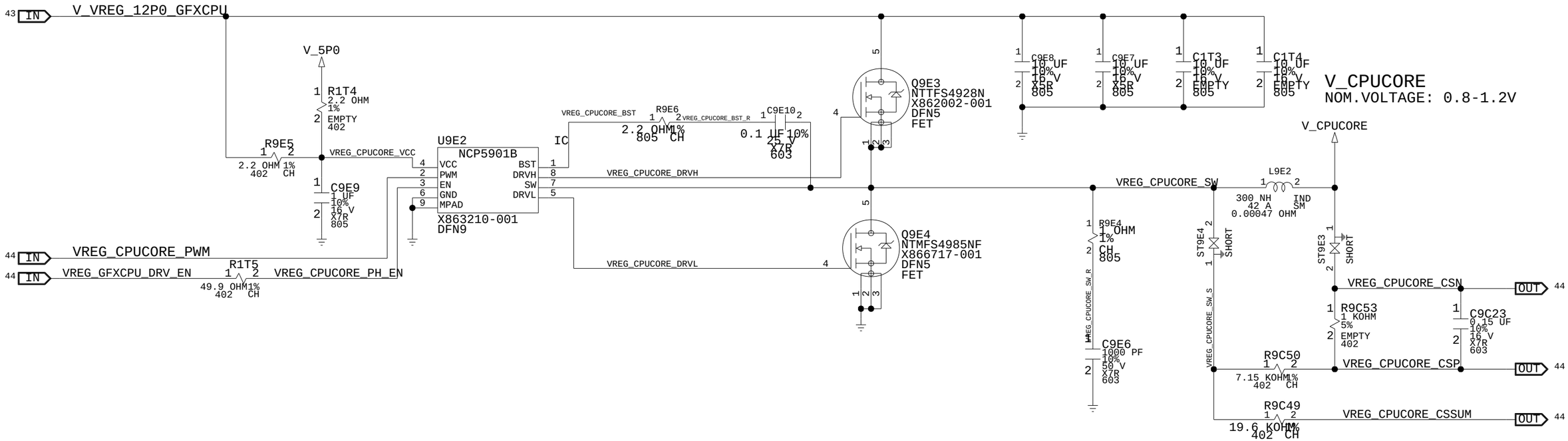


MS_PART#	MATL	REF_DES			DESCR.	BOM PROPERTY
Q9D5,Q9E1	X875319-001	FET	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A	TRUE	CPUGFX	IRF
Q9D5,Q9E1	X875313-001	FET	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A	TRUE	CPUGFX	TOS
Q9D5,Q9E1	X876425-001	FET	TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A	TRUE	CPUGFX	STM
Q9D5,Q9E1	X875316-001	FET	TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A	TRUE	CPUGFX	AOS
Q9D6,Q9E2	X875320-001	FET	TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30 V,50 A	TRUE	CPUGFX	IRF
Q9D6,Q9E2	X875312-001	FET	TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30 V,35 A	TRUE	CPUGFX	TOS
Q9D6,Q9E2	X878211-001	FET	TRA-N-CNL,SM,SO-8FL,4.5 MOHM,30 V,24 A	TRUE	CPUGFX	STM
Q9D6,Q9E2	X875317-001	FET	TRA-N-CNL,SM,SO-8FL,3.6 MOHM,30 V,32 A	TRUE	CPUGFX	AOS

DRAWING
Tue Jun 18 16:42:25 2008

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	47/72	47/72	M	1.0

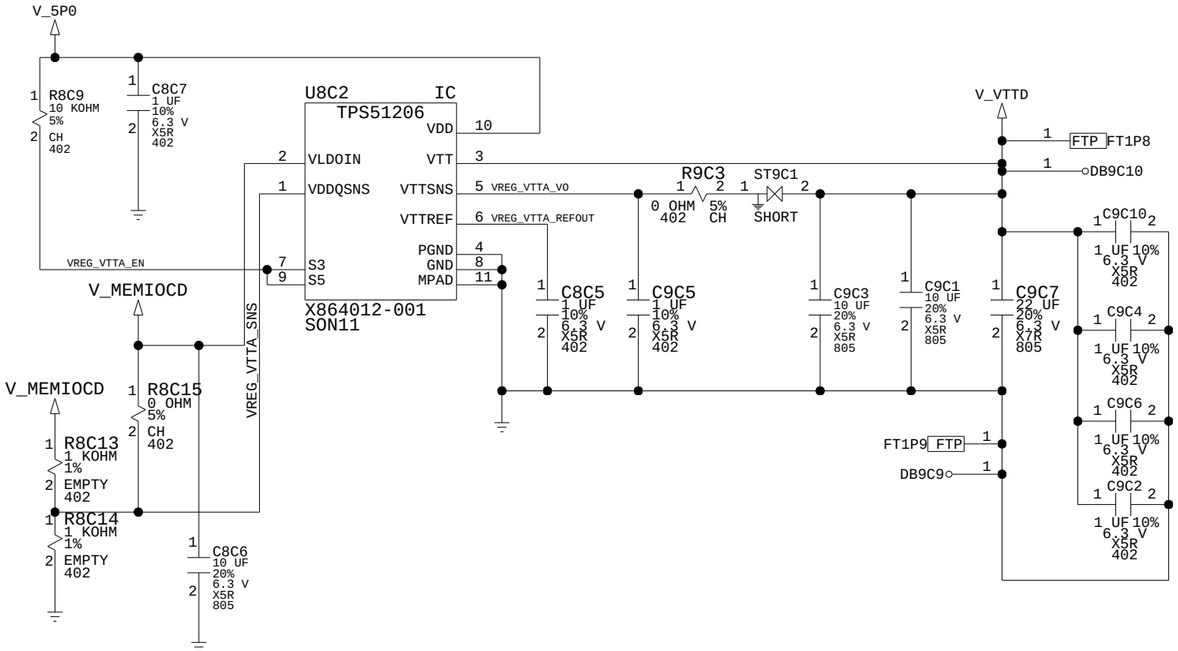
VREGS, CPUCORE OUTPUT PHASE



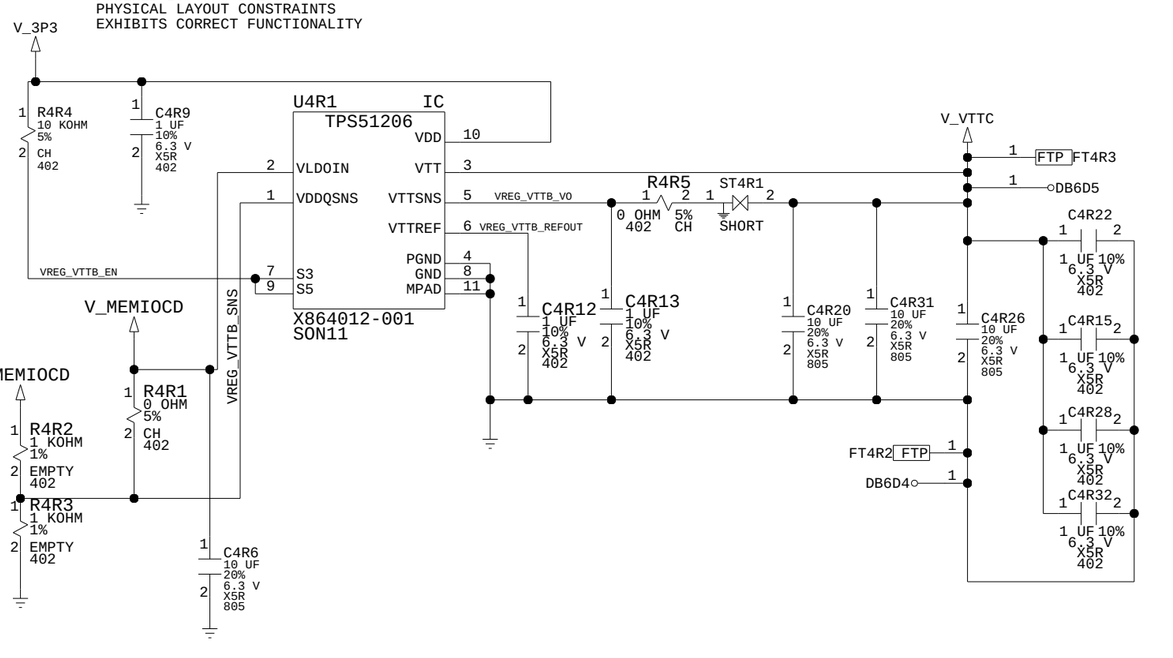
MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
Q9E3	X875319-001	FET	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A	TRUE FET_CPUGFX_IRF
Q9E3	X875313-001	FET	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A	TRUE FET_CPUGFX_TOS
Q9E3	X876425-001	FET	TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A	TRUE FET_CPUGFX_STM
Q9E3	X875316-001	FET	TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A	TRUE FET_CPUGFX_AOS
Q9E4	X875320-001	FET	TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30 V,50 A	TRUE FET_CPUGFX_IRF
Q9E4	X875312-001	FET	TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30 V,35 A	TRUE FET_CPUGFX_TOS
Q9E4	X878211-001	FET	TRA-N-CNL,SM,SO-8FL,4.5 MOHM,30 V,24 A	TRUE FET_CPUGFX_STM
Q9E4	X875317-001	FET	TRA-N-CNL,SM,SO-8FL,3.6 MOHM,30 V,32 A	TRUE FET_CPUGFX_AOS

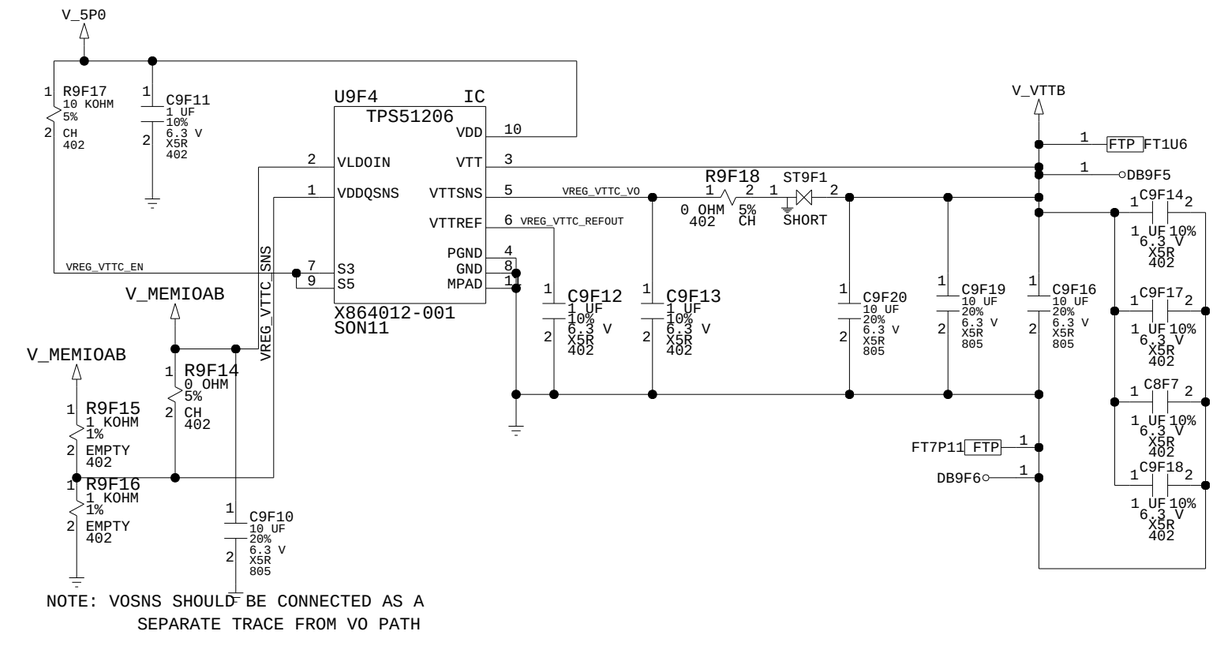
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	48/72	48/72	M	1.0

VREGS, VTT TERMINATION

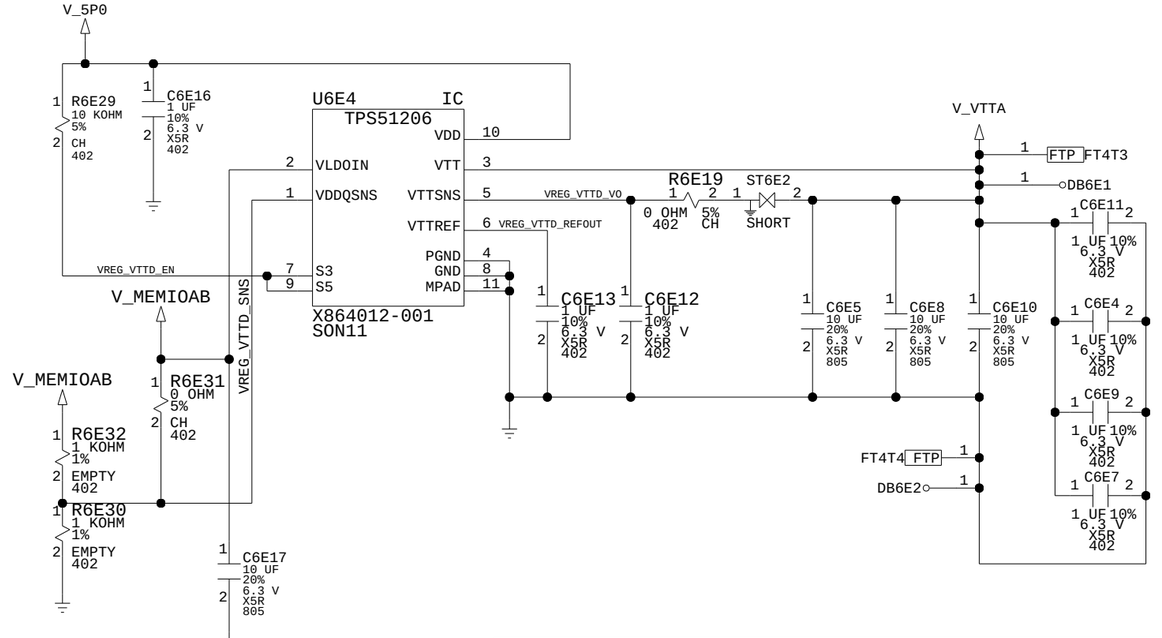


POWERED BY V3P3 DUE TO
PHYSICAL LAYOUT CONSTRAINTS
EXHIBITS CORRECT FUNCTIONALITY





NOTE: VOSNS SHOULD BE CONNECTED AS A
SEPARATE TRACE FROM VO PATH



[PAGE_TITLE=VTT TERMINATION]

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
DRAWING	GREYBULL_RETAIL	49/72	49/72	M	1.0

Tue Jun 18 16:42:25 2013

VREGS, NBCORE

MS_PART#	MATL	REF_DES		TRUE DESCR.	BOM PROPERTY
Q5F1	X875319-001	FET	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A	FET_MEMNB_	IRF
Q5F1	X875313-001	FET	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A	FET_MEMNB_	TOS
Q5F1	X876425-001	FET	TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A	FET_MEMNB_	STM
Q5F1	X875316-001	FET	TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A	FET_MEMNB_	AOS
Q5F2	X875320-001	FET	TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30 V,50 A	FET_MEMNB_	IRF
Q5F2	X875312-001	FET	TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30 V,35 A	FET_MEMNB_	TOS
Q5F2	X878211-001	FET	TRA-N-CNL,SM,SO-8FL,4.5 MOHM,30 V,24 A	FET_MEMNB_	STM
Q5F2	X875317-001	FET	TRA-N-CNL,SM,SO-8FL,3.6 MOHM,30 V,32 A	FET_MEMNB_	AOS

43 IN V_VREG_12P0_MEM_NBCORE

C5F7 10 UF 10% 16 V X5R 805
C5F8 10 UF 10% 16 V X5R 805
C5U4 10 UF 10% 16 V EMPTY 805
C5U3 10 UF 10% 16 V EMPTY 805

R5F13 2.2 OHM 1% 402 CH
C5F4 1 UF 10% 16 V X7R 805

U5F3 IC
TPS53819A

VREG_NBCORE_VCC 8 VDD
VREG_NBCORE_LDO5 9 VREG

C5F6 1 UF 10% 6.3 V X5R 402
R5F15 249 KOHM 1% 402 CH
69 26 57 51 OUT VREG_PWRGPB_PWRGD 15 PGOOD
FT5U1 FTP 1

VREG_NBCORE_ADDR 16 ADDR
FT5U2 FTP 1
DB5U1 VREG_NBCORE_ALERT_N 3 ALERT_N
R5F12 47.5 KOHM 1% 402 CH
69 26 IN VREG_PWRGPB_EN 14 EN

65 44 26 IN SMBUS_CLK 1 SCL
66 65 64 54 53 52 44 40 26 BI SMBUS_DATA 2 SDA

VREG_NBCORE_OCP 4 TRIP
R5F10 130 KOHM 1% 402 CH
7 GND
17 MPAD

X869867-001
QFN17

DRVH 11 VREG_NBCORE_DH
VBST 13 VREG_NBCORE_BST
C5F5 0.1 UF 10% 25 V X7R 603
SW 12 VREG_NBCORE_SW
DRVL 10 VREG_NBCORE_DL
VO 5 VREG_NBCORE_VO
FB 6 VREG_NBCORE_FB

Q5F1 FET NTTFS4928N X862002-001 DFN5
Q5F2 NTMFS4985NF X866717-001 DFN5 FET

L5F1 400 NH 42 A IND SM 0.00047 OHM
R5F18 1 OHM 1% EMPTY 805
VREG_NBCORE_SW
C5F9 1000 PF 10% 50 V EMPTY 603

DB3T2

R5F9 10 KOHM 1% 402 CH
R5F7 0 OHM 5% 805 EMPTY
VENABLE

ST3T3 SHORT

DB5F1
R5F8 13.3 KOHM 1% 402 CH
VREG_NBCORE_FB_R_GND
ST3T4 SHORT

FT4U2 FTP 1
DB6F2 1

V_NBCORE
NOM.VOLTAGE: 1.25V

C6F9 820 UF 20% 2.5 V POLY 2 RDL
C6F12 820 UF 20% 2.5 V POLY 2 RDL
FT4U1 FTP 1
DB6F1 1
FT4U3 FTP 1
DB6F3 1

TO INCREASE CURRENT LIMIT
INCREASE TRIP RESISTOR VALUE

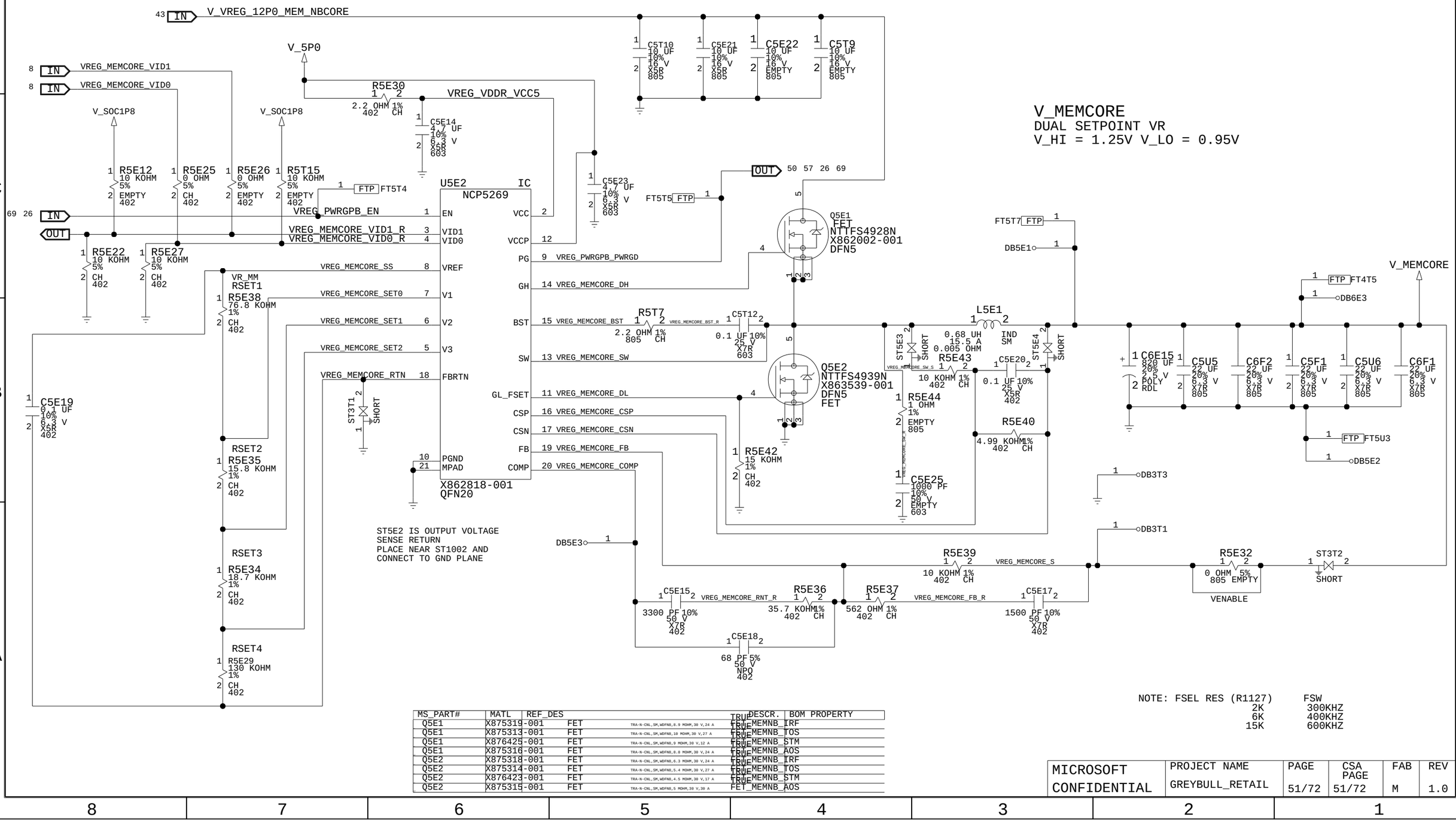
FREQUENCY = 400KHZ, SET OVER I2C
SEE DATASHEET FOR REGISTER TABLE

TPS53819 I2C ADDRESS

	0010 101	R/W	HEX
WRITE	0010 101	0	0X2A
READ	0010 101	1	0X2B

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	50/72	50/72	M	1.0

VREGS, MEMCORE



V_MEMCORE
DUAL SETPOINT VR
V_HI = 1.25V V_LO = 0.95V

ST5E2 IS OUTPUT VOLTAGE SENSE RETURN PLACE NEAR ST1002 AND CONNECT TO GND PLANE

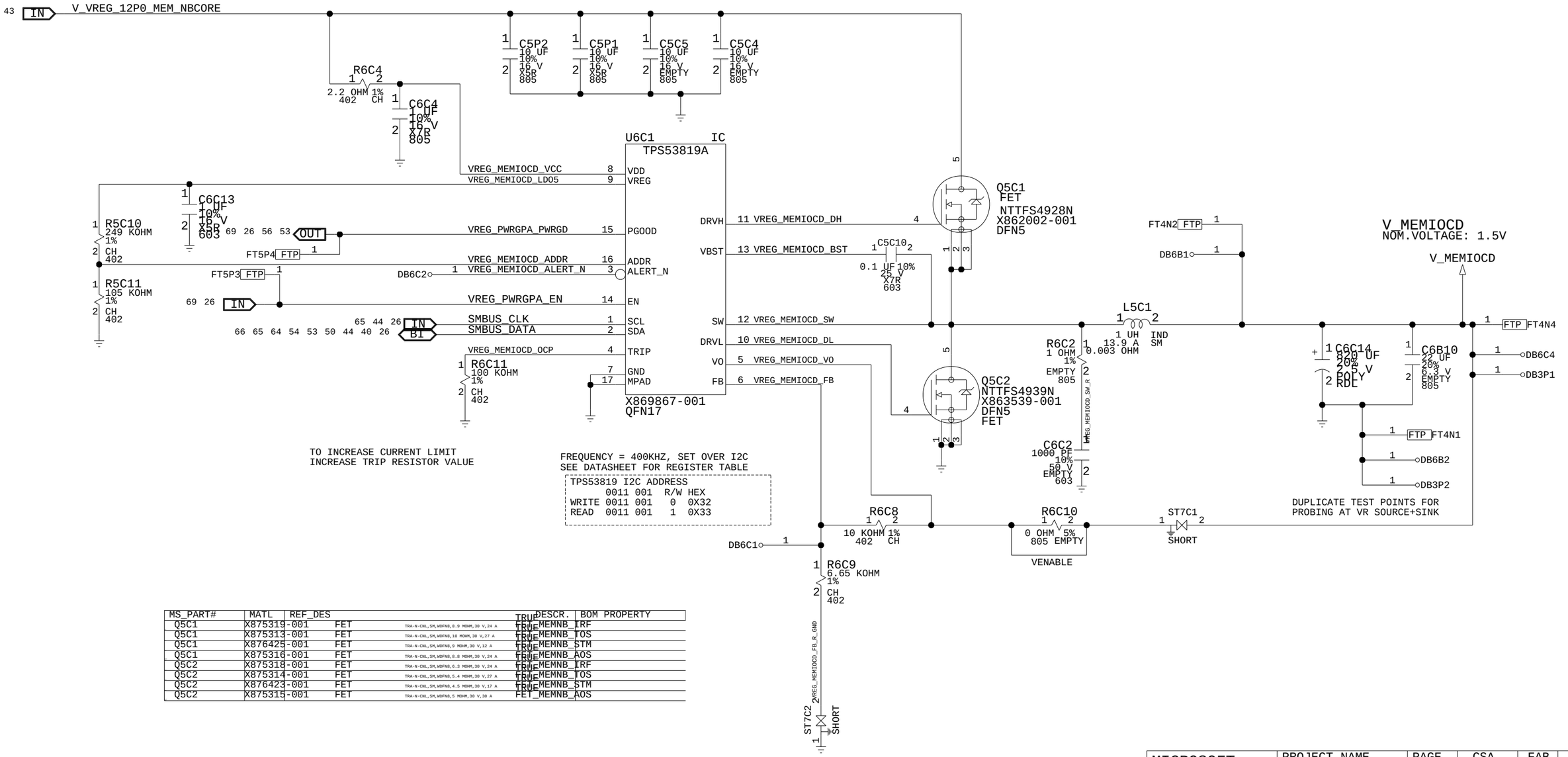
NOTE: FSEL RES (R1127) FSW

2K	300KHZ
6K	400KHZ
15K	600KHZ

MS_PART#	MATL	REF_DES		DESCR.	BOM PROPERTY
Q5E1	X875319-001	FET	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A	FET_MEMNB_IRF	TRUE
Q5E1	X875313-001	FET	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A	FET_MEMNB_TOS	TRUE
Q5E1	X876425-001	FET	TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A	FET_MEMNB_STM	TRUE
Q5E1	X875316-001	FET	TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A	FET_MEMNB_AOS	TRUE
Q5E2	X875318-001	FET	TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A	FET_MEMNB_IRF	TRUE
Q5E2	X875314-001	FET	TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A	FET_MEMNB_TOS	TRUE
Q5E2	X876423-001	FET	TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A	FET_MEMNB_STM	TRUE
Q5E2	X875315-001	FET	TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A	FET_MEMNB_AOS	TRUE

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	51/72	51/72	M	1.0

VREG, MEMIOCD



MS_PART#	MATL	REF_DES	DESCR.	BOM PROPERTY
Q5C1	X875319-001	FET TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A	TRUE	FET_MEMNB_IRF
Q5C1	X875313-001	FET TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A	TRUE	FET_MEMNB_TOS
Q5C1	X876425-001	FET TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A	TRUE	FET_MEMNB_STM
Q5C1	X875316-001	FET TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A	TRUE	FET_MEMNB_AOS
Q5C2	X875318-001	FET TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A	TRUE	FET_MEMNB_IRF
Q5C2	X875314-001	FET TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A	TRUE	FET_MEMNB_TOS
Q5C2	X876423-001	FET TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A	TRUE	FET_MEMNB_STM
Q5C2	X875315-001	FET TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A	TRUE	FET_MEMNB_AOS

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	52/72	52/72	M	1.0

VREG, MEMIOAB

43 IN V_VREG_12P0_MEM_NBCORE

C9F4 10 UF 10% 16 V X5R 805
C1U2 10 UF 10% 16 V X5R 805
C1U3 10 UF 10% 16 V EMPTY 805
C9F3 10 UF 10% 16 V EMPTY 805

R1U3 2.2 OHM 1% 402 CH
C9F8 1 UF 10% 16 V X7R 805

U9F2 IC TPS53819A

VREG_MEMIOAB_VCC 8 VDD
VREG_MEMIOAB_LDO5 9 VREG

C9F6 1 UF 10% 16 V X5R 603

R9F6 249 KOHM 1% 402 CH
R9F11 90.9 KOHM 1% 402 CH

56 52 OUT VREG_PWRGPA_PWRGD 15 PGOOD
69 26 FT1U1 FTP 1

FT1U2 FTP 1
DB9F2 1 VREG_MEMIOAB_ALERT_N 3 ALERT_N
VREG_MEMIOAB_ADDR 16 ADDR

69 26 IN VREG_PWRGPA_EN 14 EN

65 44 26 IN SMBUS_CLK 1 SCL
66 65 64 54 52 50 44 40 26 BI SMBUS_DATA 2 SDA

VREG_MEMIOAB_OCP 4 TRIP

R9F12 100 KOHM 1%

7 GND
17 MPAD

X869867-001 QFN17

DRVH 11 VREG_MEMIOAB_DH
VBST 13 VREG_MEMIOAB_BST
SW 12 VREG_MEMIOAB_SW
DRVL 10 VREG_MEMIOAB_DL
VO 5 VREG_MEMIOAB_VO
FB 6 VREG_MEMIOAB_FB

Q9F2 FET NTTFS4928N X862002-001 DFN5

C9F7 0.1 UF 10% 25 V X7R 603

Q9F1 NTTFS4939N X863539-001 DFN5 FET

R9F1 1 OHM 1% EMPTY 805
L9E3 1 UH IND 13.9 A SM 0.003 OHM
C9F2 1000 PF 10% 50 V EMPTY 603
VREG_VDD1OCD_SW_R

FT1U3 FTP 1
DB9F7 1

V_MEMIOAB
NOM.VOLTAGE: 1.5V

C9F11 820 UF 20% 2.5V 2 RDL
C9F21 22 UF 20% 6.3 V EMPTY 805

FTP FT1U4
DB9F3
DB3U1
FTP FT1U5
DB9F4
DB3U2

DUPLICATE TEST POINTS FOR PROBING AT VR SOURCE+SINK

R9F10 10 KOHM 1% 402 CH
R9F13 0 OHM 5% 805 EMPTY
VENABLE

ST7F1 SHORT

DB9F1 1
R1U5 6.65 KOHM 1% 402 CH
VREG_MEMIOAB_FB_R_GND
ST7F2 SHORT

TO INCREASE CURRENT LIMIT
INCREASE TRIP RESISTOR VALUE

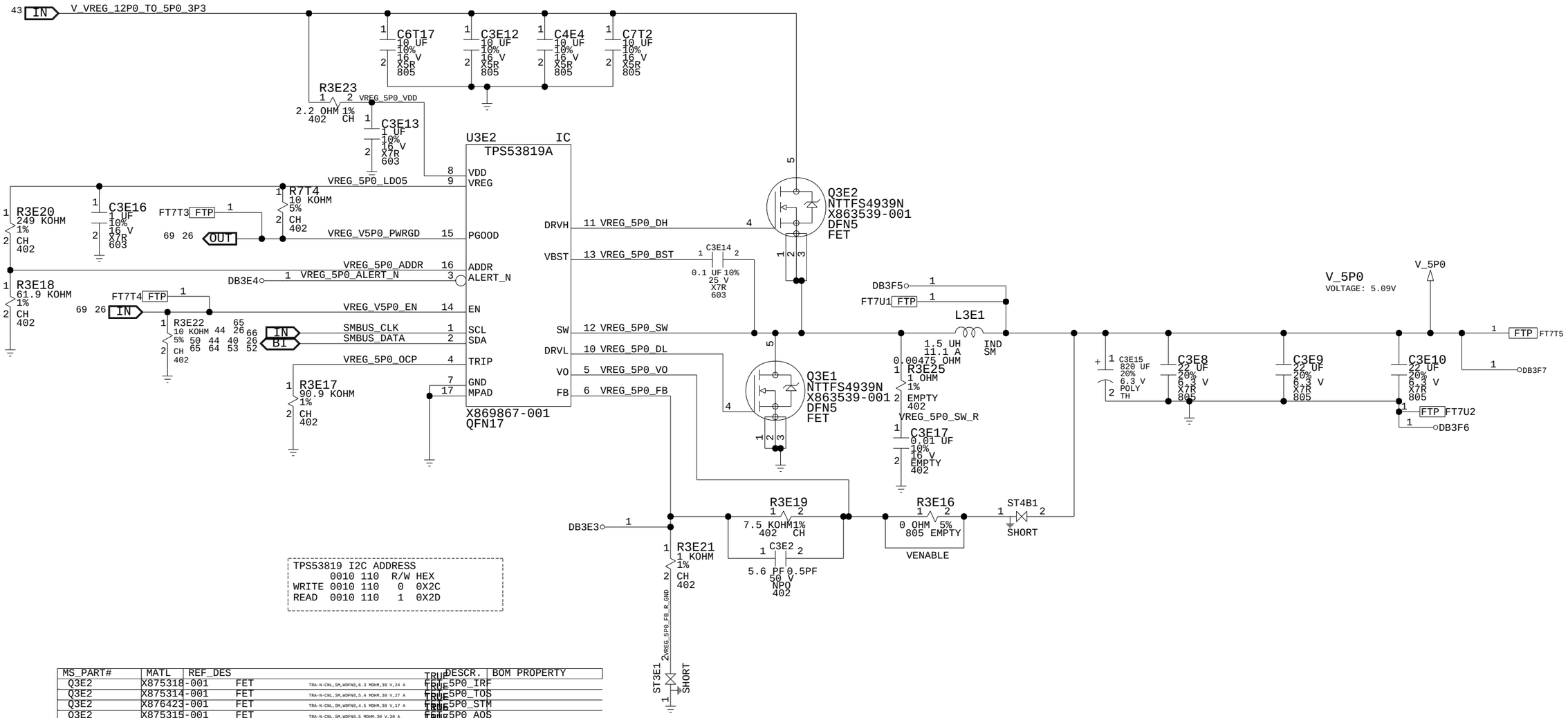
FREQUENCY = 400KHZ, SET OVER I2C
SEE DATASHEET FOR REGISTER TABLE

TPS53819 I2C ADDRESS
0011 000 R/W HEX
WRITE 0011 000 0 0X30
READ 0011 000 1 0X31

MS_PART#	MATL	REF_DES		TRU	DESCR.	BOM PROPERTY
Q9F2	X875319-001	FET	TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A	TRUE	FET_MEMNB_IRF	
Q9F2	X875313-001	FET	TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A	TRUE	FET_MEMNB_TOS	
Q9F2	X876425-001	FET	TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A	TRUE	FET_MEMNB_STM	
Q9F2	X875316-001	FET	TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A	TRUE	FET_MEMNB_AOS	
Q9F1	X875318-001	FET	TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A	TRUE	FET_MEMNB_IRF	
Q9F1	X875314-001	FET	TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A	TRUE	FET_MEMNB_TOS	
Q9F1	X876423-001	FET	TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A	TRUE	FET_MEMNB_STM	
Q9F1	X875315-001	FET	TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A	TRUE	FET_MEMNB_AOS	

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	53/72	53/72	M	1.0

VREGS, V5P0



TPS53819 I2C ADDRESS

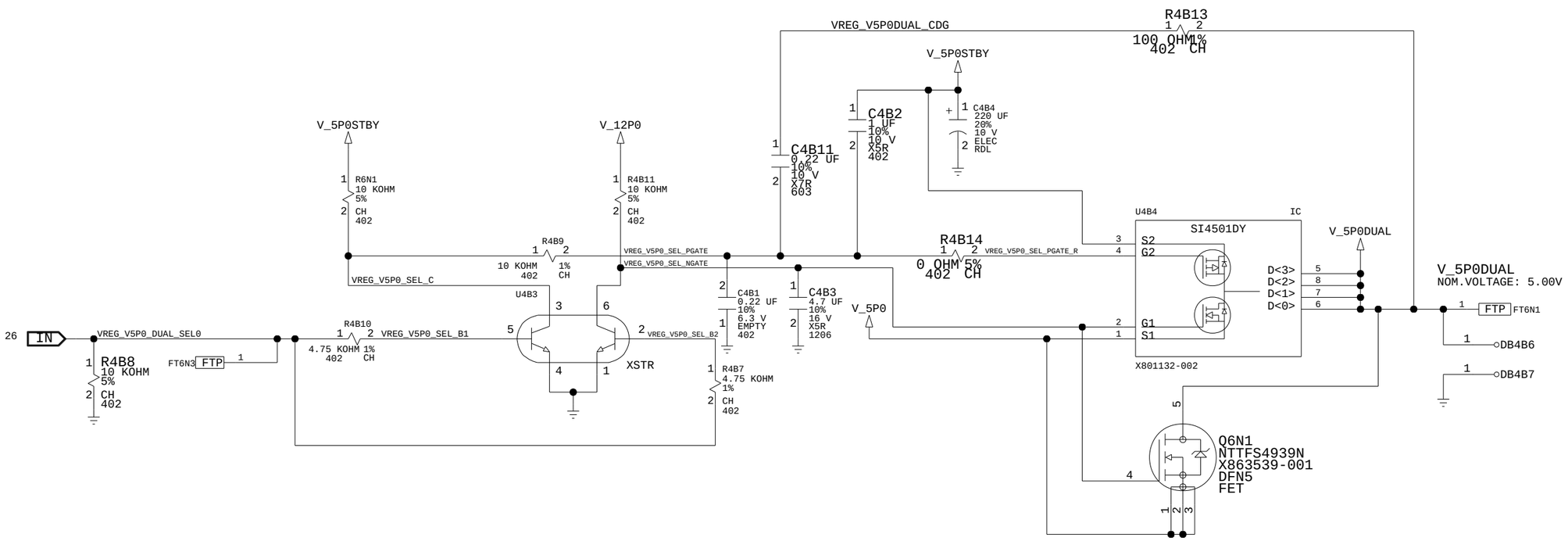
	0010 110	R/W	HEX	
WRITE	0010 110	0	0X2C	
READ	0010 110	1	0X2D	

MS_PART#	MATL	REF_DES		DESCR.	BOM PROPERTY
Q3E2	X875318-001	FET	TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A	FET_5P0_IRF	
Q3E2	X875314-001	FET	TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A	FET_5P0_TOS	
Q3E2	X876423-001	FET	TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A	FET_5P0_STM	
Q3E2	X875315-001	FET	TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A	FET_5P0_AOS	
Q3E1	X875318-001	FET	TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A	FET_5P0_IRF	
Q3E1	X875314-001	FET	TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A	FET_5P0_TOS	
Q3E1	X876423-001	FET	TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A	FET_5P0_STM	
Q3E1	X875315-001	FET	TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A	FET_5P0_AOS	

[PAGE_TITLE=VREGS, V5P0]

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	54/72	54/72	M	1.0

VREGS, V5P0 DUAL



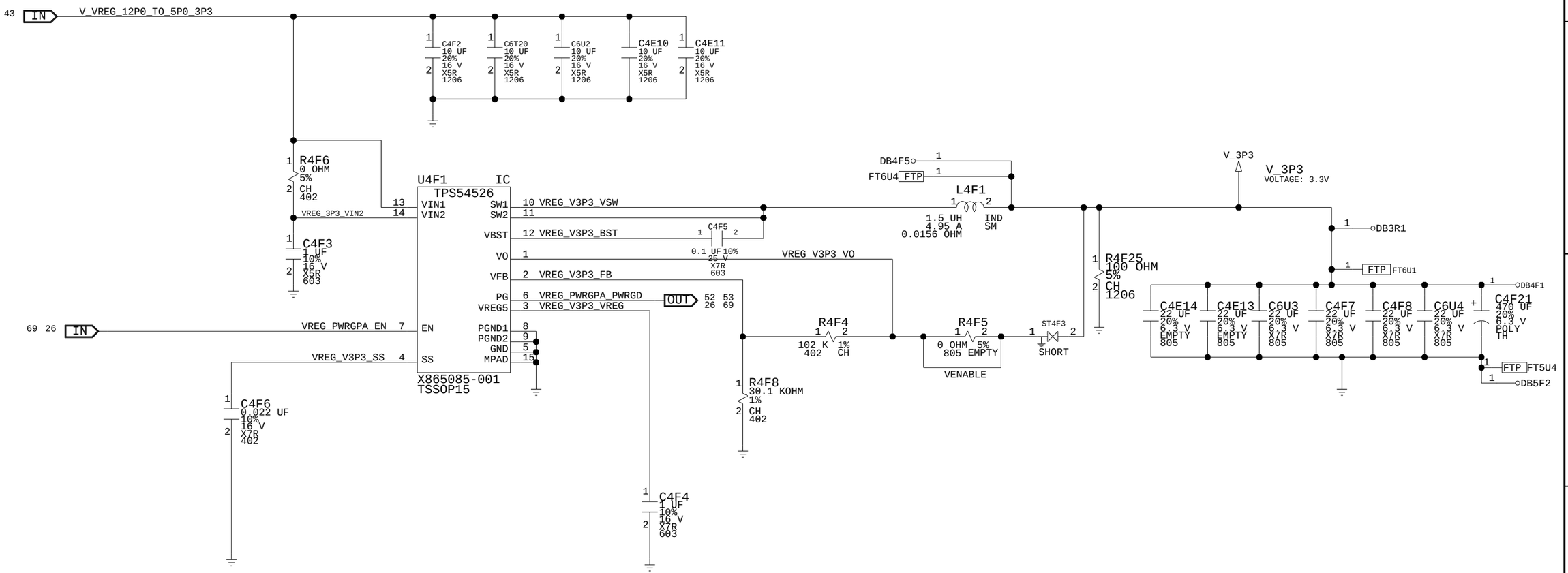
VREG_V5P0_DUAL_SEL0	VREG_5P0_SEL NGATE/PGATE	V_5P0DUAL
HIGH	LOW	V_5P0STBY
LOW	HIGH	V_5P0

[PAGE_TITLE=VREGS, V5P0 DUAL]

DRAWING
Tue Jun 18 16:42:27 2013

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	55/72	55/72	M	1.0

VREGS, V3P3

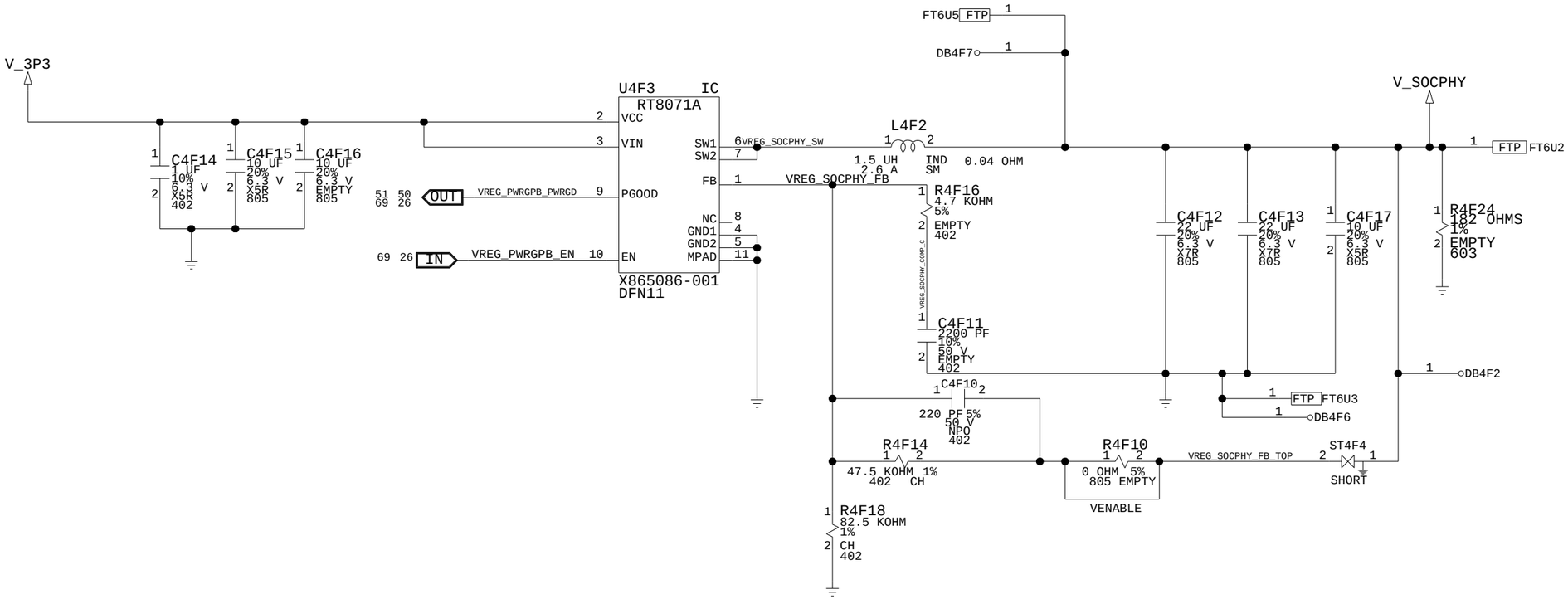


[PAGE_TITLE=VREGS, V3P3]

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	56/72	56/72	M	1.0

VREGS, VSOCPHY

V_SOCPHY
NOM.VOLTAGE: 0.95V



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	57/72	57/72	M	1.0

VREGS, VSOCPLL + VBURN + VFUSE + VBAT

I170
V_1P8STBY

R6R21
0 OHM
5%
CH
402

I169
V_BAT

V_BAT
NOM.VOLTAGE: 1.8V

R6R20
4.99 KOHM
1%
EMPTY
402

I171

V_3P3

R6C14
4.7 KOHM
5%
EMPTY
402

U6C3 IC
CAT6219
VIN VOUT
ADJ
EN GND
X865083-001
SOT23-5

VREG_BURN_EN_R

C6C5
1 UF
10%
6.3 V
X5R
402

R6C13
0 OHM
5%
CH
402

R6C15
10 KOHM
5%
CH
402

IN VREG_BURN_EN

C6C7
1 UF
10%
6.3 V
EMPTY
402

C6C8
1 UF
10%
6.3 V
EMPTY
402

VREG_BURN_OUT

R6C22
1 OHM
1%
CH
402

R6C19
887 OHM
1%
CH
402

VREG_BURN_COUT

VREG_BURN_ADJUST

C6C6
0.1 UF
10%
6.3 V
EMPTY
402

C6D1
4.7 UF
10%
6.3 V
X5R
603

R6C16
1.65 KOHM
1%
CH
402

V_BURN
NOM.VOLTAGE: 1.9V

V_BURN

DB6C3

U6C6
FET
SOT23
X857156-001

C6C10
1 UF
10%
6.3 V
X5R
402

C6D11
0.047 UF
10%
10 V
X7R
402

C6C9
10 UF
10%
6.3 V
X5R
805

R3R1
1 KOHM
1%
CH
402

VREG_BURN_OUT_G

R6D39
100 OHM 1%
402
CH

VREG_BURN_OUT_CGD

R6D38
5.49 KOHM
1%
CH
402

NOTE: BOTTOM ADJUST RESISTOR
LEFT IN TO REDUCE EFFECT OF ADJUST
PIN CURRENT WHEN USED WITH 1K DIGIPOT

FT5P6 FTP

DB5C2

V_SOC1P8
V_SOC1P8
NOM.VOLTAGE: 1.83V

C5C19
1 UF
10%
6.3 V
EMPTY
402

C5C20
1 UF
10%
6.3 V
EMPTY
402

NOTE: THIS WAS SET TO 1.83V TO ACCOMODATE DROP IN FERRITES

DB5D3

FTP FT5R4

FTP FT5R5

DB5D2

V_3P3

U5C3 IC
NCP1117
IN OUT
ADJUST/GND
X819037-001
1.8V

C5C23
1 UF
10%
6.3 V
X5R
402

R5C20
1 OHM
1%
CH
402

R5C19
200 OHM
1%
CH
603

VREG_SOC1P8_COUT

DB5D1

FTP FT5P5

VREG_SOC1P8_ADJUST

C5C18
0.1 UF
10%
6.3 V
EMPTY
402

C5C21
4.7 UF
10%
6.3 V
X5R
603

R5C15
100 OHM
1%
CH
402

NOTE: BOTTOM ADJUST RESISTOR
LEFT IN TO REDUCE EFFECT OF ADJUST
PIN CURRENT WHEN USED WITH 1K DIGIPOT

V_3P3

C6D5
1 UF
10%
6.3 V
X5R
402

U6D1 IC
CAT6241
VIN1 VIN2
VOUT1 VOUT2
EN ADJ
GND MPAD
CBYP
X865082-001
DFN9

69 26 IN VREG_PWRGPB_EN

DB6D3

VREG_FUSE_CBYP

C6D6
0.01 UF
10%
16 V
EMPTY
402

I178

V_FUSE
NOM.VOLTAGE: 0.95V

V_FUSE

FTP FT4R1

DB6D2

DB3R2

R6D8
1 OHM
1%
CH
402

C6D3
1 UF
10%
6.3 V
X5R
402

C6D4
1 UF
10%
6.3 V
X5R
402

VREG_FUSE_COUT

VREG_FUSE_ADJ

C6D2
4.7 UF
10%
6.3 V
X5R
603

R6D10
2.74 KOHM
1%
CH
402

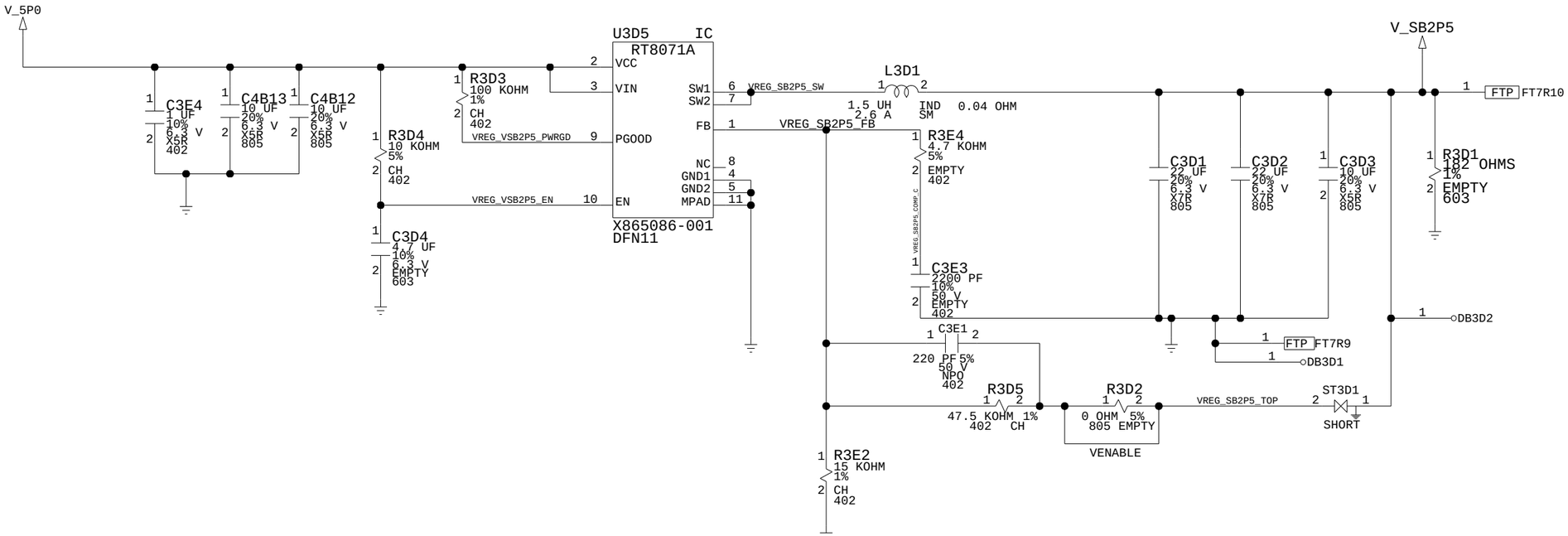
R6D11
3 KOHM
1%
CH
402

I176

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	58/72	58/72	M	1.0

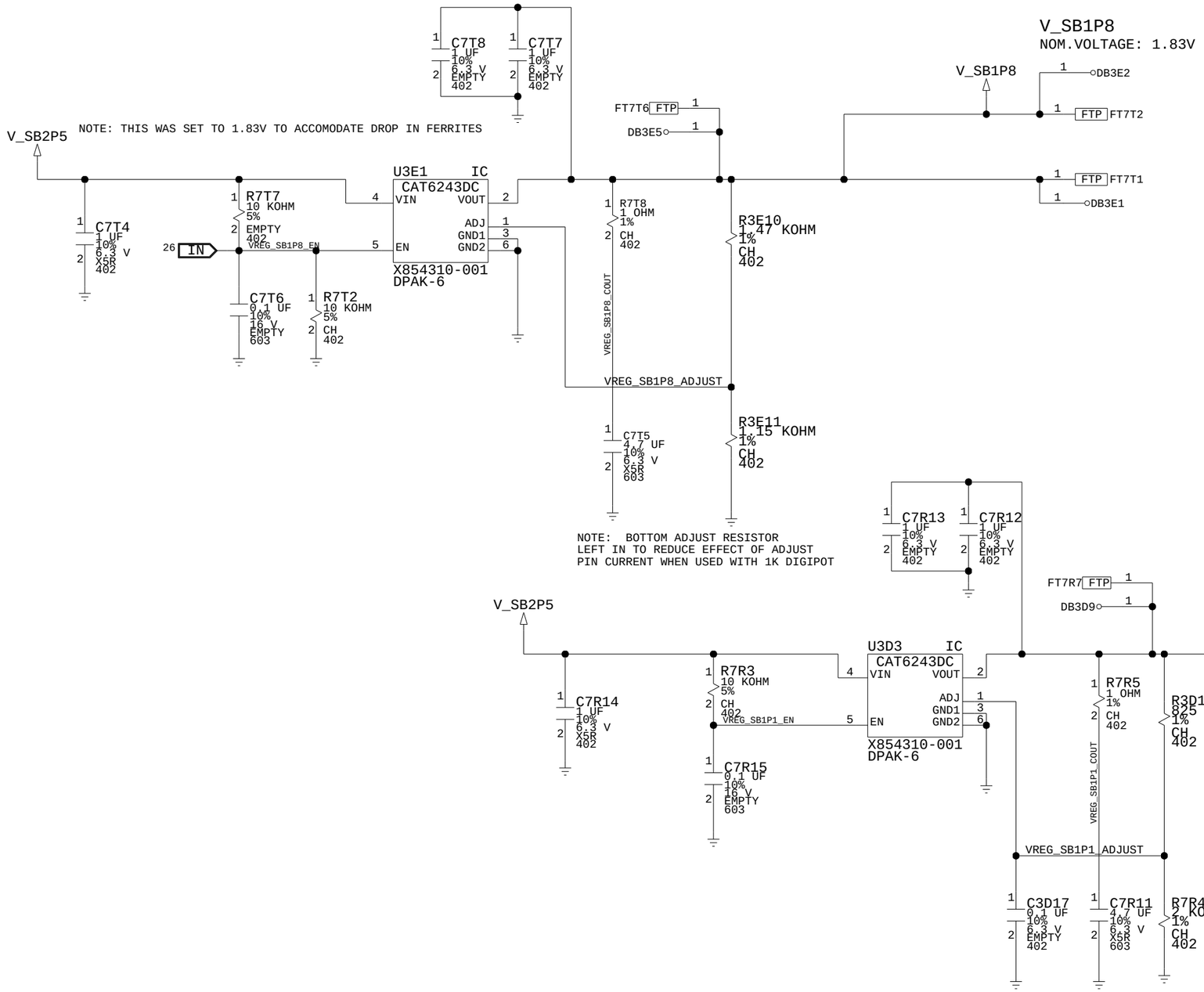
VREGS, VSB2P5

V_SB2P5
NOM.VOLTAGE: 2.5V



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	59/72	59/72	M	1.0

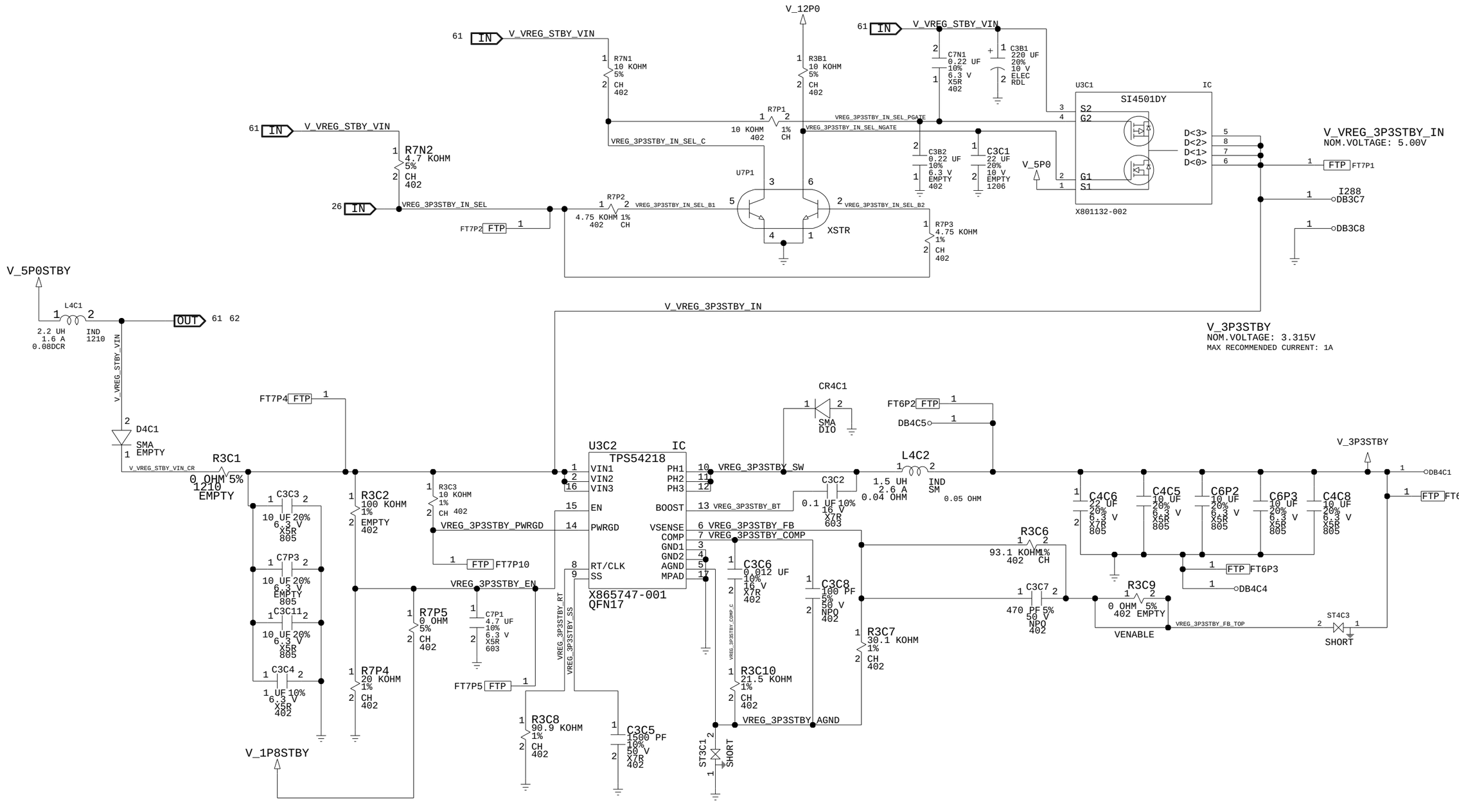
VREGS, VSB1P8PLL + VSB1P8IO + VSBCORE + VSB1P1PLL



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	60/72	60/72	M	1.0

VREGS,STANDBY SWITCHERS 3P3

VREG_3P3STBY_IN_SEL	VREG_3P3STBY_IN_SEL NGATE/PGATE	VREG_3P3STBY_VCC
HIGH	LOW	V_5P0STBY
LOW	HIGH	V_5P0

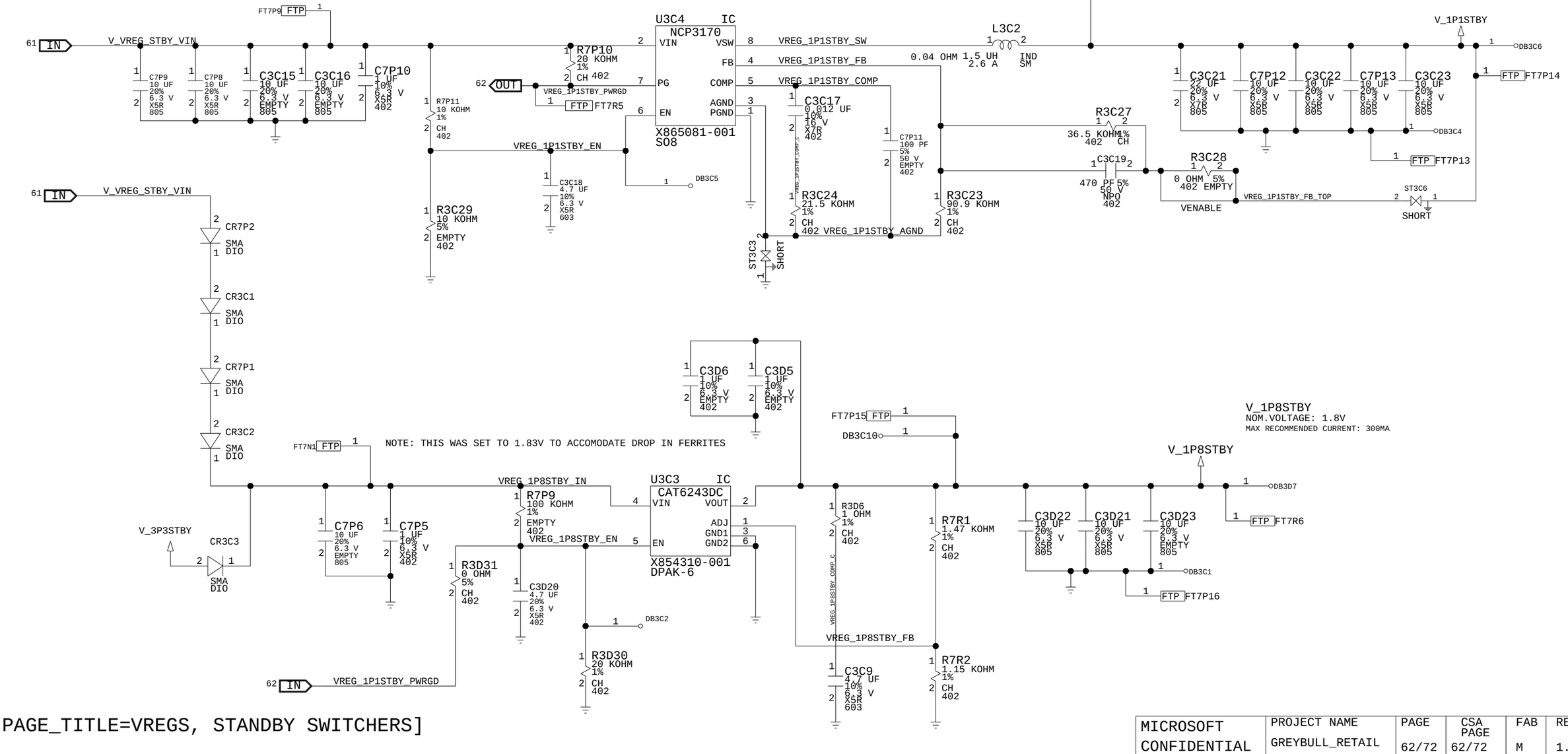


[PAGE_TITLE=VREGS, STANDBY SWITCHERS]

R5A5 SHOULD BE 182KOHM 0402, WAITING ON COMPONENT TB ADDED IN TC

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	61/72	61/72	M	1.0

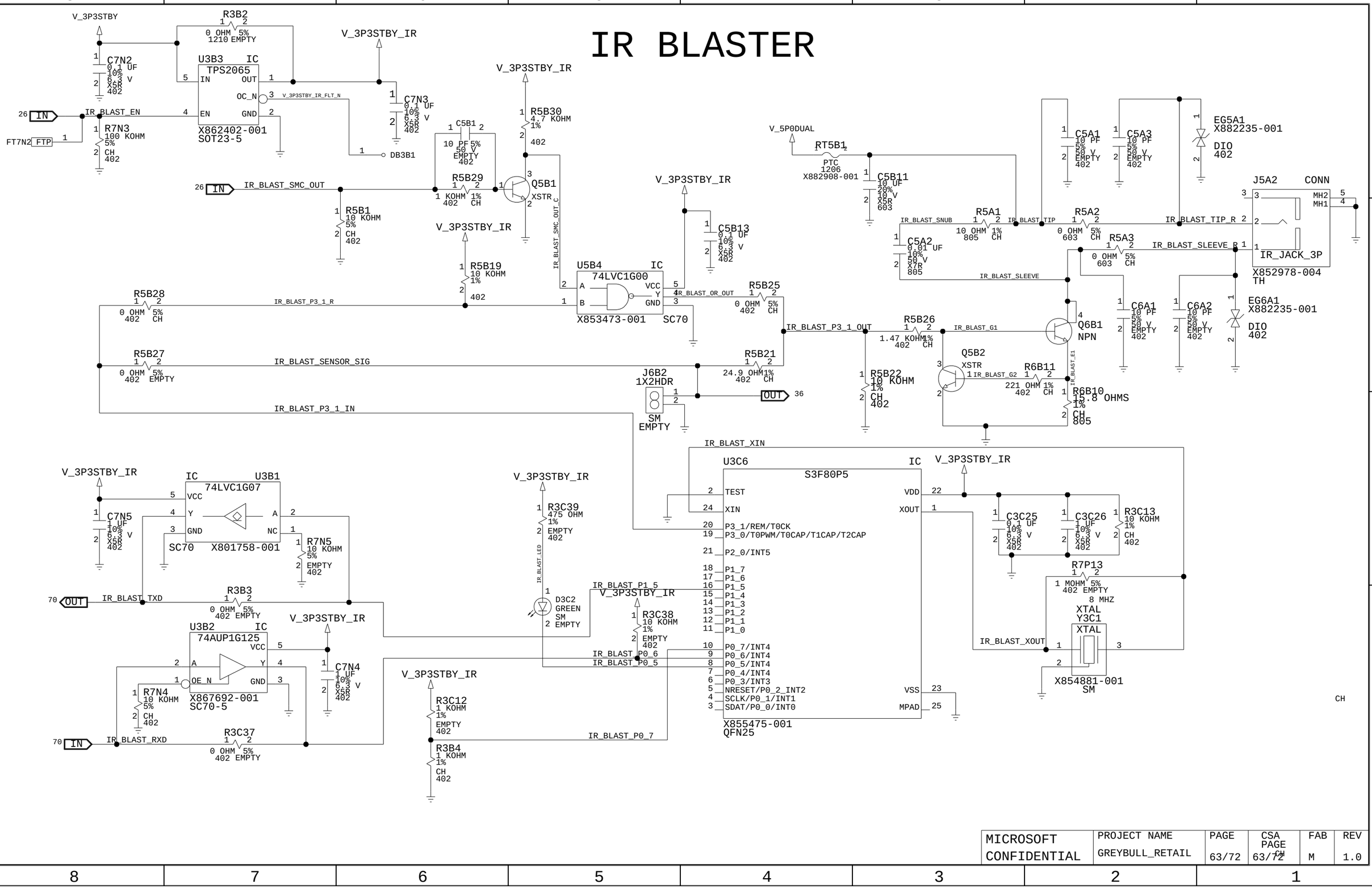
VREGS,STANDBY SWITCHERS 1P1 + 1P8



[PAGE_TITLE=VREGS, STANDBY SWITCHERS]

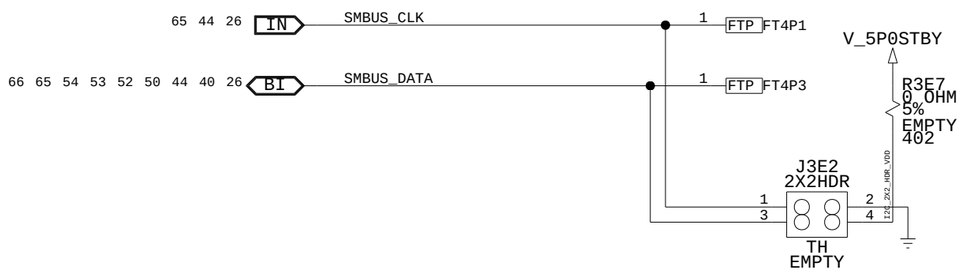
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	62/72	62/72	M	1.0

IR BLASTER



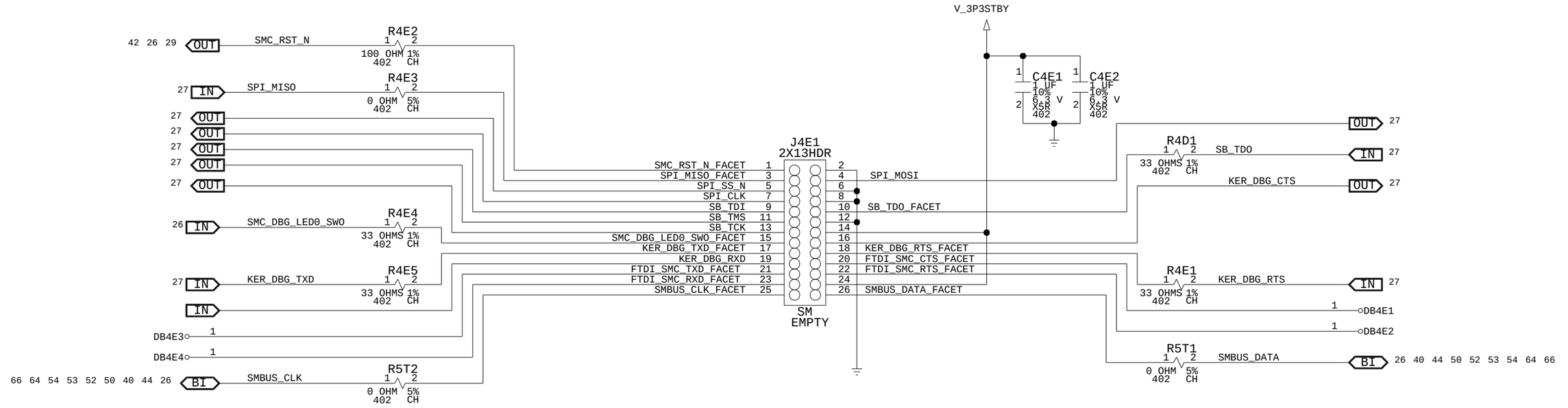
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	63/72	63/72 CH	M	1.0

I2C



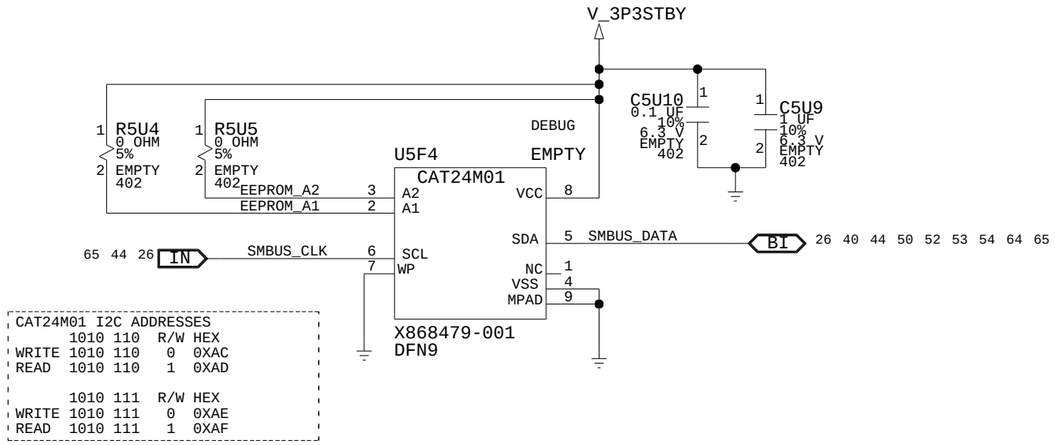
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	64/72	64/72	M	1.0

FACET, FTDI



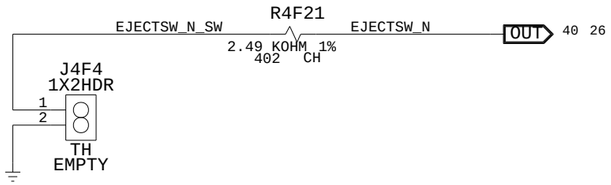
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	65/72	65/72	M	1.0

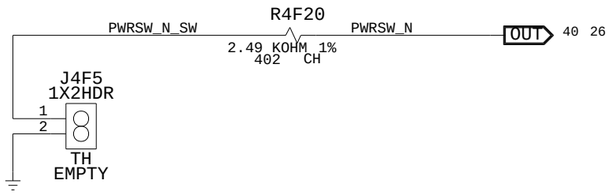
FACET, FTDI+MISC



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	66/72	66/72	M	1.0

CONN, SWITCHES



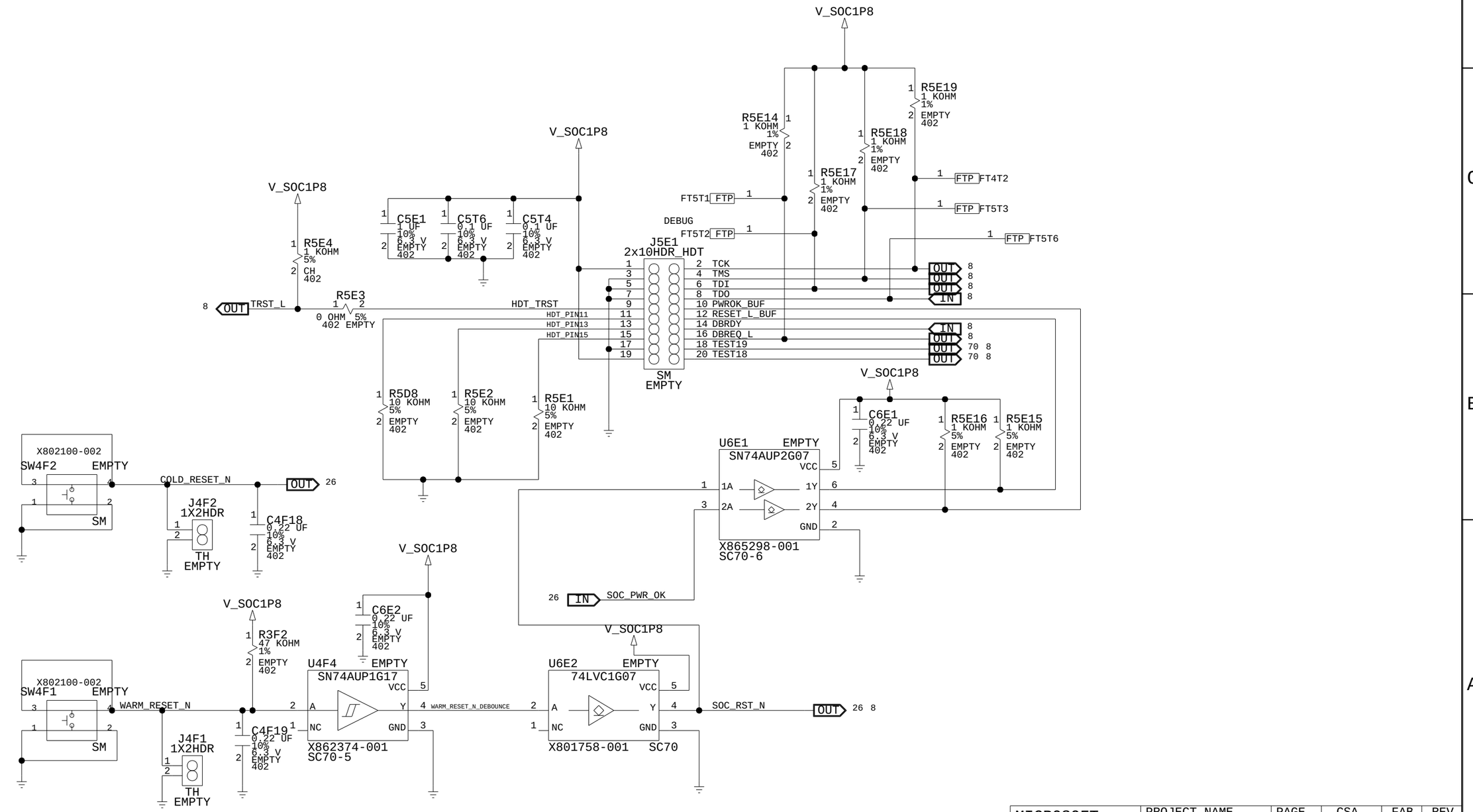






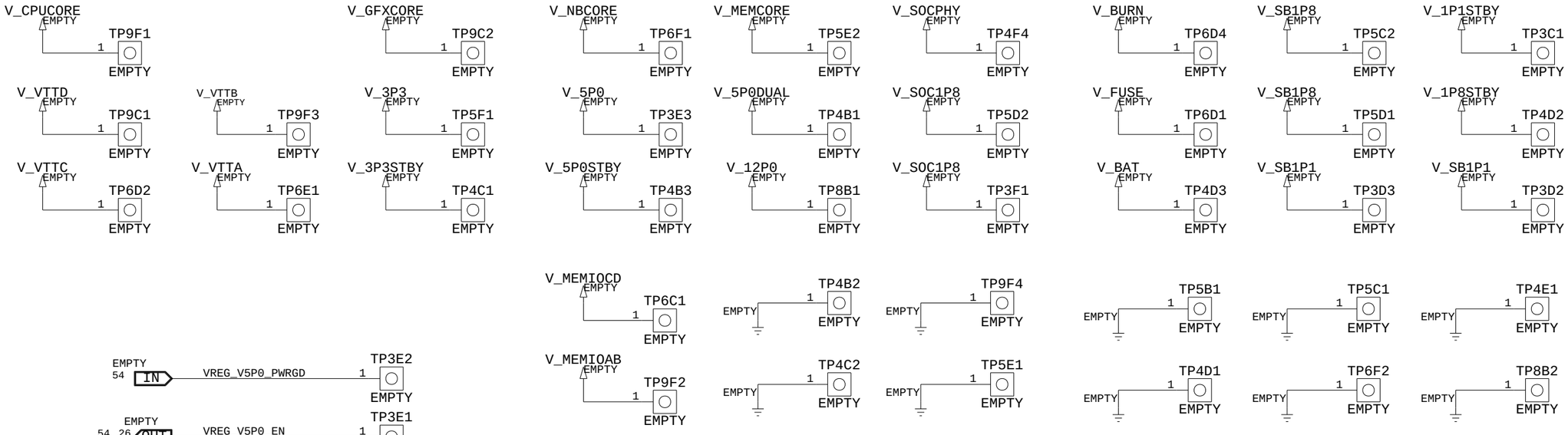
MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	67/72	67/72	M	1.0

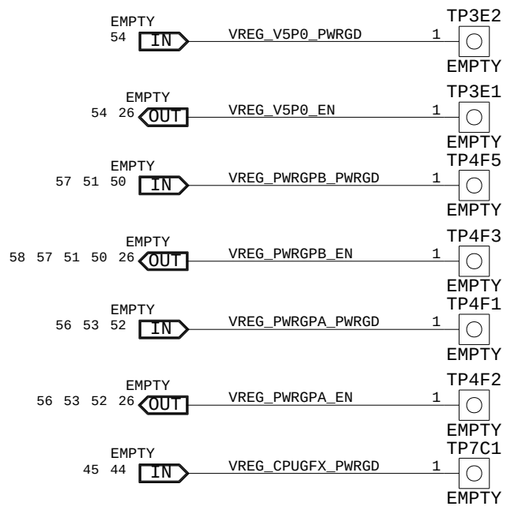
CONN, HDT



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	68/72	68/72	M	1.0

DEBUG, VR HEADERS AND TEST POINTS





MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	69/72	69/72	M	1.0

DEBUG, CONNECTORS

TITAN POWER CONNECTOR
V_3P3STBY

DEBUG

J3C2
1X2HDR
1
2
TH
EMPTY

DB4D2 FTDI_SMC_CTS

63 IN IR_BLAST_TXD

R6R8
100 OHM
5%
CH
402

R6R9
100 OHM
5%
EMPTY
402

R6R11
100 OHM
5%
CH
402

IR_BLAST_RXD OUT 63

FTDI_SMC_RTS 1 DB4D3

R6R10
100 OHM
5%
EMPTY
402

26 BI SMC_TXD
26 BI SMC_RTS
26 BI SMC_CTS
26 BI SMC_RXD

V_SOCPHY

R5F31
510 OHM
5%
CH
402

2 IN TEST25_N
2 IN TEST25_P

R5F27
510 OHM
5%
CH
402

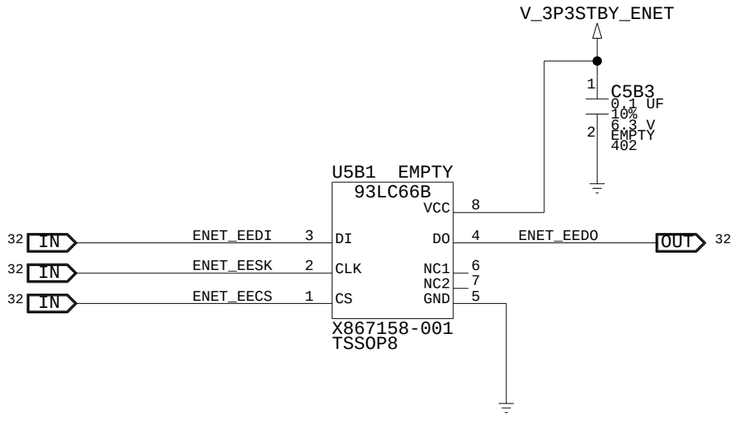
TEST19 OUT 68 8
TEST18 OUT 68 8

R5E7
1 KOHM
5%
EMPTY
402

R5E8
1 KOHM
5%
EMPTY
402

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	REV
	GREYBULL_RETAIL	70/72	70/72	1.0

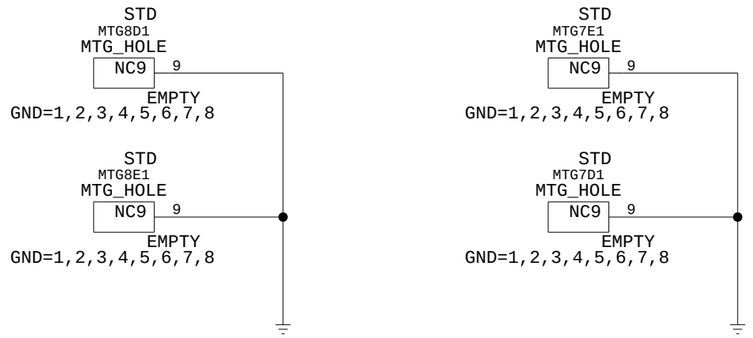
DEBUG, ENET EEPROM + MISC



MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	71/72	71/72	M	1.0

LABELS AND MOUNTING

HEAT SINK MOUNTING HOLES



INTELLIGENT SERIAL NUMBER TARGET

X801181-001
LB3F1
LABEL

1375X250_TARGET

MICROSOFT CONFIDENTIAL	PROJECT NAME	PAGE	CSA PAGE	FAB	REV
	GREYBULL_RETAIL	72/72	72/72	M	1.0